



Development of Web-Based Anonymous Feedback System for CCIS Students Using Sentiment Analysis and Algorithmic Concern Ranking

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I give thanks to God for the power to conduct this research project. I would also like to give my heartfelt thanks to Mr. Noel S. Rafanan, my instructor in CMPSC 146 Software Engineering, for his assistance in developing the project and its documentation.

The following is the development background and evaluation of ResBack for the College of Computing and Information Sciences of Mariano Marcos State University. Twenty-five students have already responded to a preliminary questionnaire using Google Forms. The additional stakeholder validation and system evaluation processes as outlined in Chapter 3 shall be done upon receiving the appropriate authorization.

CHAPTER 1 INTRODUCTION

1.1 Background of the Project

Student feedback allows the institution to gauge their experiences of the teaching process, the academic facilities and other services provided, as well as the facilities themselves. Feedback in written form will reveal issues which may be hard to identify through scoring. Richardson [8] explains the importance of the feedback tool in evaluating teaching and the effectiveness of the program. Shah et al. [9] note the need for institutions to use feedback obtained.

For the College of Computing and Information Sciences (CCIS), Mariano Marcos State University, ResBack's objective is to arrange the comments for authorized examination without compromising the anonymity of the submitter. Initially, twenty-five participants were surveyed using a Google Forms survey from September 24 to 29, 2026. All twenty-three respondents agreed or strongly agreed that they would feel more comfortable giving feedback if they could remain anonymous, and all twenty-five of them linked anonymity to truthfulness [32].

ResBack requires an account that privately links submission information with its owners for personal history and abuse protection purposes. The dashboard and feedback report for faculty lack any information about the identity of the submitter. As such, anonymity in this project is synonymous with identity suppression in the review process. However, the link through accounts stays in the database, while comments could reveal identities.

Sentiment analysis classifies the comments into positive, neutral, and negative sentiments. Reviews by Kastrati et al. [4], Dalipi et al. [5], and Shaik et al. [7] show that

sentiment analysis has been used in the field of education and faces difficulties such as context understanding, negation handling, opinion inconsistency, and language diversity. Sentiment analysis in the Philippines has been done using English and Filipino comment analysis by Balahadia et al. [2], academic-service analytics by Justo [3], and multilingual faculty assessment by Bonifacio et al. [1].

ResBack is a combination of web-browser-based concern submission and pretrained Gemma inference with the Google Gemini API. It validates the output from the model, normalizes language and concern categories, and provides summaries and report downloads. Concerns are ranked according to a programmed equation that takes into account negative volume, negative ratio, model output certainty, and recency of concerns. This project does not use model training on any language dataset.

The project combines identity protection, multi-language analysis, topic classification, concern prioritization, and role-based reporting for CCIS. It helps authorized users analyze the feedback received. Investigation, disciplinary action, and concern resolution are still the responsibility of the organization.

1.2 Statement of the Problem and Project Gap

The key issue that arises is how to gather CCIS feedback, preserve anonymity in the submission interface for the submitter, and assist the designated person in finding recurring problems. For this, there needs to be consistent labeling of sentiments and languages used, as well as summaries of topics.

Previous works have shown analytics for student feedback [2][3]. ResBack fills a gap in local integration through the use of English, Tagalog, Ilocano, and bilingual processing

together with submission of data using private accounts and a ranked rule. This will be evaluated for CCIS.

The study addresses the following questions.

1. How is a web-based system able to create a feedback mechanism for authenticated CCIS users but exclude the identities of the submitters on the faculty dashboards and feedback reports?
2. How is a pretrained language model able to classify feedback into positive, neutral, or negative and provide analysis results that are valid and actionable?
3. How is the system able to identify and categorize English, Tagalog, Ilocano, and mixed language feedback types?
4. How can submissions be categorized in terms of concern topics so that repetitive concerns can be analyzed together?
5. How can a ranking rule be developed based on negative volume, negative ratio, confidence, and recency but without making the score seem like urgency?
6. How can filtered dashboards, Excel files, and PDF reports help facilitate review by faculty and administration?
7. What evaluation criteria will be used in order to ensure functionality, analysis quality, usability, privacy and ranking behavior?

1.3 Objectives of the Project

1.3.1 Main Objective

To develop and assess ResBack, an online CCIS feedback tool which maintains anonymity of the submitter in the interface for review and incorporates pre-trained sentiment analysis, multilingual categorization, topic clustering, and explicit concern prioritization.

1.3.2 Specific Objectives

1. Describe the CCIS implementation's feedback, access, privacy, and reporting standards and verify them with the appropriate stakeholders.
2. Offer registration, login, private feedback submission, and feedback history by user with access controls from the server side.
3. Include pretrained Gemma inference and verify its results for sentiment, confidence, keywords, language, and topic.
4. Standardize the English, Tagalog, Ilocano, Taglish, Iloclish, Taglocano, Trilingual, and Other Language categories.
5. Categorize the feedback by standardized concern names and compute the weighted ranking with the number of supporting evidence samples and selected examples.
6. Include dashboards with filtering options based on date and language and generate Excel and PDF reports without identifying the submitter.
7. Evaluate the application through functional and failure-path testing, independent feedback annotations, ranking verification, and stakeholder usability study.

8. Iteratively refine the application through Agile Data Science 2.0 and develop the respective design documents, test cases, and operating instructions.

1.4 Scope and Limitations of the Project

This web application is used by CCIS students, faculty, administrative staff members, and a privileged super administrator. The users can provide their feedback and check their history. The faculty members and administrative users can use the analytics and reports facility, while the administrative users handle the user accounts and remove feedback. The mobile application is out of the scope of this project.

The feedback is allowed up to 10–2,000 characters only in the active CCIS category. The controls available are duplicate detection over a 24-hour period, a maximum of five submission per account in ten minutes, and a rejectable dictionary. The rejected submission stays saved, is exempted from model evaluation, and does not appear in faculty analytics/reporting. This does not validate any complaint.

Analysis covers overall sentiment, model-reported confidence, up to five keywords, concern topics, detected languages, and language confidence. Topics include internet access, classrooms, facilities, cleanliness, teaching, schedules, enrollment, the library, safety, equipment, administration, and other concerns. A submission may receive several topics, so topic counts can exceed submission counts.

The languages to be covered by the system include English, Tagalog, Ilocano, and any combination of these languages. Accuracy will depend on the language, sarcasm, regional variations, spelling, and sentiment. Every individual comment gets a single sentiment score regardless of the number of topics being discussed. Pretrained Gemma will be used.

Development relies on Laravel, PHP, XAMPP Database Server for the local environment, browsers and Google APIs. The model needs to have internet connection for analysis as well as access to API. Delay may happen during synchronous analysis when the API is slow. Failure of the analysis does not give any sentiment label.

Concern Ranking reflect the filtered information but not the severity of the issue itself. It is necessary to examine the weighting system along with the thirty-day recency half-life.

Emergency response, automation of problem-solving process, and the appeals process are out of scope for now.

1.5 Significance of the Project

For students, using ResBack include having a common medium through which feedback can be provided, having their identity protected in the reviewing interface, and access to their submission history. Local language and multilingual support is intended to facilitate participation.

For faculty and administrators, Sentiment summaries, topic groupings, filters, and example content may help identify patterns. Reports can be downloaded for review and discussion, while comment access will ensure that the context is retained.

For CCIS and the university, the project provides documented access rules, system diagrams, and evaluation procedures that can inform future expansion. Any improvement in institutional responsiveness will depend on how personnel use and act on the feedback.

Researchers can build on the documented model integration, safeguards, and ranking through language-specific evaluation and sensitivity analysis.

CHAPTER 2 REVIEW OF RELATED LITERATURE

2.1 Review of Related Literature and Systems

This chapter reviews student feedback, privacy, sentiment analysis, multilingual processing, concern ranking, and dashboards. It relates findings from these areas to ResBack's design and compares similar educational systems.

2.1.1 Student Feedback and Institutional Improvement

Feedback tools for the purposes of teaching, satisfaction with the program, and quality perception are discussed by Richardson [8], along with constraints of interpretation and application. The feedback approaches in Australia and Scotland are examined by Shah et al. [9].

ResBack supports this process through grouped topics and reports. The current application does not record a complete case-resolution dialogue or notify students of every administrative action. Closing the feedback loop remains an institutional responsibility and a possible system extension.

2.1.2 Identity Protection Privacy and Anonymity

The concepts of consent, de-identification, surveillance, and interpretation of student data are addressed by Slade and Prinsloo [22]. Privacy as a factor is linked to transparency, trust, and accountability by Pardo and Siemens [23]. Such principles would be conducive to implementing separate access controls on accounts and feedback reports.

In their study of university peer assessment, Fraile et al. [24] state that 73.6% of students favored anonymity, although this preference did not always enhance perception

measurement and identifiable discussion was still useful. The above finding can help to address the issue of user preferences; it cannot be automatically applied to CCIS feedback.

ResBack keeps identities hidden from faculty views and exports but maintains private user connections. It is important for users to be aware of this situation and have access to data at the back end. As per the implementation rules of the Philippine Data Privacy Act, transparency, legitimate purpose, and proportionality are highlighted [29].

2.1.3 Sentiment Analysis in Educational Feedback

Pang, Lee, and Vaithyanathan [13] provide an example of sentiment classification based on movie reviews, forming a base for sentiment-based text organization. The application of the technique in multilingual educational feedback analysis needs to be evaluated specifically.

Kastrati et al. [4] survey 92 papers on NLP and deep learning in student feedback. The paper of Dalipi et al. [5] focuses on sentiment analysis in MOOCs, while Shaik et al. [7] talk about different analytical levels, negation, and polarity. In their turn, Grimalt-Álvaro and Usart [6] pinpoint some limitations in the use of sentiment analysis tools in education.

In their analysis of lexicon-based and machine learning systems for student feedback Dervenis, Kanakis and Fitsilis [31], offer evidence that comparisons should be conducted on a similar task. The ResBack method will be evaluated against independently generated labels, and an alternative model/lexicon used as a benchmark.

2.1.4 Philippine Applications and Multilingual Feedback

Balahadia et al. [2] analyze English and Filipino teacher-evaluation comments and display sentiment results graphically. Justo [3] applies sentiment analytics to academic services

among computer studies students at Laguna State Polytechnic University. These applications support ResBack's coverage of teaching and broader CCIS services.

Bonifacio et al. [1] introduce ClasSentiments, which uses English, Tagalog, and Bisaya texts for evaluating teaching quality at a Philippine State University. The labeling of the comments, visualization of the data and human oversight make for an extremely similar functional comparison. Their models and evaluation results cannot be compared to those of ResBack's Gemma.

Bautista [15] mentions Tagalog-English code-switching as a linguistic tool, while Herrera, Aich, and Parde [16] present TweetTaglish for analyzing this language context. The two references mentioned above accept multilingual input, though neither gives a CCIS sentiment corpus or classifies Ilocano.

Samanta et al. [17] study sentiment detection in code-mixed text, whereas Conneau et al. [18] show multilingual representation learning using XLM-R. These studies lead us to perform experiments in different languages and mixed languages. ResBack will test language identification and sentiment classification independently.

2.1.5 Pretrained Language Models and Structured Analysis

The Gemma [12] and Gemma 4 [30] technical reports describe the pretrained model families and their benchmark performance. They establish model provenance but do not measure accuracy on CCIS feedback.

Kuzmanova et al. [10] use Gemini in feedback provided by students at the university level, which involves linguistic variability, slang, and sentiment. The work done by Nguyen et al.

[11] includes EduPulse, which is used for feedback in Vietnam that involves sentiment, categorization, suggestion detection, and summarization.

Requests made by ResBack require structured JSON and validation of field values prior to storage. This is done to ensure usable data format but human readable labels are necessary for meaning assessment.

2.1.6 Concern Topics Confidence and Prioritization

Pontiki et al. [14], there is a difference between identifying aspects and finding sentiments towards the aspects. ResBack provides topic labels to multi-topics reviews but only provides one single sentiment label, which makes it less effective than aspect based sentiment analysis.

Guo et al. [25], neural-network confidence does not necessarily equate to the accuracy of the predictions made by neural networks. Calibration results provided by the researchers cannot be used as proof of the validity of Gemma's calculated confidence figures.

ResBack's weighting system for its ranking (40 percent negative volume, 25 percent negative ratio, 20 percent confidence, and 15 percent recency) is a project heuristic. Their appropriateness will be evaluated by means of controlled cases, faculty evaluations, and sensitivity testing.

2.1.7 Dashboards Reporting and Human Review

Schwendimann et al. [19] discuss educational dashboards and information visualization. Kaliisa et al. [20] cover the design of teacher-oriented dashboards, stakeholder participation, privacy, and evaluation. Masiello et al. [21] summarize the existing body of work on dashboards and their limitations.

These studies deal mostly with information about learning activities, while ResBack deals with the feedback that has been submitted. The design principle which is applicable here is helping users to answer real-life review questions. This principle is implemented by ResBack through date filtering, language filtering, comments, and export.

2.1.8 Comparison of Related Systems and Synthesis

Table 1 compares the reported features, methods, and language coverage of related systems. Unreported features are not assumed to be absent.

Table 1 Comparison of related feedback systems

Related system	Setting and capability	Connection and distinction
Balahadia et al. [2]	Comments on English and Filipino teacher evaluations; opinion extraction.	Facilitates feedback categorization. ResBack incorporates identity protection based on accounts and ranking of topics.
Justo [3]	Academic services of LSPU; Sentiment analysis and Data Analytics.	Service feedback analysis support. CCIS language capabilities and ranking need local assessment.
Bonifacio et al. [1]	Evaluation process of multilingual faculty using transformer model.	Close Philippine comparison. The published work cannot be directly applied to ResBack.
Kuzmanova et al. [10]	Analysis of LLM-based feedback by students through Gemini.	Facilitates LLM Analysis Studies. ResBack enables validation of structured results and deals with failed queries.

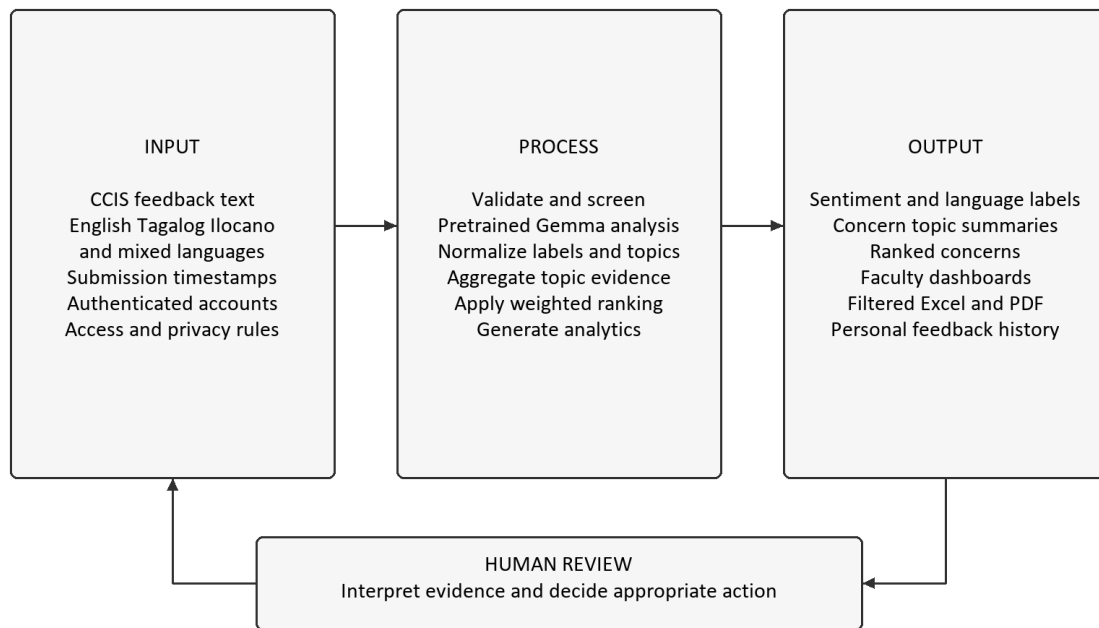
Related system	Setting and capability	Connection and distinction
Nguyen et al. [11]	Vietnamese educational feedback; LLM-assisted opinion mining.	Facilitates practical integration. Specific language and deployment environments require individual testing.
Dervenis et al. [31]	Comparison of student feedback sentiment analysis techniques.	Supports measuring model suitability rather than assuming one method performs best.
ResBack project	Feedback from CCIS; Gemma's conclusion, topic summary, report, and weighted ranking.	Initial questionnaire completed; models and task-oriented systems yet to be evaluated.

The literature supports a central feedback channel, privacy controls, automated analysis evaluation, multilingual topic categorization, and clear reports. The ResBack system incorporates all these elements for CCIS and suggests evaluation of these for this environment.

2.2 Conceptual Framework

The theoretical framework takes the input-process-output format but is human-aided. The inputs consist of comments, timestamps, accounts, and access policies. The process entails validation, screening, pretrained inference, label standardization, aggregation, ranking, and reporting. The outputs consist of classified feedback, topics summary, ranked concerns, dashboards, and downloaded reports.

ResBack conceptual framework



Interpretation and evaluation inform refinement; outcomes are not automatically guaranteed.

Figure 1 ResBack conceptual framework

Figure 1 shows that comments provide analytical content, timestamps provide recency scores, topic tags allow aggregation, and access policies control usage. The screening comes before inference and validation of the model's responses. There are privacy controls across the entire process.

Authorizations are required for reviewing the summaries along with comments. The feedback received from evaluation is used for changing interfaces, prompts, safety measures, and ranking criteria. The return arrow stands for human intervention along with system improvement and not for concern resolution through automation.

CHAPTER 3 METHODOLOGY

3.1 Project and Research Design

The project follows a developmental software engineering design to build and evaluate ResBack for the College of Computing and Information Sciences (CCIS). Development combines requirements gathering, iterative implementation, functional testing, and planned evaluation of analysis quality and usability. The preliminary student requirements questionnaire has been completed. Model evaluation and stakeholder usability testing remain proposed activities; this chapter defines their procedures rather than reporting completed performance results.

Agile Data Science 2.0, described by Journey [26], guides iterative development through Records, Charts, Reports, Recommendations, and Actions. For ResBack, these stages connect feedback collection to visual summaries, downloadable reports, concern prioritization, and authorized human review. Findings from testing and review may require changes to earlier stages. The framework is adapted to the existing Laravel application and does not require the project to adopt every technology used in the book.

ResBack integrates a pretrained machine learning model through the Google Gemini API and applies programmed rules for input protection, language and topic normalization, and concern ranking. The team does not train or fine-tune Gemma. Evaluation will therefore assess the configured model on CCIS feedback, verify the ranking calculations, and observe users completing system tasks. Questionnaire responses about expected usefulness cannot establish model accuracy or application usability.

3.2 Method and Approach

3.2.1 Setting Stakeholders and Preliminary Requirements Questionnaire

The study is situated in CCIS at Mariano Marcos State University. Students submit feedback and review their own submission history. Faculty members may submit feedback and review authorized analytics and reports. Admin and Super Admin users review analytics, manage accounts within their permissions, and delete feedback. Technical evaluators will examine access control, failure handling, reporting consistency, and maintainability.

The preliminary Google Forms questionnaire contains consent and profile questions, 25 statements rated on a five-point agreement scale, and five open-ended questions [32]. The source dataset records 25 voluntary student respondents from September 24 to 29, 2026: 17 from Computer Science and eight from Information Technology. All selected the 18–25 age bracket. The dataset contains no faculty or administrator respondents, so its findings represent the participating students rather than all stakeholder groups.

Table 2 links the project requirements to implementation areas and verification activities. These links will support traceability when features, prompts, or analytical rules are revised.

Table 2 Requirements and implementation traceability

ID	Requirement	Implementation or verification focus
R1	Authenticated private submission and history.	Account controls, FeedbackController, ownership checks.
R2	Suppress submitter identity in review outputs.	Dashboard and export field review; privacy task questions.
R3	Accept only active CCIS feedback categories.	StoreFeedbackRequest; category access tests.
R4	Screen abuse and limit repeated submissions.	FeedbackSubmissionGuard and FeedbackAutoRejectService.
R5	Validate sentiment and multilingual analysis.	SentimentService and LanguageCategoryService; annotated corpus.
R6	Normalize topics and calculate concern ranking.	ConcernTopicService and ConcernRankingService; arithmetic and sensitivity tests.
R7	Provide filtered dashboards and matching reports.	DashboardController, Excel and PDF controllers; record reconciliation.
R8	Restrict administrative and account actions.	Role middleware and account authorization; hierarchy tests.
R9	Expose analysis status and retain failed submissions.	Pending, analyzed, failed, and rejected lifecycle tests.
R10	Support repeatable evaluation and maintenance.	Test records, diagrams, modular services, and documented configuration.

The 25 students provided 625 ratings across the 25 agreement statements. Each item mean is the sum of its ratings divided by 25. Table 3 reports the unweighted mean of the five item means within each questionnaire area. These descriptive results summarize this preliminary sample and do not constitute a usability or model-performance score.

Table 3 Preliminary questionnaire descriptive areas

Descriptive area	Items	Mean rating from 1 to 5
Perceived value of feedback	1–5	4.39
Anonymity and expression	6–10	4.62
Sentiment categorization	11–15	4.23
Concern ranking	16–20	4.18
Willingness and perceived usefulness	21–25	4.27

Anonymity has the highest area mean, at 4.62 on the five-point scale. The source summary records 23 of the 25 respondents, or 92 percent, as more comfortable providing feedback through this approach; the corresponding item mean is 4.68. The item about anonymity supporting honest expression has a mean of 4.64. An item mean alone does not establish that every respondent selected an agreement response.

The open-ended questionnaire responses were grouped descriptively by recurring themes, including safer expression, institutional concerns, seriousness of issues, students affected, accessibility, and progress updates. One respondent raised concerns about irrelevant submissions and jokes. Blank responses and entries without substantive

content were excluded from the thematic summary. These observations come from questionnaire responses, not completed interviews or a formal qualitative study.

The findings support identity protection, personal feedback history, summary information, and submission safeguards. Requests for resolution updates identify a limitation: the application records submission and analysis status but does not track a complete institutional resolution process. The questionnaire does not measure classification accuracy, validate ranking weights, or determine how many students are affected by a submitted concern.

Proposed faculty and administrator interviews and walkthroughs will examine access requirements, language coverage, dashboard interpretation, and institutional review procedures. Recruitment, permission, and consent will be documented. The preliminary questionnaire sample will be reported separately from the later usability sample, even if some participants take part in both activities.

3.2.2 Agile Data Science 2.0 Development Process

The Records stage establishes accounts, feedback, categories, analysis results, and rankings with explicit ownership, validation, status, and timestamp rules. Deliverables include the data model, migrations, access specifications, and sample records for testing. Feedback text and its analysis are stored separately so that a saved submission can remain available when external analysis fails.

The Charts stage presents sentiment distributions, language categories, trends, and concern summaries from stored records. Verification will reconcile displayed totals with controlled data and check how pending, failed, or missing analysis is represented. The

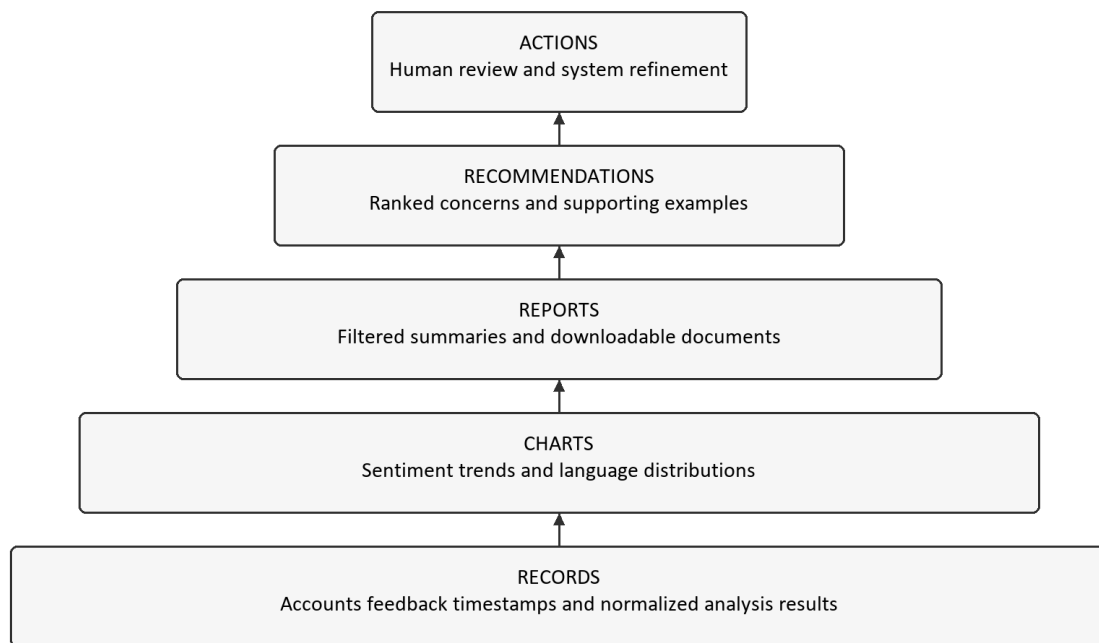
same selected date and language scope must apply to the displayed records and analytical summaries.

The Reports stage combines feedback tables, Excel exports, and PDF analytics reports using the same supported filters. Tests will compare the selected records, statuses, and applicable totals with controlled data. Review outputs must exclude rejected submissions and identifying account fields, while preserving the feedback content required for interpretation.

The Recommendations stage groups feedback into concern topics and applies the weighted ranking procedure in Section 3.2.6. Counts, negative ratios, model-reported confidence, recency, and supporting examples make the basis of each ranking visible. Controlled cases, faculty review, and sensitivity analysis will assess the weights and low-evidence indicators. A high score supports review; it does not establish urgency, severity, or the truth of a complaint.

The Actions stage supports authorized users examining concerns and deciding how the institution should respond. The research team will use evaluation findings to refine records, prompts, interfaces, and ranking rules. This adaptation does not add automated concern resolution. Appendix B presents the revised Gantt chart with retained scheduling dates, overlapping development work, and progress marked For confirmation until evidence is verified.

Agile Data Science 2.0 applied to ResBack



Iterative value building; recommendations are heuristic and actions remain with people.

Figure 2 Agile Data Science 2.0 stages adapted to ResBack

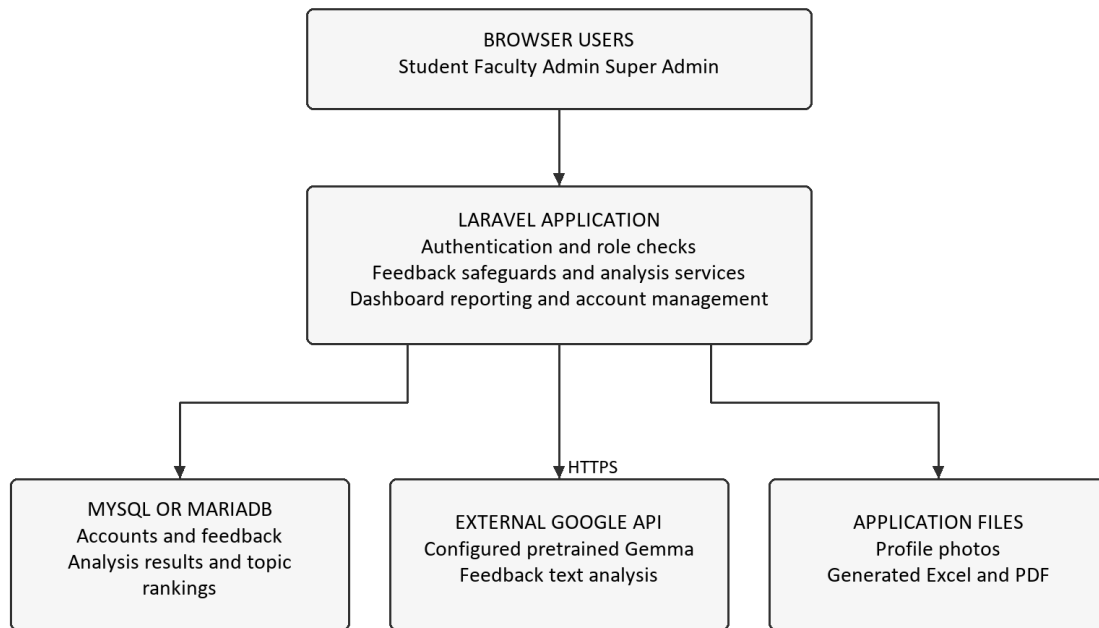
3.2.3 Application Architecture and Role Permissions

ResBack uses Laravel routes, controllers, services, Eloquent models, and Blade views. Authentication, active-account checks, and role restrictions govern protected requests. Controllers coordinate submission and reporting, while services handle screening, external analysis, normalization, and ranking. Chart.js supports charts, PhpSpreadsheet generates Excel files, and Dompdf generates PDF reports.

Figure 3 shows the local development architecture using PHP and Laravel, XAMPP MySQL or MariaDB, local application files, and the Google Gemini API over HTTPS. The browser accesses records through Laravel. Figure 31 in Appendix I presents the proposed online deployment using Docker on Render, Supabase PostgreSQL, and Supabase Storage for profile photos. This target requires database compatibility tests, storage integration,

configuration, and operational verification before it can be described as a completed deployment.

ResBack application and deployment architecture



Local web and database services are distinct from the external model inference service.

Figure 3 Local development architecture

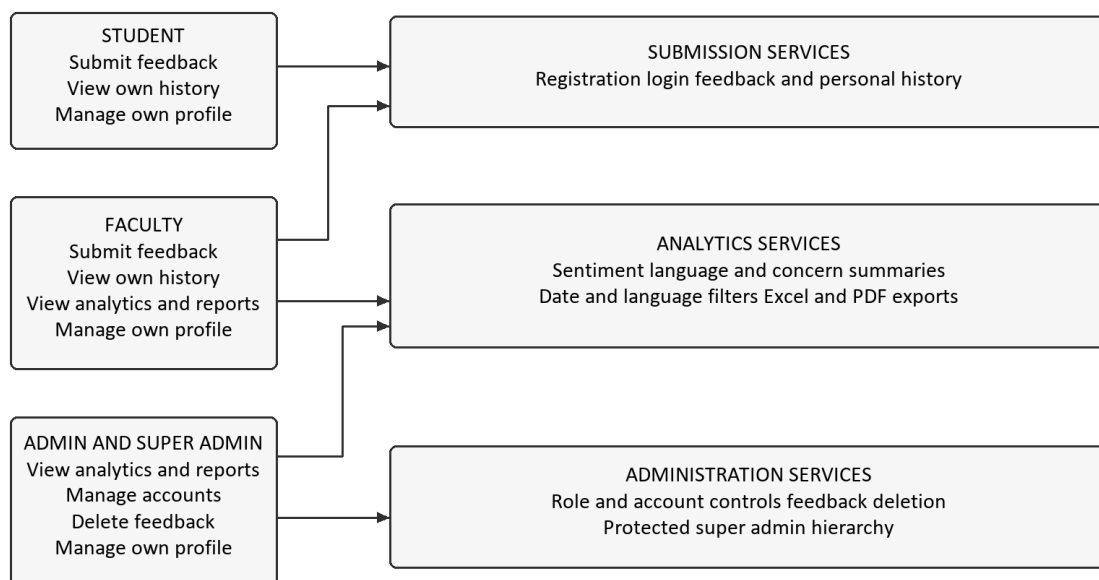
Figure 28 in Appendix G groups the application into Web Presentation, Identity and Accounts, Feedback Management, Sentiment Analysis, Concern Ranking, and Dashboard and Reporting. These are logical components within one Laravel application. Their responsibilities cover interfaces, account access, submission, analysis, scoring, and review outputs.

The component diagram uses provided and required interfaces to represent service contracts and delegation connectors to link application boundaries with internal components. External inference is supplied through the Google Gemini API. Dompdf and PhpSpreadsheet execute within the application, and the current profile-photo implementation uses the Laravel public filesystem. Relational storage holds application

records and the configured database-backed session and cache data. Accepted feedback is analyzed synchronously; rejected feedback skips external inference.

Student users can submit feedback, view only their own feedback history, and edit their profiles. Faculty users have these submission capabilities and can also access dashboards and reports. Admin and Super Admin users can review analytics and reports, administer accounts within their role hierarchy, and delete feedback, but they cannot submit feedback. Administrative controls protect the signed-in account from self-deletion, self-role changes, and self-deactivation.

ResBack roles and use case overview



Administrative roles do not submit feedback; server side access checks enforce permissions.

Figure 4 Roles and use case overview

The appendices follow the required order: Appendix A presents the Fishbone Diagram; Appendix B, the Gantt Chart; Appendix C, the Use Case Diagram; Appendix D, the Use Case Descriptions; Appendix E, the Activity Diagrams; Appendix F, the Class Diagrams; Appendix G, the Component Diagram; Appendix H, the Package Diagram; and Appendix I, the Local

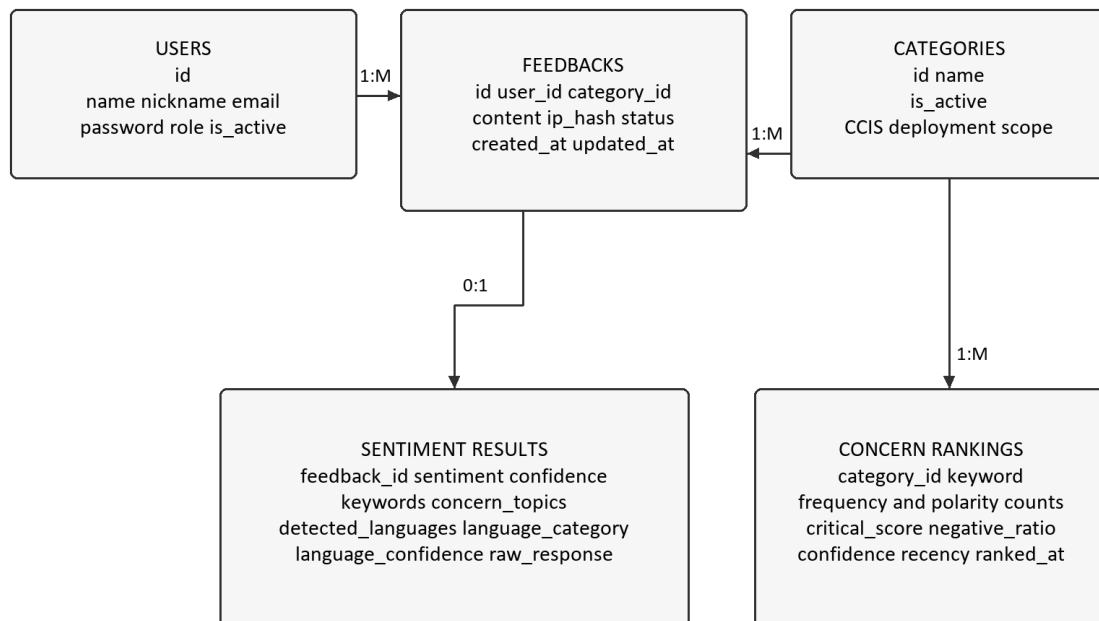
and Online Deployment Diagrams. The use case diagram contains 17 use cases, supported by 17 descriptions. Twelve activity diagrams describe workflows, and six class views describe the application structure.

3.2.4 Data Model and Feedback Lifecycle

The core entities are User, Category, Feedback, SentimentResult, and ConcernRanking. A user may own multiple feedback records, and each feedback record belongs to a category. Successful analysis creates or updates the related sentiment result. A category may have multiple topic rankings. The CCIS category defines the submission scope, while concern topics identify issues within analyzed feedback.

Feedback records store the original content, private user ownership, a hashed network address, processing status, and timestamps. Analysis records store sentiment, model-reported confidence, keywords, normalized concern topics, language information, and the provider response. Ranking records store counts and scoring metrics for each category and topic. In the current ranking model, the field named keyword stores the normalized topic label.

ResBack core data relationships



1:M means one to many; 0:1 means an optional single analysis result.

Figure 5 Core data relationships

The feedback lifecycle has four statuses. Pending identifies a saved submission awaiting analysis. Analyzed identifies a submission with a usable stored result. Failed identifies a saved submission whose analysis could not produce a usable result. Rejected identifies content matched by the configured screening rules; it skips inference and is excluded from review analytics and exports. Failed and rejected records do not receive a neutral sentiment label.

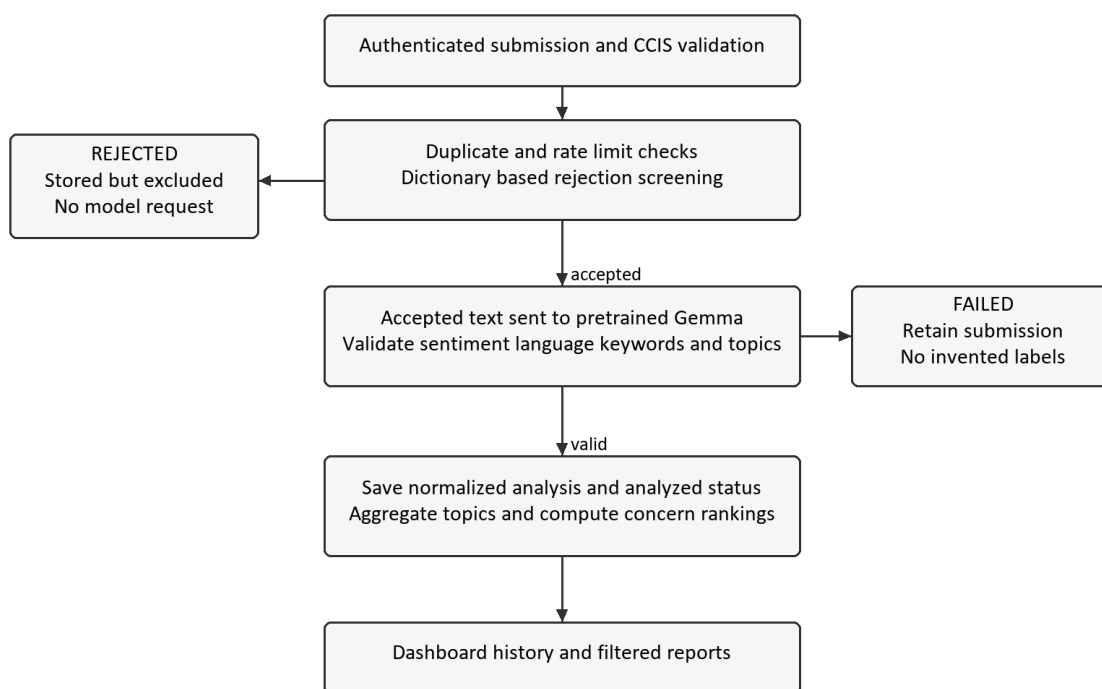
3.2.5 Submission Safeguards and Analysis Procedure

The server accepts only the active CCIS category and feedback containing 10–2,000 characters. Duplicate detection compares the same account and category over the preceding 24 hours after normalizing case and whitespace. A separate account-based limit allows five saved submissions within a ten-minute rate-limit window. Validation, duplicate, or rate-limit failures prevent the new record from being saved.

The configured prohibited-content dictionary screens validated feedback before inference. A matching submission is stored as rejected; otherwise it is stored as pending. The original feedback text is retained, and a network address hash is recorded. Both accepted and rejected saved submissions count toward the submission limit. Dictionary screening is a programmed safeguard and does not establish whether a complaint is valid. SentimentService sends feedback with classification instructions requesting sentiment, confidence, keywords, topics, languages, and language_confidence in JSON. The prompt instructs the model to treat the feedback as content rather than executable instructions. The response parser excludes thought-marked parts, parses the returned JSON, checks usable sentiment, confidence, keywords, and language fields, and clamps numeric confidence values to the interval from zero to one. Topic and language outputs are normalized before storage.

LanguageCategoryService maps detected languages to English, Tagalog, Ilocano, Taglish, Iloclish, Taglocano, Trilingual, or Other Language. ConcernTopicService maps concern labels to the supported topic set and uses keywords or feedback content as fallback evidence when needed. A usable result is stored, the feedback becomes analyzed, and its category rankings are rebuilt. Missing configuration, connection failure, an unsuccessful API response, or an unusable analysis result marks the saved feedback failed.

Feedback processing and failure paths



Duplicate and rate limited requests stop before a new feedback record is created.

Figure 6 Feedback processing and failure paths

The repository default model identifier is gemma-4-26b-a4b-it; the deployed configuration may override it. The current request uses a temperature of 0.1 and a maximum output allowance of 256 tokens. Evaluation records will identify the model actually requested, prompt version, request settings, evaluation date, and observed failures. Model or prompt changes will require evaluation on a controlled set before their results are compared.

3.2.6 Concern Ranking Procedure

ConcernRankingService groups stored analysis results by normalized topic within the selected category, date, and language scope. It counts sentiments and calculates four factors for each topic. A comment assigned to several topics contributes its single overall

sentiment to every assigned topic. This is topic grouping with comment-level sentiment; it is not a separate sentiment prediction for each aspect.

Let n be a topic's negative count, t its total analyzed count, and $n_{\max}$ the largest negative count among the selected topics, with a minimum of one. The volume factor is $V = \ln(1 + n) / \ln(1 + n_{\max})$, with $V = 0$ when $n = 0$. The negative ratio is $R = n / t$. The confidence factor C is the mean model-reported confidence of negative comments. The recency factor T is their mean $\exp(-\ln(2) \times \text{age in days} / 30)$, using non-negative ages relative to the calculation time. If there are no negative comments, C and T are zero.

The concern score is $S = 100 \times (0.40V + 0.25R + 0.20C + 0.15T)$, rounded to one decimal place. Logarithmic volume normalization reduces the influence of very large counts, while the ratio reflects the share of negative comments. Each comment's recency contribution halves after 30 days. The weights and half-life are project heuristics requiring evaluation, and model-reported confidence is not an established probability that a classification is correct [25].

For example, $V = 1.00$, $R = 0.75$, $C = 0.80$, and $T = 0.50$ produce a score of 82.3. This is an illustrative calculation. A topic with no negative comments receives a score of zero. The current low-evidence indicator is set when a topic has fewer than three total analyzed comments, regardless of sentiment. Scores are ordered from highest to lowest; ties are resolved by negative count, then topic name. Up to three recent negative examples support review.

Volume normalization depends on the largest negative count in the selected dataset. A topic score may therefore change when filters or other topic counts change, even if that topic's own records are unchanged. Verification will fix the calculation time and selected

scope for reproducibility. Sensitivity analysis will compare alternative weights and recency half-lives, including a calculation without the confidence factor.

3.2.7 Proposed Evaluation Participants and Data

The proposed usability sample targets approximately 30 CCIS students, five faculty members, and two administrative users through purposive sampling, subject to permission and availability. Student recruitment will seek variation in year level, browser access, and language use. Reviewers will have responsibilities relevant to the tested tasks. These are recruitment targets, and the achieved sample and any overlap with preliminary respondents will be reported separately.

The proposed frozen evaluation corpus contains 210 comments without direct identifiers: ten comments for each of the three sentiment classes within each of seven language categories. These categories are English, Tagalog, Ilocano, Taglish, Iloclish, Taglocano, and Trilingual. Other Language, ambiguous content, screening matches, and service failures will be examined in additional test cases. The balanced design supports coverage and does not reproduce the natural frequency of submitted feedback.

Comments may be contributed with consent or authored as explicitly identified test scenarios by contributors competent in the relevant languages. Source type will be recorded, duplicates removed, and unnecessary identifiers removed before external processing. Natural and authored cases will be reported separately where sample size permits. Prompt-development examples will be separate from the 210 evaluation comments; the evaluation set will be frozen before the final prompt is assessed.

Two independent annotators with the necessary language competence will label sentiment, languages, and concern topics. An annotation guide will define positive,

neutral, negative, mixed sentiment, borrowed words, and multiple topics. Agreement will be recorded before disagreements are adjudicated to form the reference labels. The guide and adjudication record will make the evaluation procedure reproducible.

3.2.8 Functional Testing and Quality Evaluation

Functional testing will cover authentication, history ownership, role permissions, input validation, submission safeguards, analysis success and failure, normalization, filters, account administration, deletion, and exports. Existing PHPUnit tests provide an implementation baseline. Each evaluation run will record its environment, expected and observed outcomes, failures, corrections, and retests. Passing tests will be reported only for runs that have actually been executed.

Selected product-quality criteria will be organized using ISO/IEC 25010:2023 [27]. The project will examine functional suitability, performance efficiency, reliability, security, compatibility, and maintainability. Interoperability will be considered within compatibility through API and export behavior. Maintainability checks will examine modular services, traceability, and repeatable tests. Role-based task observations and SUS will address usability. These selected measures do not constitute certification or a complete assessment of every characteristic in the standard.

End-to-end tests will use isolated test accounts and a separate test database. External model calls will be mocked for deterministic application tests and evaluated separately through controlled inference requests. Failure cases include missing configuration, connection timeouts, unsuccessful responses, malformed JSON, and invalid labels. Export checks will reconcile selected records and applicable totals and confirm that identifying account fields and rejected feedback are excluded.

Performance records will include submission latency, dashboard latency, export duration, analysis failure rate, dataset size, environment, and API conditions. Repeated requests will be used to report latency distributions rather than a single measurement. External inference duration will be recorded separately from local processing where instrumentation permits, because synchronous model requests affect the total submission response time.

Table 4 Proposed evaluation measures and records

Area	Proposed measure	Evidence to record
Functional behavior	Expected outcomes for permissions, safeguards, filters, and exports.	Case results, failures, fixes, and retests.
Sentiment analysis	Accuracy, class precision and recall, macro-F1.	Confusion matrix; labels; ambiguous and failed attempts.
Language and topics	Language accuracy; topic micro- and macro-F1.	Category counts, missed or extra labels, annotator agreement.
Concern ranking	Arithmetic checks, rank agreement, leading-concern agreement, sensitivity.	Controlled data; independent faculty rankings and comments.
Usability	Task completion, errors, assistance, time, and SUS.	Role-specific observations; scores and achieved sample.
Performance and reliability	Latency distributions, export duration, failure rate.	Environment, dataset size, API conditions, repeated runs.
Security and privacy	Ownership, role, export identity, and data handling checks.	Access test results; disclosures; approved data procedures.

3.2.9 Proposed Usability and Analysis Measures

Usability tasks will follow role permissions. Students will submit feedback, inspect the result status, and locate their own history. Faculty members will filter analytics, inspect supporting examples, and generate reports. Administrative users will perform permitted account and feedback actions on test data. Task completion, errors, assistance, and completion time will be recorded before participants answer the ten-item System Usability Scale (SUS) [28].

SUS uses ten items rated on a five-point response scale. For odd-numbered items, the contribution is the response minus one. For even-numbered items, the contribution is five minus the response. The ten contributions are summed and multiplied by 2.5 to produce a score from zero to 100. This score is a usability index, not a percentage of completed tasks.

Sentiment evaluation will compare predictions with adjudicated reference labels using accuracy, per-class precision and recall, macro-F1, and a confusion matrix. Language-category accuracy will be reported overall and by category. Topic assignment will use micro-F1 and macro-F1 with examples of missed and extra labels. Unusable responses and failed requests will be reported separately, together with valid-response coverage; accuracy among valid results alone must not conceal operational failures.

Controlled records will verify ranking arithmetic, tie-breaking, date effects, filtering, and low-evidence indicators. Faculty reviewers will prioritize the same anonymous evidence independently before seeing the system ranking. Comparison will examine rank agreement and agreement on leading concerns, with the chosen rank statistic and tie

procedure specified before evaluation. Disagreements will be retained for analysis rather than treated as proof that either ranking is correct.

Acceptance criteria will be agreed with stakeholders before evaluation. Ownership and role checks must pass, exports must match the selected scope, and processing failures must remain visible. Model and usability reports will include errors, sample limitations, and achieved coverage. A benchmark suitable for the tested language subset may be evaluated on the same reference cases; its language coverage and configuration must be documented before comparison. Comparisons between manual and dashboard-assisted review require a separately specified procedure.

3.2.10 Ethical Procedures and Data Management

Participation will be voluntary and subject to institutional approval. Information sheets will explain the study purpose, tasks, data use, withdrawal conditions, and external model processing. Participation or refusal will not affect grades or ordinary account access. The proposed research procedures and the preliminary questionnaire will be distinguished in the study record.

Research testing will use test accounts, limited identifiers, restricted research-file access, and approved retention and deletion periods. Sensitive allegations will not be solicited as test material. Identity masking in a dashboard does not prevent identification through free text or erase the application's private ownership link. The institution will review data collection, access, disclosure, and external processing against the applicable Data Privacy Act principles [29].

3.3 Machine Learning Integration and Evaluation

The machine learning workflow below connects the implemented Gemma integration to data acquisition, preprocessing, feature extraction, model evaluation, and model utilization. Detailed safeguards, data specifications, and metrics remain in Sections 3.2.5–3.2.9. The project uses an externally pretrained model [12][30]; its planned experiments evaluate inference and application behavior rather than training new model weights.

3.3.1 Data Acquisition

Operational input consists of feedback submitted by authenticated Student and Faculty users, with category and timestamp information. The application retains a private ownership link while omitting identifying account fields from review outputs. Research data will consist of the separately documented, consented or authored evaluation comments described in Section 3.2.7. The requirements questionnaire informs feature design and is not a labeled sentiment-training dataset.

3.3.2 Preprocessing and Data Cleaning

Application preprocessing validates the category and text length, compares normalized text for duplicates, enforces submission limits, and screens configured prohibited terms. Original accepted text is passed to the model without a separate stemming, stop-word-removal, or TF-IDF pipeline. Research preparation will remove unnecessary identifiers and duplicates and record language and reference labels. Negation, mixed language, and wording that conveys sentiment will be retained so that cleaning does not change the intended meaning.

3.3.3 Feature Extraction Using the Pretrained Model

Gemma uses representations learned during pretraining. ResBack does not extract internal embeddings or build a manually trained feature vector. It receives sentiment, self-reported confidence, keywords, concern topics, and language information.

Validation and normalization turn these outputs into analytical attributes for filtering, aggregation, and ranking. The programmed ranking factors are derived from stored results and timestamps.

3.3.4 Model Configuration and Performance Evaluation

Each experiment will fix the model, prompt, settings, and evaluation cases. Two annotators will establish reference labels independently. The measures in Section 3.2.9 cover sentiment accuracy, precision, recall, and macro-F1; language accuracy; and topic micro-F1 and macro-F1. Results will include sample counts, coverage, failures, and examples. Separate prompt-development data will reduce evaluation leakage. Training loss and epochs are outside the implemented inference workflow.

3.3.5 Utilization of the Model

Accepted feedback undergoes synchronous inference, validation, normalization, result storage, and ranking reconstruction. Dashboards and reports present the selected records for authorized review. Rejected content skips inference, and failed analysis retains its visible status. Reviewers examine original comments and supporting evidence before deciding on institutional action. Evaluation findings guide further Agile Data Science 2.0 iterations and reassessment of revised configurations.

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APPENDIX A FISHBONE DIAGRAM

ResBack | Detailed Fishbone Diagram

CCIS feedback collection, analysis and institutional review | 8 categories / 15 subcategories / 24 candidate causes

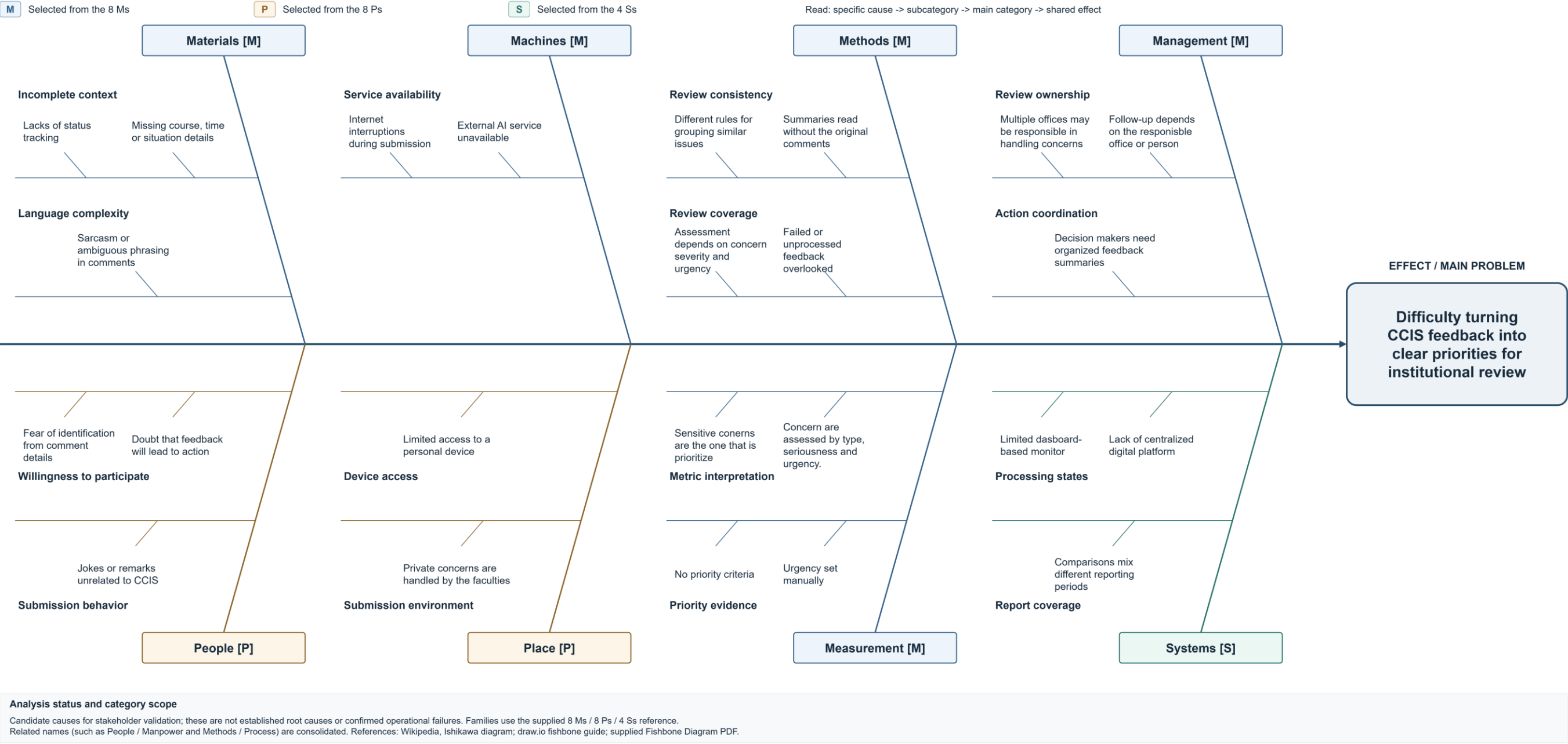


Figure 7 ResBack fishbone diagram

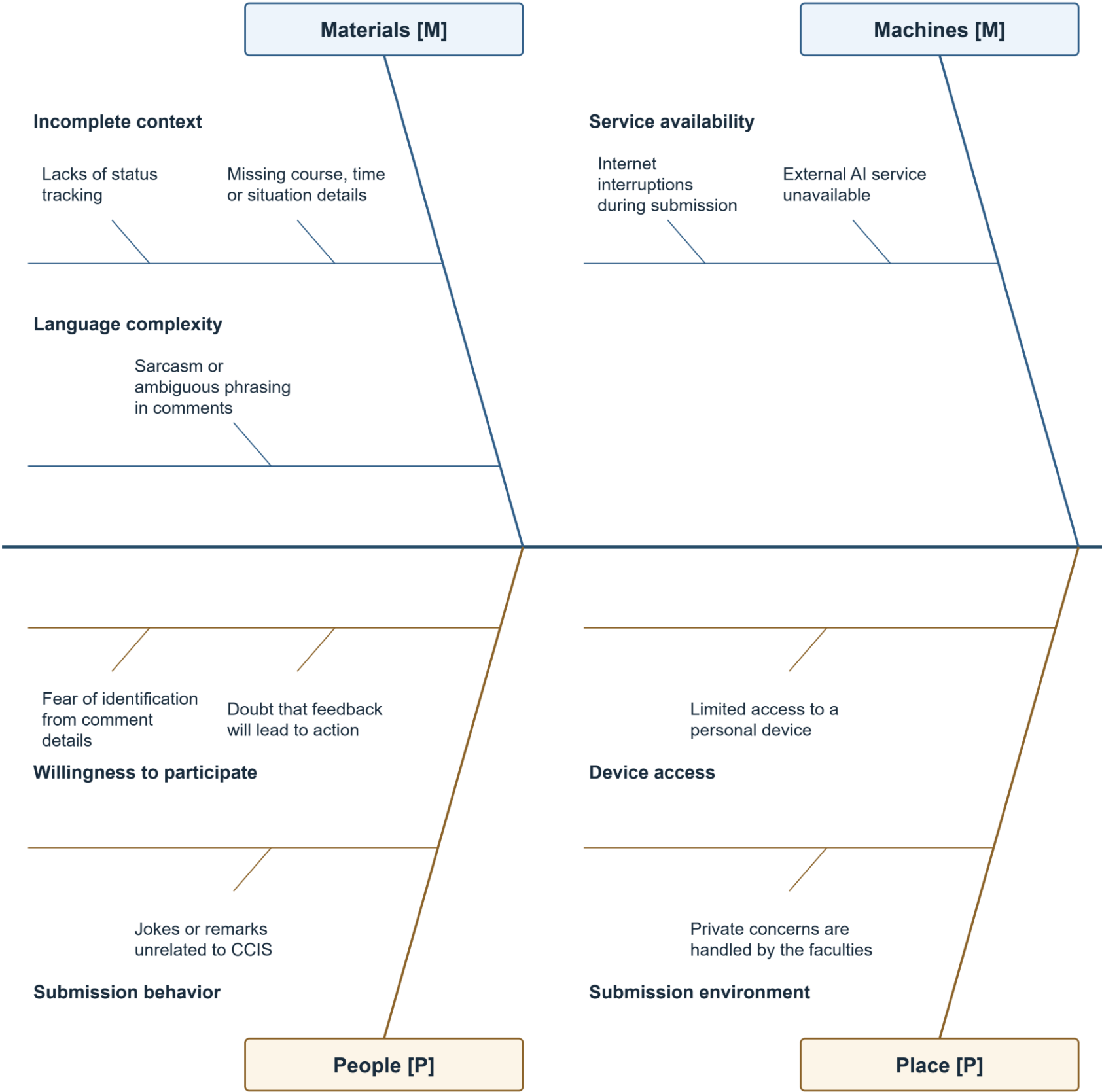


Figure 7 ResBack fishbone diagram enlarged continuation panel 1 of 2

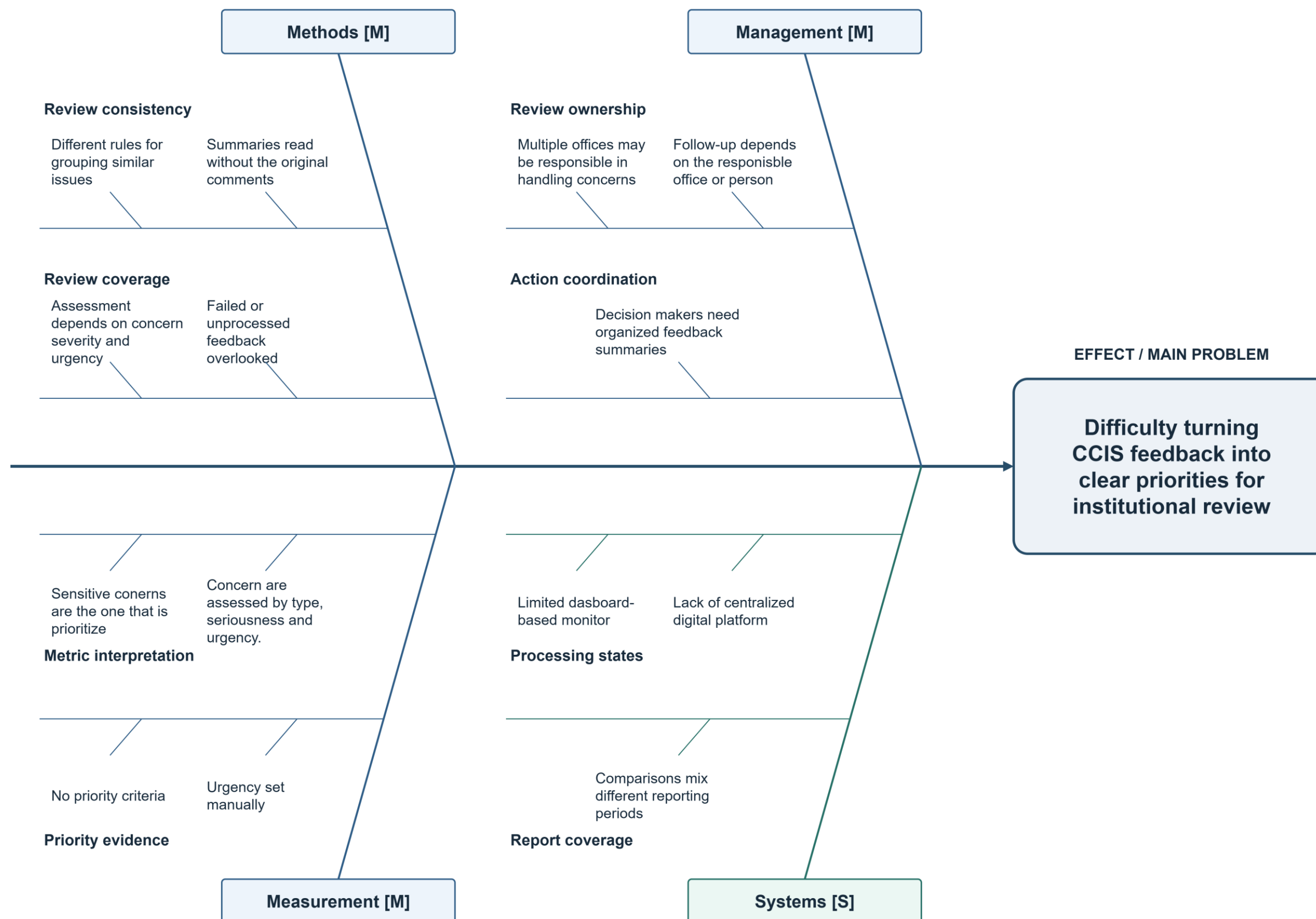


Figure 7 ResBack fishbone diagram enlarged continuation panel 2 of 2

APPENDIX B GANTT CHART

Figure 8 presents the ResBack schedule adapted to Agile Data Science 2.0. It retains the original task dates and assigned team members from August 3 to October 31, 2026.

Preparation supports the five data-value stages of Records, Charts, Reports, Recommendations, and Actions. Testing, iteration, documentation, and delivery support work across these stages.

All task progress entries are marked For confirmation. Scheduled bars indicate planned task windows and do not establish completed work. The chart retains the standard template formatting. Enlarged continuation panels repeat the task names so that task details can be matched to their timeline and deliverables.

W1 covers August 3–9. Each following week begins on Monday; W5 begins August 31, W9 begins September 28, and W13 covers October 26–31. The final partial week is included so that tasks ending October 31 remain visible.

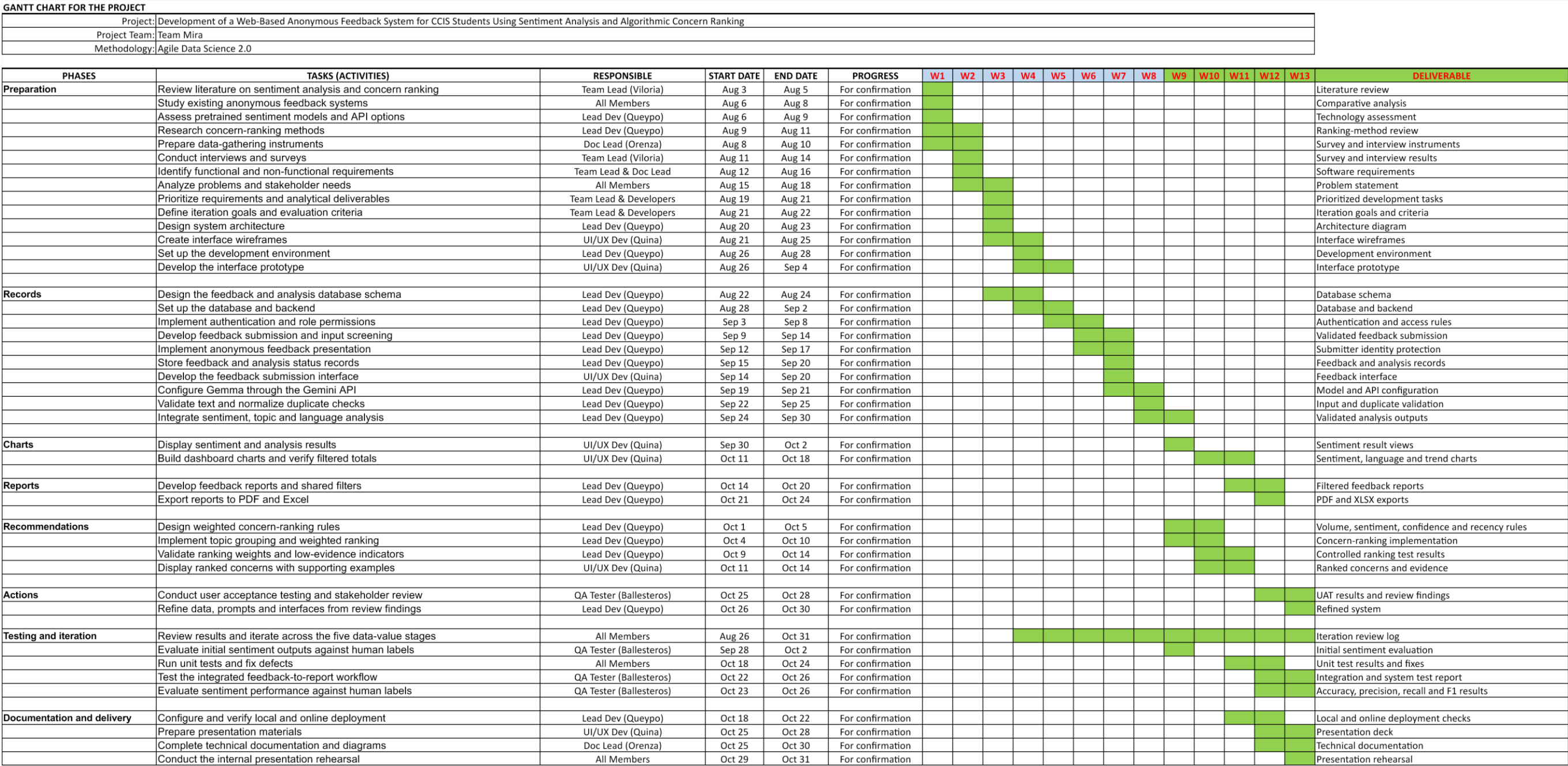


Figure 8 ResBack Gantt chart using Agile Data Science 2.0

Preparation Task details

PHASES	TASKS (ACTIVITIES)	RESPONSIBLE	START DATE	END DATE	PROGRESS
Preparation	Review literature on sentiment analysis and concern ranking	Team Lead (Viloria)	Aug 3	Aug 5	For confirmation
	Study existing anonymous feedback systems	All Members	Aug 6	Aug 8	For confirmation
	Assess pretrained sentiment models and API options	Lead Dev (Queypo)	Aug 6	Aug 9	For confirmation
	Research concern-ranking methods	Lead Dev (Queypo)	Aug 9	Aug 11	For confirmation
	Prepare data-gathering instruments	Doc Lead (Orenza)	Aug 8	Aug 10	For confirmation
	Conduct interviews and surveys	Team Lead (Viloria)	Aug 11	Aug 14	For confirmation
	Identify functional and non-functional requirements	Team Lead & Doc Lead	Aug 12	Aug 16	For confirmation
	Analyze problems and stakeholder needs	All Members	Aug 15	Aug 18	For confirmation
	Prioritize requirements and analytical deliverables	Team Lead & Developers	Aug 19	Aug 21	For confirmation
	Define iteration goals and evaluation criteria	Team Lead & Developers	Aug 21	Aug 22	For confirmation
	Design system architecture	Lead Dev (Queypo)	Aug 20	Aug 23	For confirmation
	Create interface wireframes	UI/UX Dev (Quina)	Aug 21	Aug 25	For confirmation
	Set up the development environment	Lead Dev (Queypo)	Aug 26	Aug 28	For confirmation
	Develop the interface prototype	UI/UX Dev (Quina)	Aug 26	Sep 4	For confirmation

Timeline and deliverables Same tasks and order

TASKS (ACTIVITIES)	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	DELIVERABLE
Review literature on sentiment analysis and concern ranking														Literature review
Study existing anonymous feedback systems														Comparative analysis
Assess pretrained sentiment models and API options														Technology assessment
Research concern-ranking methods														Ranking-method review
Prepare data-gathering instruments														Survey and interview instruments
Conduct interviews and surveys														Survey and interview results
Identify functional and non-functional requirements														Software requirements
Analyze problems and stakeholder needs														Problem statement
Prioritize requirements and analytical deliverables														Prioritized development tasks
Define iteration goals and evaluation criteria														Iteration goals and criteria
Design system architecture														Architecture diagram
Create interface wireframes														Interface wireframes
Set up the development environment														Development environment
Develop the interface prototype														Interface prototype

Figure 8 ResBack Gantt chart enlarged continuation panel 1 of 3

Records to Recommendations Task details

PHASES	TASKS (ACTIVITIES)	RESPONSIBLE	START DATE	END DATE	PROGRESS
Records	Design the feedback and analysis database schema	Lead Dev (Queypo)	Aug 22	Aug 24	For confirmation
	Set up the database and backend	Lead Dev (Queypo)	Aug 28	Sep 2	For confirmation
	Implement authentication and role permissions	Lead Dev (Queypo)	Sep 3	Sep 8	For confirmation
	Develop feedback submission and input screening	Lead Dev (Queypo)	Sep 9	Sep 14	For confirmation
	Implement anonymous feedback presentation	Lead Dev (Queypo)	Sep 12	Sep 17	For confirmation
	Store feedback and analysis status records	Lead Dev (Queypo)	Sep 15	Sep 20	For confirmation
	Develop the feedback submission interface	UI/UX Dev (Quina)	Sep 14	Sep 20	For confirmation
	Configure Gemma through the Gemini API	Lead Dev (Queypo)	Sep 19	Sep 21	For confirmation
	Validate text and normalize duplicate checks	Lead Dev (Queypo)	Sep 22	Sep 25	For confirmation
	Integrate sentiment, topic and language analysis	Lead Dev (Queypo)	Sep 24	Sep 30	For confirmation
Charts	Display sentiment and analysis results	UI/UX Dev (Quina)	Sep 30	Oct 2	For confirmation
	Build dashboard charts and verify filtered totals	UI/UX Dev (Quina)	Oct 11	Oct 18	For confirmation
Reports	Develop feedback reports and shared filters	Lead Dev (Queypo)	Oct 14	Oct 20	For confirmation
	Export reports to PDF and Excel	Lead Dev (Queypo)	Oct 21	Oct 24	For confirmation
Recommendations	Design weighted concern-ranking rules	Lead Dev (Queypo)	Oct 1	Oct 5	For confirmation
	Implement topic grouping and weighted ranking	Lead Dev (Queypo)	Oct 4	Oct 10	For confirmation
	Validate ranking weights and low-evidence indicators	Lead Dev (Queypo)	Oct 9	Oct 14	For confirmation
	Display ranked concerns with supporting examples	UI/UX Dev (Quina)	Oct 11	Oct 14	For confirmation

Timeline and deliverables Same tasks and order

TASKS (ACTIVITIES)	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	DELIVERABLE
Design the feedback and analysis database schema														Database schema
Set up the database and backend														Database and backend
Implement authentication and role permissions														Authentication and access rules
Develop feedback submission and input screening														Validated feedback submission
Implement anonymous feedback presentation														Submitter identity protection
Store feedback and analysis status records														Feedback and analysis records
Develop the feedback submission interface														Feedback interface
Configure Gemma through the Gemini API														Model and API configuration
Validate text and normalize duplicate checks														Input and duplicate validation
Integrate sentiment, topic and language analysis														Validated analysis outputs
Display sentiment and analysis results														Sentiment result views
Build dashboard charts and verify filtered totals														Sentiment, language and trend charts
Develop feedback reports and shared filters														Filtered feedback reports
Export reports to PDF and Excel														PDF and XLSX exports
Design weighted concern-ranking rules														Volume, sentiment, confidence and recency rules
Implement topic grouping and weighted ranking														Concern-ranking implementation
Validate ranking weights and low-evidence indicators														Controlled ranking test results
Display ranked concerns with supporting examples														Ranked concerns and evidence

Figure 8 ResBack Gantt chart enlarged continuation panel 2 of 3

Actions Testing and Delivery Task details

PHASES	TASKS (ACTIVITIES)	RESPONSIBLE	START DATE	END DATE	PROGRESS
Actions	Conduct user acceptance testing and stakeholder review	QA Tester (Ballesteros)	Oct 25	Oct 28	For confirmation
	Refine data, prompts and interfaces from review findings	Lead Dev (Queypo)	Oct 26	Oct 30	For confirmation
Testing and iteration	Review results and iterate across the five data-value stages	All Members	Aug 26	Oct 31	For confirmation
	Evaluate initial sentiment outputs against human labels	QA Tester (Ballesteros)	Sep 28	Oct 2	For confirmation
	Run unit tests and fix defects	All Members	Oct 18	Oct 24	For confirmation
	Test the integrated feedback-to-report workflow	QA Tester (Ballesteros)	Oct 22	Oct 26	For confirmation
	Evaluate sentiment performance against human labels	QA Tester (Ballesteros)	Oct 23	Oct 26	For confirmation
Documentation and delivery	Configure and verify local and online deployment	Lead Dev (Queypo)	Oct 18	Oct 22	For confirmation
	Prepare presentation materials	UI/UX Dev (Quina)	Oct 25	Oct 28	For confirmation
	Complete technical documentation and diagrams	Doc Lead (Orenza)	Oct 25	Oct 30	For confirmation
	Conduct the internal presentation rehearsal	All Members	Oct 29	Oct 31	For confirmation

Timeline and deliverables Same tasks and order

TASKS (ACTIVITIES)	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	DELIVERABLE
Conduct user acceptance testing and stakeholder review														UAT results and review findings
Refine data, prompts and interfaces from review findings														Refined system
Review results and iterate across the five data-value stages														Iteration review log
Evaluate initial sentiment outputs against human labels														Initial sentiment evaluation
Run unit tests and fix defects														Unit test results and fixes
Test the integrated feedback-to-report workflow														Integration and system test report
Evaluate sentiment performance against human labels														Accuracy, precision, recall and F1 results
Configure and verify local and online deployment														Local and online deployment checks
Prepare presentation materials														Presentation deck
Complete technical documentation and diagrams														Technical documentation
Conduct the internal presentation rehearsal														Presentation rehearsal

Figure 8 ResBack Gantt chart enlarged continuation panel 3 of 3

APPENDIX C USE CASE DIAGRAM

Figure 9 shows the 17 ResBack use cases and their actor associations. Student, Faculty, Admin, and Super Admin permissions are represented within the system boundary.

Super Admin inherits administrative capabilities, while feedback screening and accepted-feedback analysis are associated with submission.

Appendix D provides the corresponding use case descriptions. Figure numbers identify each diagram's position in this paper; labels within the source diagram sheets remain unchanged.

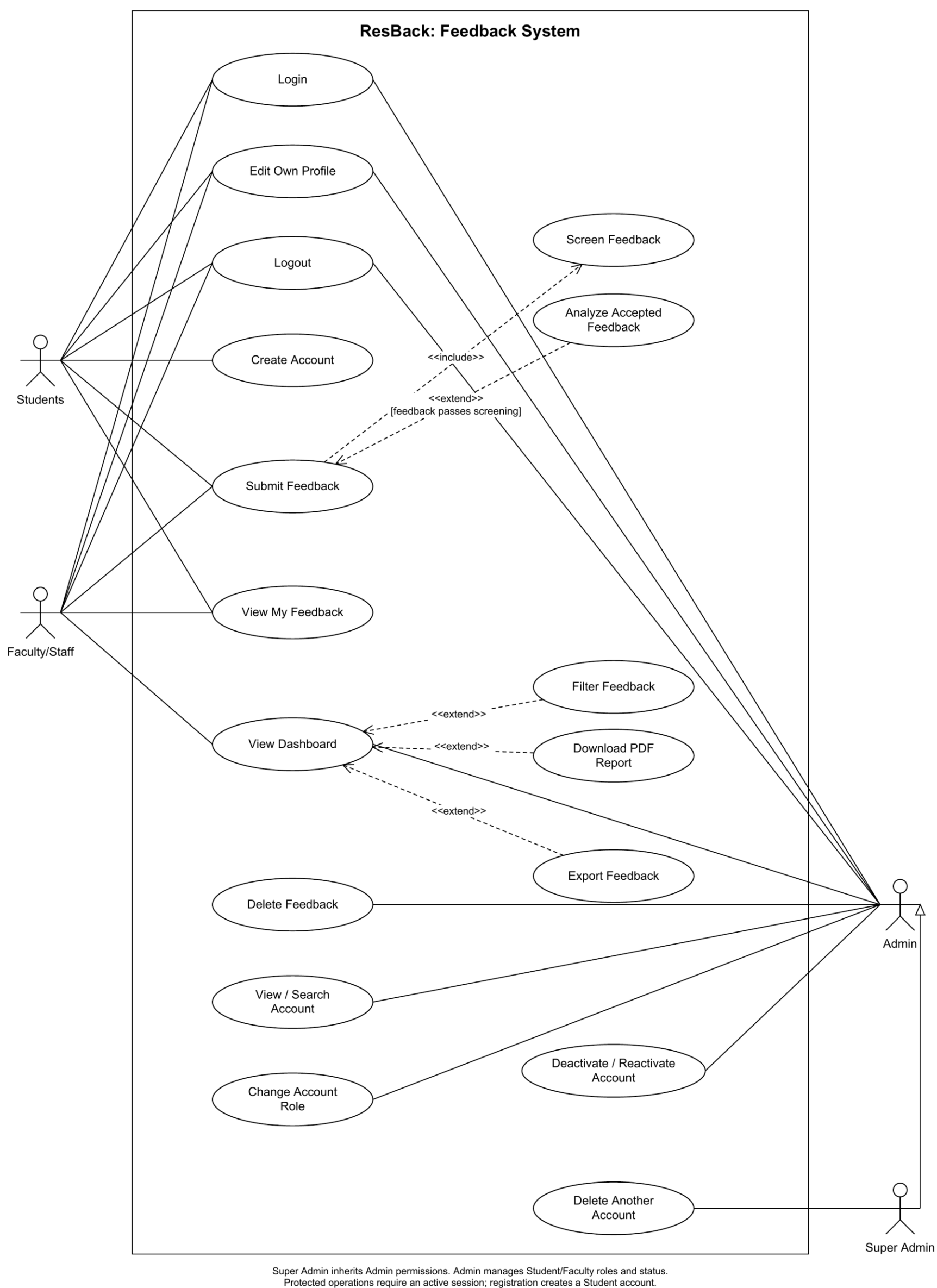


Figure 9 ResBack UML use case diagram

APPENDIX D USE CASE DESCRIPTIONS

The following 17 descriptions correspond to the use cases in Figure 9. Each table identifies the actors, preconditions, trigger, main steps, alternate and exception flows, postconditions, and non-functional requirements. The tables retain the previously prepared Lab 5 description format.

UC-01 Create Account

UC-02 Login

UC-03 Submit Feedback

UC-04 Screen Feedback

UC-05 Edit Own Profile

UC-06 Analyze Accepted Feedback

UC-07 View My Feedback

UC-08 View Dashboard

UC-09 Filter Feedback

UC-10 Download PDF Report

UC-11 Export Feedback

UC-12 Delete Feedback

UC-13 View / Search Account

UC-14 Delete Another Account

UC-15 Change Account Role

UC-16 Deactivate / Reactivate Account

UC-17 Logout

Table 5 UC-01 Create Account

Use Case Name	Create Account	
Use Case ID	UC-01	
Use Case Description	Allows a new Student to register an account and access the feedback submission form.	
Primary Actor	Student registering a new account	
Secondary Actor	None	
Precondition	The Student is not signed in and can access the registration page.	
Postcondition	A Student account is saved and an authenticated session opens the feedback form.	
Normal Flow (Main Flow)	User Action	System Response
	1. Selects Create Account.	2. Displays the registration form.
	3. Enters a nickname, email, password and password confirmation; submits the form.	4. Validates the required fields, uniqueness and password confirmation.
	5. None	6. Saves the account with the Student role and a hashed password.
	7. None	8. Starts an authenticated session and displays the feedback form.
Alternate Flow	1a. A Student who already has an account selects Login (UC-02) instead of registering again.	
Exception Flow	4a. Invalid or duplicate details prevent account creation. The system identifies the fields to correct. A storage failure prevents successful registration.	
Non-Functional Requirements	Passwords must be hashed. Validation must not reveal stored passwords. The authenticated session must be regenerated.	

Table 6 UC-02 Login

Use Case Name	Login	
Use Case ID	UC-02	
Use Case Description	Allows an existing user to access ResBack according to the account role.	
Primary Actor	Student, Faculty/Staff, Admin or Super Admin	
Secondary Actor	None	
Precondition	The user has an active account and is not currently signed in.	
Postcondition	The user is authenticated and reaches the feedback form or dashboard allowed for the role.	
Normal Flow (Main Flow)	User Action	System Response
	1. Opens Login.	2. Displays the email and password fields.
	3. Enters the registered email and password; selects Login.	4. Validates the input, verifies the credentials and checks that the account is active.
	5. None	6. Starts an authenticated session and regenerates the session identifier.
	7. None	8. Opens the Student feedback form or the Faculty/Staff, Admin or Super Admin dashboard according to the account role.
Alternate Flow	1a. A Student without an account selects Create Account (UC-01). Successful registration signs the Student in automatically.	
Exception Flow	4a. Incorrect credentials or an inactive account prevents login. Missing or invalid fields return validation errors.	
Non-Functional Requirements	Password entry must be masked. Authentication errors must not expose stored credentials. Sessions must be protected from fixation.	

Table 7 UC-03 Submit Feedback

Use Case Name	Submit Feedback	
Use Case ID	UC-03	
Use Case Description	Allows a Student or Faculty/Staff member to submit confidential CCIS feedback for screening and, when accepted, automatic analysis.	
Primary Actor	Student or Faculty/Staff	
Secondary Actor	None	
Precondition	The submitter is signed in with an active Student or Faculty account and can access the active CCIS feedback category. Admin and Super Admin cannot submit feedback.	
Postcondition	The feedback is saved with its screening and analysis result. The submitter receives the submission result.	
Normal Flow (Main Flow)	User Action	System Response
	1. Opens Submit Feedback.	2. Displays the CCIS feedback form.
	3. Enters feedback and selects Submit.	4. Validates the content, category, duplicate submission and submission limit.
	5. None	6. Performs Screen Feedback (UC-04) and saves accepted feedback as pending.
	7. None	8. Performs Analyze Accepted Feedback (UC-06) and displays the stored status and available analysis.
Alternate Flow	6a. If screening detects prohibited content, the system saves the feedback as rejected, displays the rejection result and skips UC-06.	
Exception Flow	4a. Invalid content/category, duplicate feedback or exceeding the submission limit prevents saving. 8a. Analysis failure leaves the saved feedback marked failed.	
Non-Functional Requirements	Feedback history must remain private to its owner. The result must distinguish an unsaved submission from a saved record whose analysis failed.	

Table 8 UC-04 Screen Feedback

Use Case Name	Screen Feedback	
Use Case ID	UC-04	
Use Case Description	Checks submitted feedback against configured prohibited-content rules before analysis.	
Primary Actor	Student or Faculty/Staff through Submit Feedback (UC-03)	
Secondary Actor	None	
Precondition	Feedback has passed submission validation and duplicate and submission-limit checks.	
Postcondition	The screening decision is returned to Submit Feedback; rejected content is excluded from AI analysis.	
Normal Flow (Main Flow)	User Action	System Response
	1. None	2. Receives the validated feedback text from Submit Feedback (UC-03).
	3. None	4. Normalizes the text and configured blocked terms and their variants.
	5. None	6. Checks the normalized text for prohibited matches.
	7. None	8. Returns an accepted decision to UC-03 when no prohibited match is found; UC-03 saves the feedback as pending and continues to UC-06.
Alternate Flow	6a. A prohibited match returns a rejected decision to UC-03. The system saves the record as rejected, displays the rejection result and skips UC-06.	
Exception Flow	4a/6a. A screening error prevents successful completion; the system must not report the submission as successfully analyzed.	
Non-Functional Requirements	Screening must apply the configured rules consistently. It must preserve the original feedback text and keep rejected content out of dashboard reports.	

Table 9 UC-05 Edit Own Profile

Use Case Name	Edit Own Profile	
Use Case ID	UC-05	
Use Case Description	Allows a signed-in user to update the user's own nickname and optional profile photo.	
Primary Actor	Student, Faculty/Staff, Admin or Super Admin	
Secondary Actor	None	
Precondition	The user is signed in with an active account.	
Postcondition	Valid profile changes are saved and a confirmation is displayed.	
Normal Flow (Main Flow)	User Action	System Response
	1. Opens the profile page.	2. Displays the current nickname and profile photo.
	3. Changes the nickname or selects a new photo; selects Save.	4. Validates nickname uniqueness and the permitted photo type and size.
	5. None	6. Stores or removes the profile photo as requested and updates only the signed-in user's profile.
	7. None	8. Displays the updated profile and a confirmation message.
Alternate Flow	3a. The user chooses to remove the existing photo before selecting Save. The user may leave the page without saving; existing details remain.	
Exception Flow	4a. A duplicate/invalid nickname or an invalid photo prevents the update. 6a. A photo-storage failure prevents successful completion.	
Non-Functional Requirements	Users must only edit their own profiles. Nicknames protect real-name privacy. Photo uploads must be validated before storage.	

Table 10 UC-06 Analyze Accepted Feedback

Use Case Name	Analyze Accepted Feedback	
Use Case ID	UC-06	
Use Case Description	Classifies accepted feedback by sentiment, language and concern topics using Gemma through the Gemini API.	
Primary Actor	Student or Faculty/Staff through Submit Feedback (UC-03)	
Secondary Actor	None	
Precondition	Screen Feedback (UC-04) accepted the content; the feedback is saved as pending and the analysis service is configured.	
Postcondition	A valid analysis result is saved, the feedback is marked analyzed and its category concern rankings are rebuilt.	
Normal Flow (Main Flow)	User Action	System Response
	1. None	2. Starts analysis for the accepted feedback record saved as pending by UC-03.
	3. None	4. Sends the feedback to the Gemini API with the required classification instructions.
	5. None	6. Receives and validates the returned sentiment, confidence, keywords, topics and language data.
	7. None	8. Saves the result, marks the feedback analyzed and rebuilds the concern rankings; returns control to UC-03 to display the result.
Alternate Flow	4a. A temporary request failure may trigger a retry before the final result is determined.	
Exception Flow	4a/6a. Missing configuration, connection failure, API error or unusable output marks the saved feedback failed; it does not create a successful analysis result.	
Non-Functional Requirements	API credentials must remain confidential and transport must use TLS. Returned data must be validated. A service failure must preserve the submitted record.	

Table 11 UC-07 View My Feedback

Use Case Name	View My Feedback	
Use Case ID	UC-07	
Use Case Description	Allows a submitter to review only the feedback records belonging to the signed-in account.	
Primary Actor	Student or Faculty/Staff	
Secondary Actor	None	
Precondition	The Student or Faculty/Staff member is signed in with an active account. Feedback history is restricted to the account owner.	
Postcondition	The user sees the personal feedback history with processing status and available analysis.	
Normal Flow (Main Flow)	User Action	System Response
	1. Selects View My Feedback.	2. Checks that the signed-in user is permitted to access personal feedback history.
	3. None	4. Retrieves only the user's feedback records with their categories and available analysis, newest first.
	5. None	6. Displays the personal feedback history, processing statuses and available analysis in a paginated list.
	7. Selects the next or previous history page when available.	8. Displays the requested page of the same user's records.
Alternate Flow	4a. If the user has no submissions, the system displays an empty history. 6a. Records without analysis show their current status.	
Exception Flow	1a. An expired session requires Login (UC-02). A retrieval failure prevents the history from being displayed.	
Non-Functional Requirements	History queries must enforce ownership. Pagination must not expose another user's records. Status labels must distinguish rejected, pending, failed and analyzed feedback.	

Table 12 UC-08 View Dashboard

Use Case Name	View Dashboard	
Use Case ID	UC-08	
Use Case Description	Allows authorized users to review CCIS feedback summaries, sentiment and language distributions, trends and ranked concerns.	
Primary Actor	Faculty/Staff, Admin or Super Admin	
Secondary Actor	None	
Precondition	The user is signed in with an active account and a role permitted to access the dashboard.	
Postcondition	The dashboard displays the available reportable feedback and analytics for the current scope.	
Normal Flow (Main Flow)	User Action	System Response
	1. Opens View Dashboard.	2. Checks dashboard access and retrieves reportable CCIS feedback.
	3. None	4. Calculates feedback counts, sentiment and language distributions and trends.
	5. None	6. Builds the leading concern rankings and retrieves the paginated feedback list.
	7. None	8. Displays the dashboard summaries, charts, concerns and feedback, with Filter Feedback (UC-09), Download PDF Report (UC-10) and Export Feedback (UC-11).
Alternate Flow	2a. With no reportable records, the dashboard shows zero totals and an empty list. After step 8, the user may activate UC-09, UC-10 or UC-11.	
Exception Flow	1a. A Student or other unauthorized user is denied dashboard access. Invalid scope values or a retrieval failure prevent the requested view.	
Non-Functional Requirements	Access must be role controlled. Rejected feedback must be excluded from reports. Dashboard records must not disclose submitters' email addresses or real names.	

Table 13 UC-09 Filter Feedback

Use Case Name	Filter Feedback	
Use Case ID	UC-09	
Use Case Description	Narrows dashboard feedback and analytics by supported language and date-range criteria.	
Primary Actor	Faculty/Staff, Admin or Super Admin	
Secondary Actor	None	
Precondition	View Dashboard (UC-08) is open and the user has dashboard permission.	
Postcondition	Dashboard records and analytics reflect the selected valid filter criteria.	
Normal Flow (Main Flow)	User Action	System Response
	1. Opens the dashboard filter controls.	2. Displays supported language and date-range options for CCIS.
	3. Selects the desired criteria and applies the filters.	4. Validates the selected criteria.
	5. None	6. Retrieves matching reportable feedback and recalculates analytics for the selected scope.
	7. None	8. Displays the filtered feedback and analytics; retains the scope for PDF and spreadsheet downloads.
Alternate Flow	3a. The user clears optional filters to restore the default CCIS view. 6a. No matching records produces an empty result.	
Exception Flow	4a. Unsupported language/category values or invalid date ranges prevent applying the requested filter. A retrieval failure prevents refreshed results.	
Non-Functional Requirements	The same criteria must be applied consistently to records, analytics and downloads. Filter use must not bypass access restrictions.	

Table 14 UC-10 Download PDF Report

Use Case Name	Download PDF Report	
Use Case ID	UC-10	
Use Case Description	Produces a downloadable PDF containing CCIS feedback summaries, concern rankings and reportable feedback for the selected scope.	
Primary Actor	Faculty/Staff, Admin or Super Admin	
Secondary Actor	None	
Precondition	The user has dashboard permission and the requested language and date filters are valid.	
Postcondition	A PDF report is generated and returned for download.	
Normal Flow (Main Flow)	User Action	System Response
	1. Selects Download PDF Report for the current dashboard scope.	2. Checks access and validates the requested language and date filters.
	3. None	4. Retrieves matching reportable CCIS feedback and calculates summaries and concern rankings.
	5. None	6. Renders the PDF report with the selected scope and page numbering.
	7. None	8. Returns the PDF for download with a descriptive filename.
Alternate Flow	1a. With no optional filters, the report uses the default CCIS scope. 4a. With no matching feedback, the report contains the available zero/empty summaries.	
Exception Flow	2a. Invalid filters or unauthorized access prevents download. 4a/6a. A retrieval or PDF-generation failure prevents a completed report.	
Non-Functional Requirements	The PDF must be readable with consistent pagination. It must reflect the selected scope, exclude rejected feedback and avoid submitters' identifying account data.	

Table 15 UC-11 Export Feedback

Use Case Name	Export Feedback	
Use Case ID	UC-11	
Use Case Description	Exports matching reportable CCIS feedback and available analysis to an Excel spreadsheet.	
Primary Actor	Faculty/Staff, Admin or Super Admin	
Secondary Actor	None	
Precondition	The user has dashboard permission and the requested filters are valid.	
Postcondition	An XLSX spreadsheet is generated and returned for download.	
Normal Flow (Main Flow)	User Action	System Response
	1. Selects Export Feedback for the current dashboard scope.	2. Checks access and validates the requested language and date filters.
	3. None	4. Retrieves matching reportable CCIS feedback and available analysis.
	5. None	6. Writes feedback, dates, statuses and analysis into an XLSX spreadsheet with column headings and formatted values.
	7. None	8. Returns the XLSX file for download with a descriptive filename.
Alternate Flow	1a. With no optional filters, the export uses the default CCIS scope. 4a. No matching feedback produces a spreadsheet containing the column headings.	
Exception Flow	2a. Invalid filters or unauthorized access prevents export. 4a/6a. A retrieval or spreadsheet-generation failure prevents a completed download.	
Non-Functional Requirements	The export must preserve the selected scope and exclude rejected feedback and identifying account fields. Feedback text must be written as text rather than executable spreadsheet formulas.	

Table 16 UC-12 Delete Feedback

Use Case Name	Delete Feedback	
Use Case ID	UC-12	
Use Case Description	Allows an administrator to permanently remove a selected feedback record and its associated analysis.	
Primary Actor	Admin or Super Admin	
Secondary Actor	None	
Precondition	The administrator is signed in with an active account and the selected feedback exists.	
Postcondition	The feedback and associated analysis are removed; the dashboard displays a deletion confirmation.	
Normal Flow (Main Flow)	User Action	System Response
	1. Locates a feedback record on the dashboard.	2. Displays the record and the permitted delete control.
	3. Selects Delete Feedback.	4. Presents the deletion confirmation.
	5. Confirms deletion.	6. Checks administrator permission and permanently deletes the record and associated analysis.
	7. None	8. Displays confirmation and the remaining feedback records.
Alternate Flow	5a. The administrator cancels confirmation; the feedback remains unchanged.	
Exception Flow	6a. A Faculty/Staff or Student user cannot delete feedback. A missing record or database failure prevents successful deletion.	
Non-Functional Requirements	Deletion must enforce administrator authorization and request protection. Confirmation must make the permanent effect clear. Associated analysis must not remain orphaned.	

Table 17 UC-13 View / Search Account

Use Case Name	View / Search Account	
Use Case ID	UC-13	
Use Case Description	Allows an administrator to view registered accounts and search them by nickname.	
Primary Actor	Admin or Super Admin	
Secondary Actor	None	
Precondition	The administrator is signed in with an active account and has account-management access.	
Postcondition	The account list or nickname search results are displayed.	
Normal Flow (Main Flow)	User Action	System Response
	1. Opens View / Search Account.	2. Displays accounts with nickname, role, status and registration date.
	3. Enters a nickname or part of a nickname and submits the search.	4. Validates the search text.
	5. None	6. Retrieves accounts matching the nickname search.
	7. None	8. Displays the paginated results while retaining the search criteria.
Alternate Flow	3a. An empty search displays the full paginated list. 6a. No matching nickname displays an empty result. After step 8, the administrator may select another result page or an authorized account action (UC-14, UC-15 or UC-16).	
Exception Flow	1a. An unauthorized user is denied access. Invalid search input or a retrieval failure prevents the requested result.	
Non-Functional Requirements	Account lists must use nicknames and omit email addresses and real names. Search must enforce administrator access and validate input.	

Table 18 UC-14 Delete Another Account

Use Case Name	Delete Another Account	
Use Case ID	UC-14	
Use Case Description	Allows a Super Admin to permanently delete an account other than the currently signed-in account.	
Primary Actor	Super Admin	
Secondary Actor	None	
Precondition	The Super Admin is signed in with an active account; the target account exists and belongs to another user.	
Postcondition	The target account is deleted and the updated account list shows a confirmation.	
Normal Flow (Main Flow)	User Action	System Response
	1. Finds the target using View / Search Account (UC-13).	2. Displays the account and its permitted delete control.
	3. Selects Delete Another Account.	4. Presents confirmation identifying the target by nickname.
	5. Confirms the deletion.	6. Verifies Super Admin permission and checks that the target is not the signed-in account.
	7. None	8. Deletes the target account and displays the updated account list with a success message.
Alternate Flow	5a. The Super Admin cancels confirmation; the target account remains unchanged.	
Exception Flow	6a. A regular Admin or any other role cannot delete accounts. Self-deletion is blocked. A missing target or database failure prevents successful completion.	
Non-Functional Requirements	Super Admin authorization and request protection must be enforced. Confirmation must make permanent deletion clear without exposing email addresses or real names.	

Table 19 UC-15 Change Account Role

Use Case Name	Change Account Role	
Use Case ID	UC-15	
Use Case Description	Allows an administrator to assign a permitted role to another registered account.	
Primary Actor	Admin or Super Admin	
Secondary Actor	None	
Precondition	The administrator is signed in with an active account and has opened View / Search Account (UC-13). The target account exists and is not the signed-in account.	
Postcondition	The permitted role is saved for the target account and a confirmation is displayed.	
Normal Flow (Main Flow)	User Action	System Response
	1. Opens View / Search Account (UC-13).	2. Displays account nicknames, current roles, statuses and permitted role controls.
	3. Selects another account, chooses a permitted role and selects Save.	4. Checks administrator authority and blocks self-role changes. Admin may manage only Student/Faculty targets; Super Admin may also manage administrator targets.
	5. None	6. Validates the selected role. Admin may assign Student or Faculty; Super Admin may also assign Admin or Super Admin.
	7. None	8. Updates the target account role and displays confirmation.
Alternate Flow	3a. The administrator leaves the page without saving; the current role remains unchanged.	
Exception Flow	4a. Self-role changes or unauthorized management of an administrator account are blocked. 6a. An invalid or disallowed role prevents the update. A missing target or database failure prevents successful completion.	
Non-Functional Requirements	Authorization must be enforced on the server. Requests must be protected against forgery. Account controls must identify users by nickname and protect the signed-in account.	

Table 20 UC-16 Deactivate / Reactivate Account

Use Case Name	Deactivate / Reactivate Account	
Use Case ID	UC-16	
Use Case Description	Allows an administrator to deactivate or reactivate another account within the administrator's authority.	
Primary Actor	Admin or Super Admin	
Secondary Actor	None	
Precondition	The administrator is signed in with an active account and has opened View / Search Account (UC-13). The target account exists and is not the signed-in account.	
Postcondition	The target account status is changed and a confirmation is displayed. Deactivated accounts cannot log in or access operations that require an active account.	
Normal Flow (Main Flow)	User Action	System Response
	1. Opens View / Search Account (UC-13).	2. Displays account nicknames, roles, statuses and permitted status controls.
	3. Selects Deactivate for an active target or Reactivate for an inactive target.	4. Checks administrator authority and blocks changes to the signed-in account. Admin may manage only Student/Faculty targets; Super Admin may also manage administrator targets.
	5. None	6. Changes the target's active status to the opposite state and saves it.
	7. None	8. Displays the updated status and a confirmation message.
Alternate Flow	3a. For an inactive account, Reactivate restores active status. For an active account, Deactivate removes active status.	
Exception Flow	4a. Changing the signed-in account's status or unauthorized management of an administrator account is blocked. A missing target or database failure prevents successful completion.	
Non-Functional Requirements	Status updates must enforce role restrictions and request protection. Active-account checks must prevent a deactivated user from accessing protected operations.	

Table 21 UC-17 Logout

Use Case Name	Logout	
Use Case ID	UC-17	
Use Case Description	Allows a signed-in actor to end the current authenticated session.	
Primary Actor	Student, Faculty/Staff, Admin or Super Admin	
Secondary Actor	None	
Precondition	The actor has an authenticated session.	
Postcondition	The actor is signed out, the session is invalidated and the Login page is displayed.	
Normal Flow (Main Flow)	User Action	System Response
	1. Selects Logout.	2. Ends authentication for the current session.
	3. None	4. Invalidates the session and clears its stored data.
	5. None	6. Regenerates the request-protection token.
	7. None	8. Redirects the actor to Login (UC-02).
Alternate Flow	1a. If the session has already expired, access to the authenticated logout route redirects the actor to Login.	
Exception Flow	2a. An invalid request-protection token prevents the logout request from being processed; the actor must reload the page and submit a valid request.	
Non-Functional Requirements	Logout must use a protected POST request. Invalidated sessions must not authorize further access to authenticated pages.	

APPENDIX E ACTIVITY DIAGRAMS

Figures 10–21 show the twelve ResBack activities. Swimlanes separate user and system responsibilities. Every activity has one final endpoint; additional exits in the corrected diagrams loop back to an action. Larger workflows include continuation panels for readability.

Figure 10. Registration validates account details, handles duplicate values, creates an active Student account, and starts an authenticated session.

Figure 11. Login checks credentials and account status before routing the user according to role. Extra endings return to an action through a loop.

Figure 12. Submission covers confirmation, validation, duplicate and rate checks, screening, storage, and accepted or rejected results.

Figure 13. Personal feedback history displays the signed-in user's submissions, processing status, and pagination.

Figure 14. Dashboard review validates date and language filters, retrieves matching feedback, summarizes sentiment, and ranks concerns.

Figure 15. Account management covers search, protected-account checks, role and status changes, and permitted account deletion. Disallowed or canceled actions return to the relevant action.

Figure 16. Analysis performs synchronous Gemma inference, validates the response, normalizes language and concern topics, stores results, and rebuilds rankings. Failed analysis remains visible as a failed status.

Figure 17. Excel export applies the selected filters and downloads feedback without submitter identities or rejected submissions.

Figure 18. PDF reporting uses the same filters and presents summaries and supporting feedback, including an empty state when appropriate.

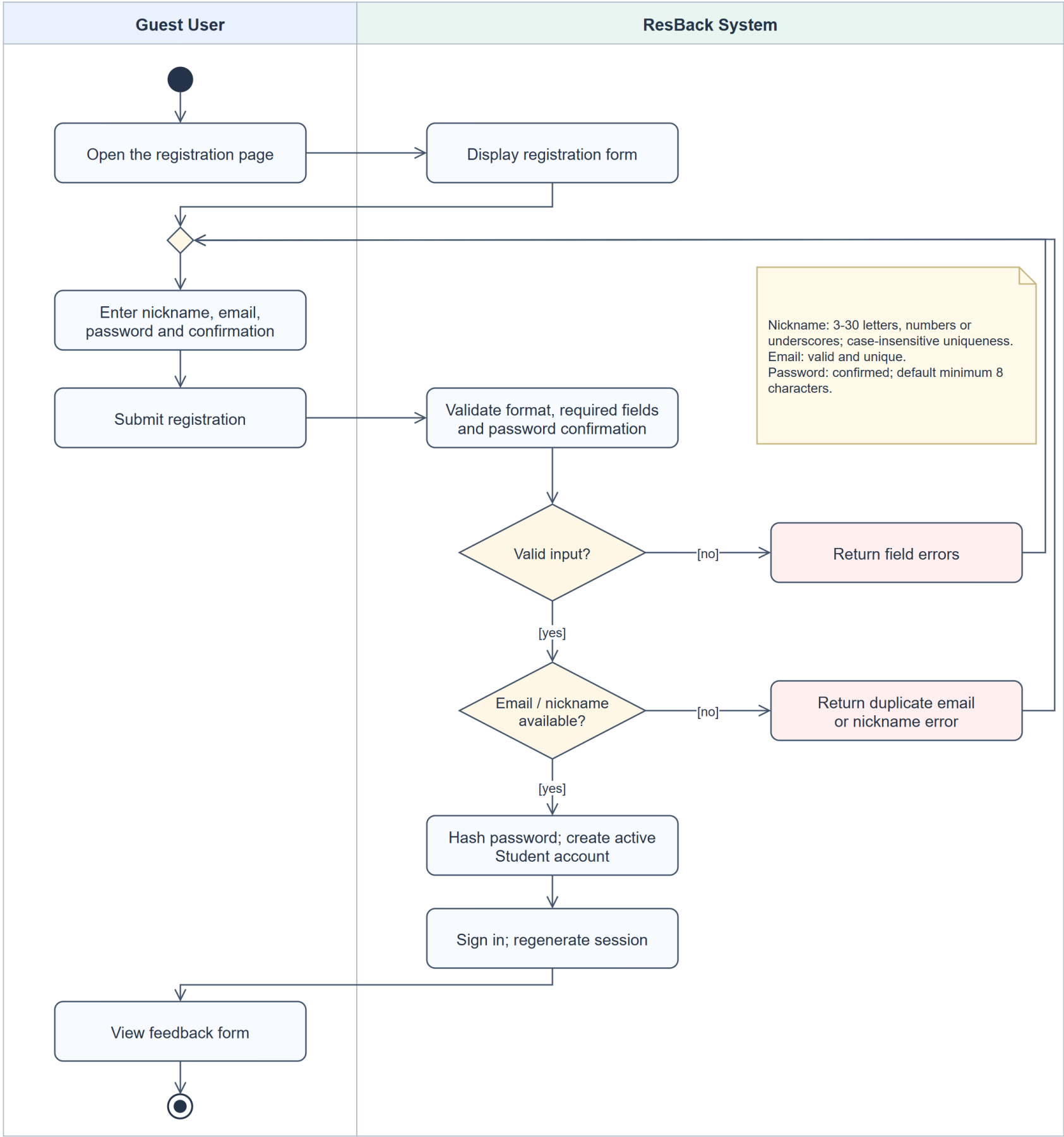
Figure 19. Feedback deletion checks permission and the selected record. Cancellation preserves the record and returns to the previous action.

Figure 20. Profile editing validates the nickname and photo, handles photo replacement or removal, and saves permitted changes.

Figure 21. Logout invalidates the authenticated session, refreshes the CSRF token, and redirects the user.

act 01 | Register an Account

Precondition: user is signed out. Registration creates an active Student account and signs the user in.

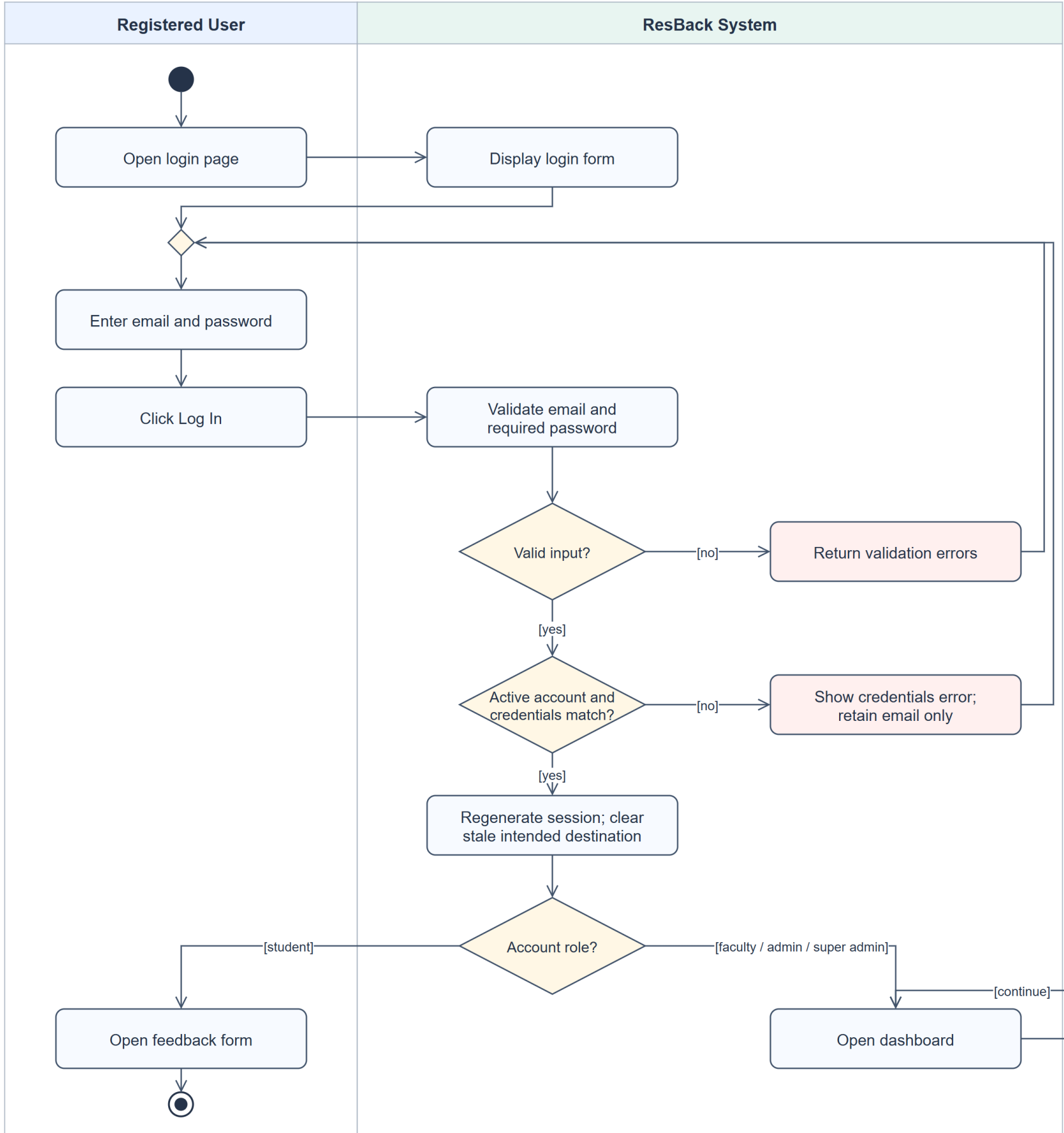


Lab 6 p.1 | Source: RegisterController.php, User.php, auth/register.blade.php
ResBack / Team Mira | Current implementation: 25 September 2026 | Guards in [brackets]; diamonds = decision / merge
Notation reference: <https://www.drawio.com/docs/diagram-types/uml/activity-diagrams/>

Figure 10 Activity diagram for register an account

act 02 | Log In

Precondition: user is signed out. Only active accounts can authenticate.



Lab 6 p.2 | Source: LoginController.php, HomeController.php
ResBack / Team Mira | Current implementation: 25 September 2026 | Guards in [brackets]; diamonds = decision / merge
Notation reference: <https://www.drawio.com/docs/diagram-types/uml/activity-diagrams/>

Figure 11 Activity diagram for log in

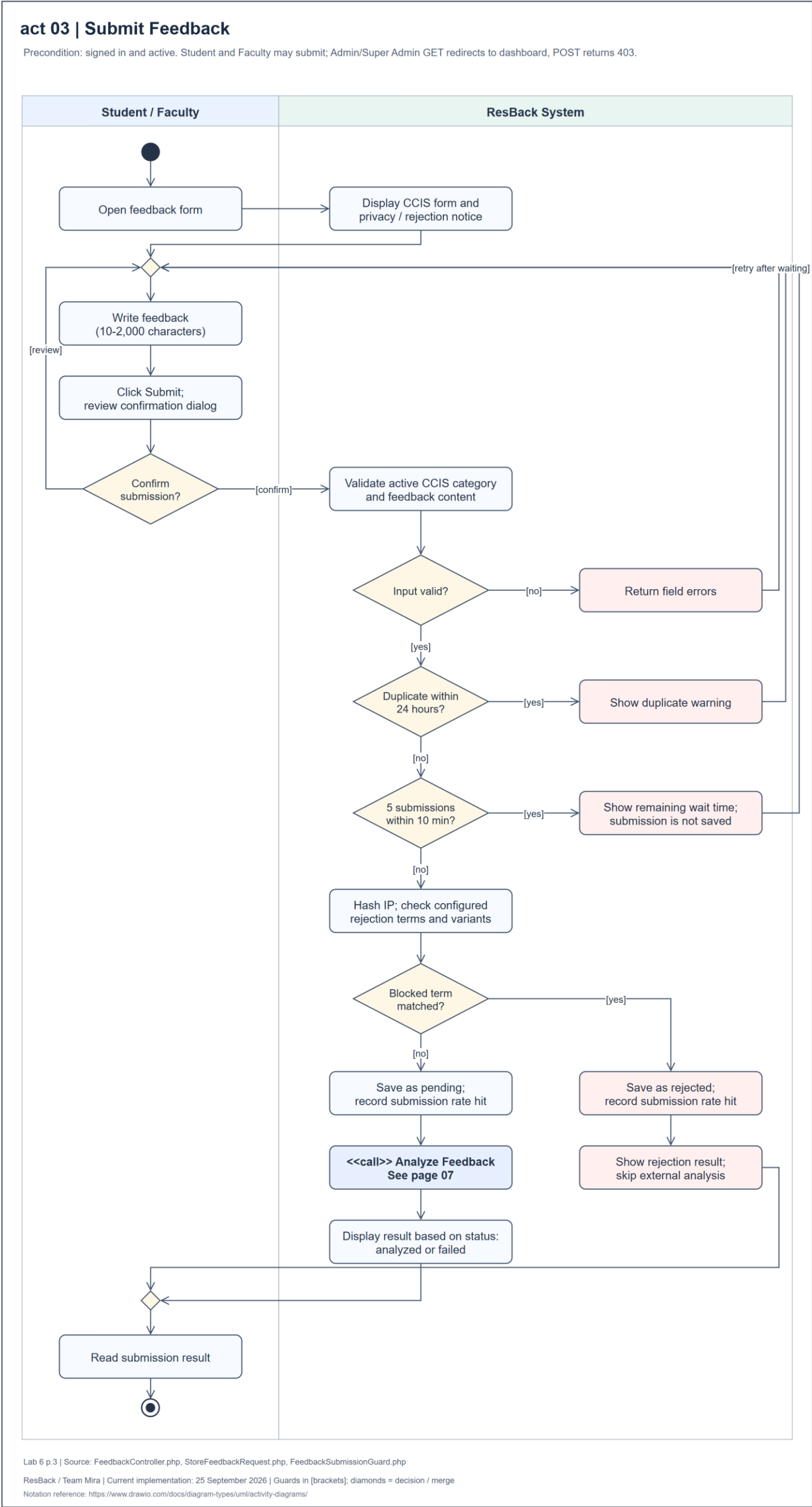


Figure 12 Activity diagram for submit feedback

act 03 | Submit Feedback

Precondition: signed in and active. Student and Faculty may submit; Admin/Super Admin GET redirects to dashboard, POST returns 403.

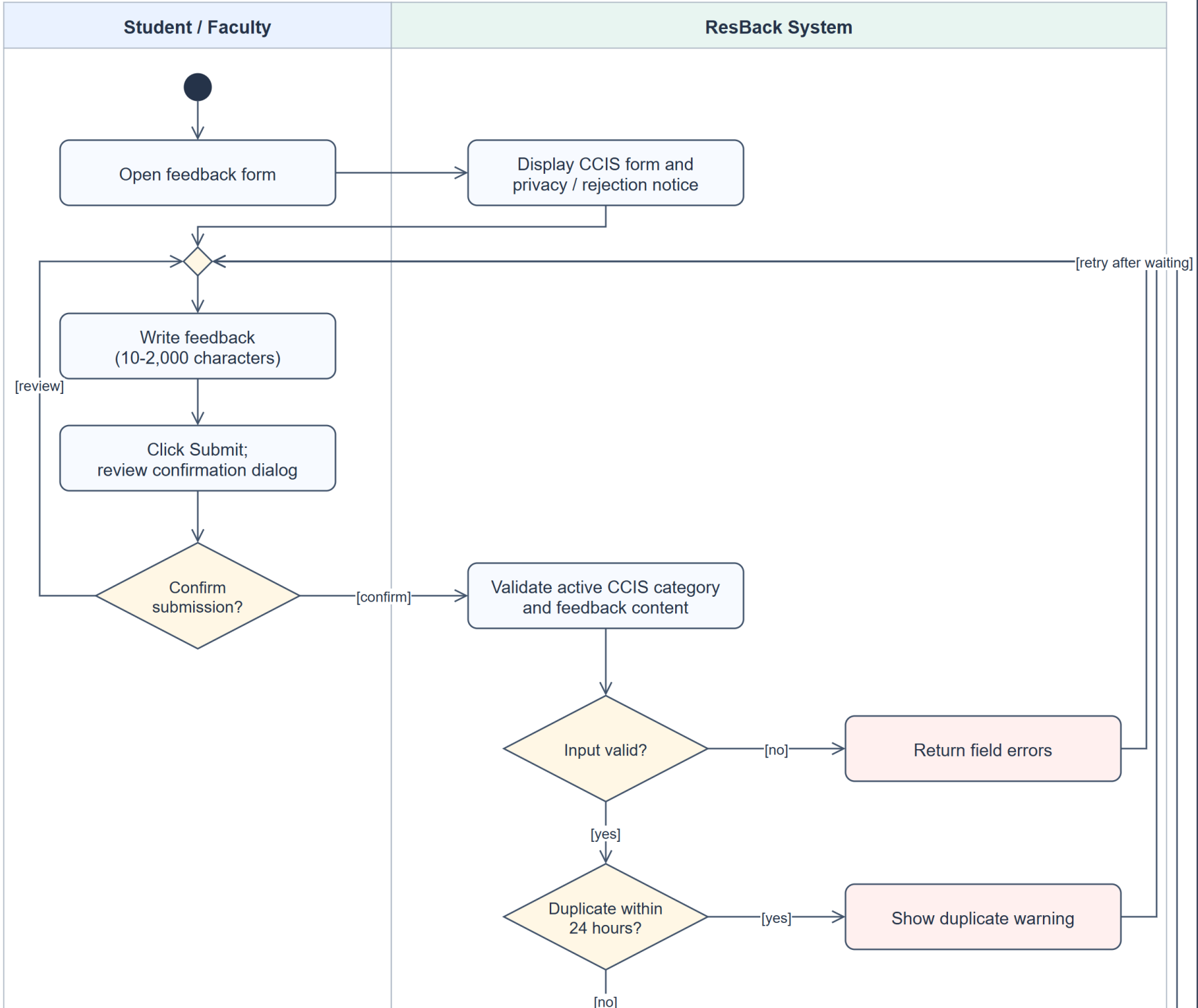


Figure 12 Activity diagram for submit feedback enlarged continuation panel 1 of 2

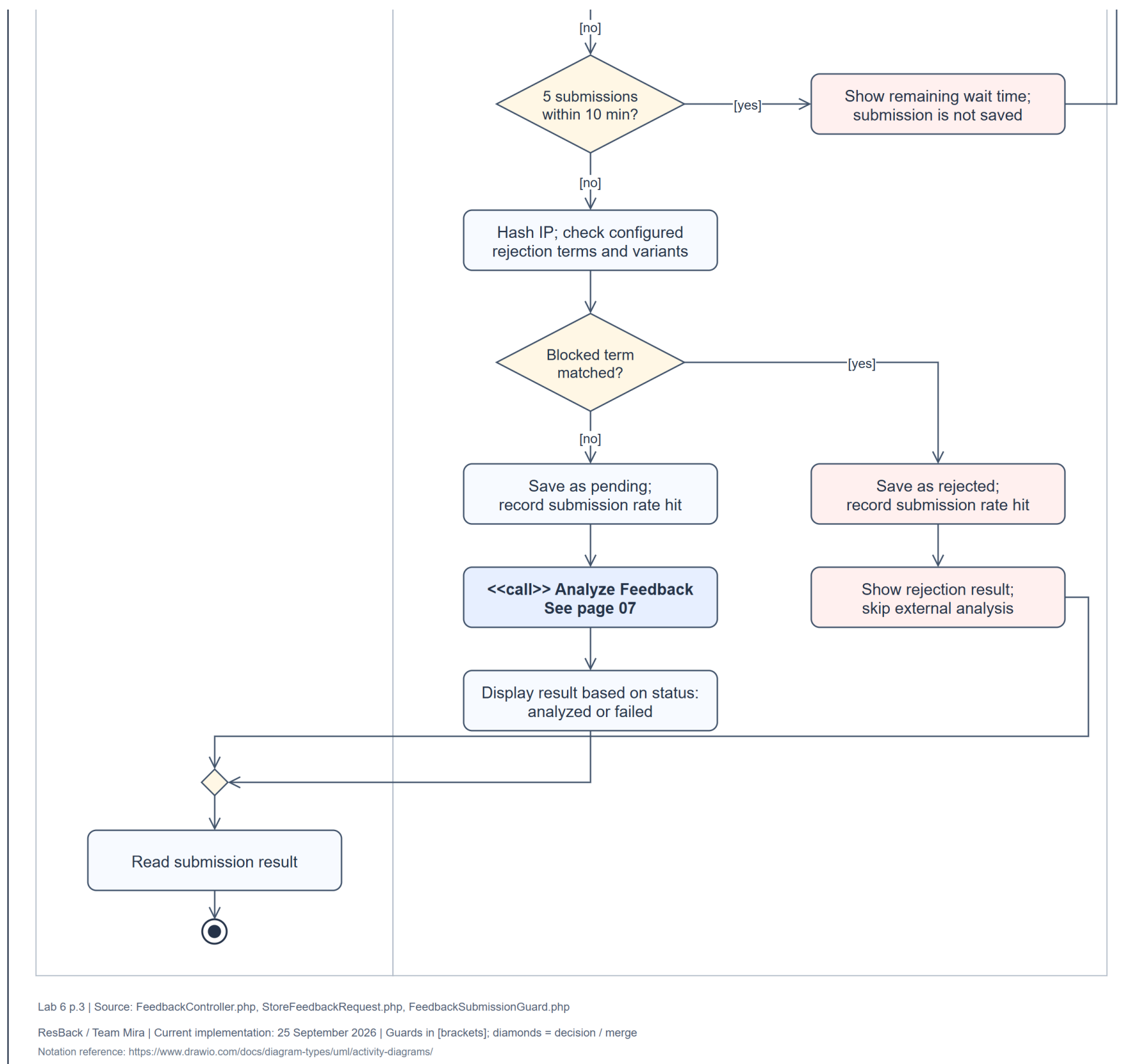
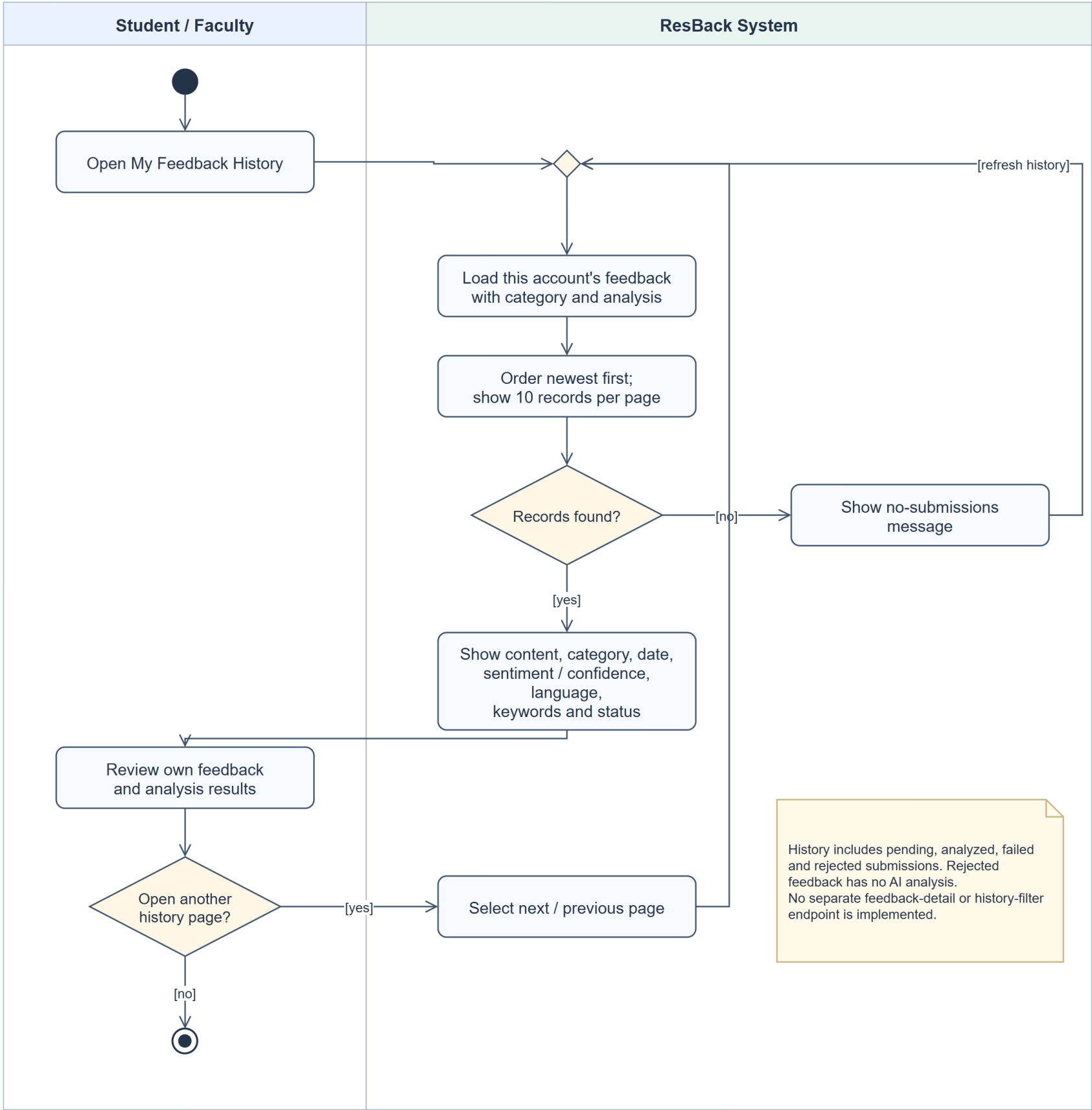


Figure 12 Activity diagram for submit feedback enlarged continuation panel 2 of 2

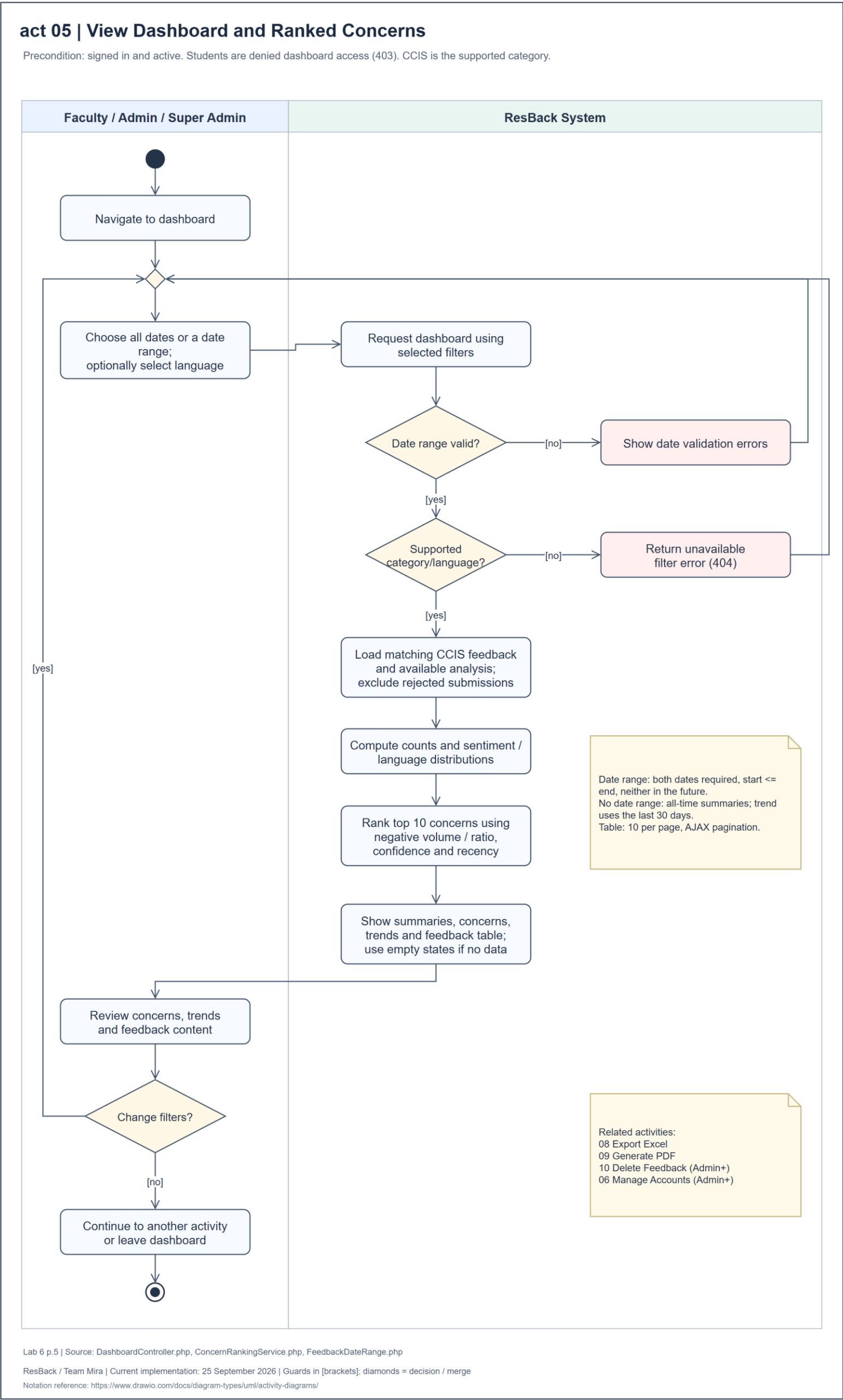
act 04 | View My Feedback History

Precondition: signed in and active; Student/Faculty access. History is private to the submitting account.



Lab 6 p.4 (private history) | Source: FeedbackController::history, feedback/history.blade.php
ResBack / Team Mira | Current implementation: 25 September 2026 | Guards in [brackets]; diamonds = decision / merge
Notation reference: <https://www.drawio.com/docs/diagram-types/uml/activity-diagrams/>

Figure 13 Activity diagram for view my feedback history



act 05 | View Dashboard and Ranked Concerns

Precondition: signed in and active. Students are denied dashboard access (403). CCIS is the supported category.

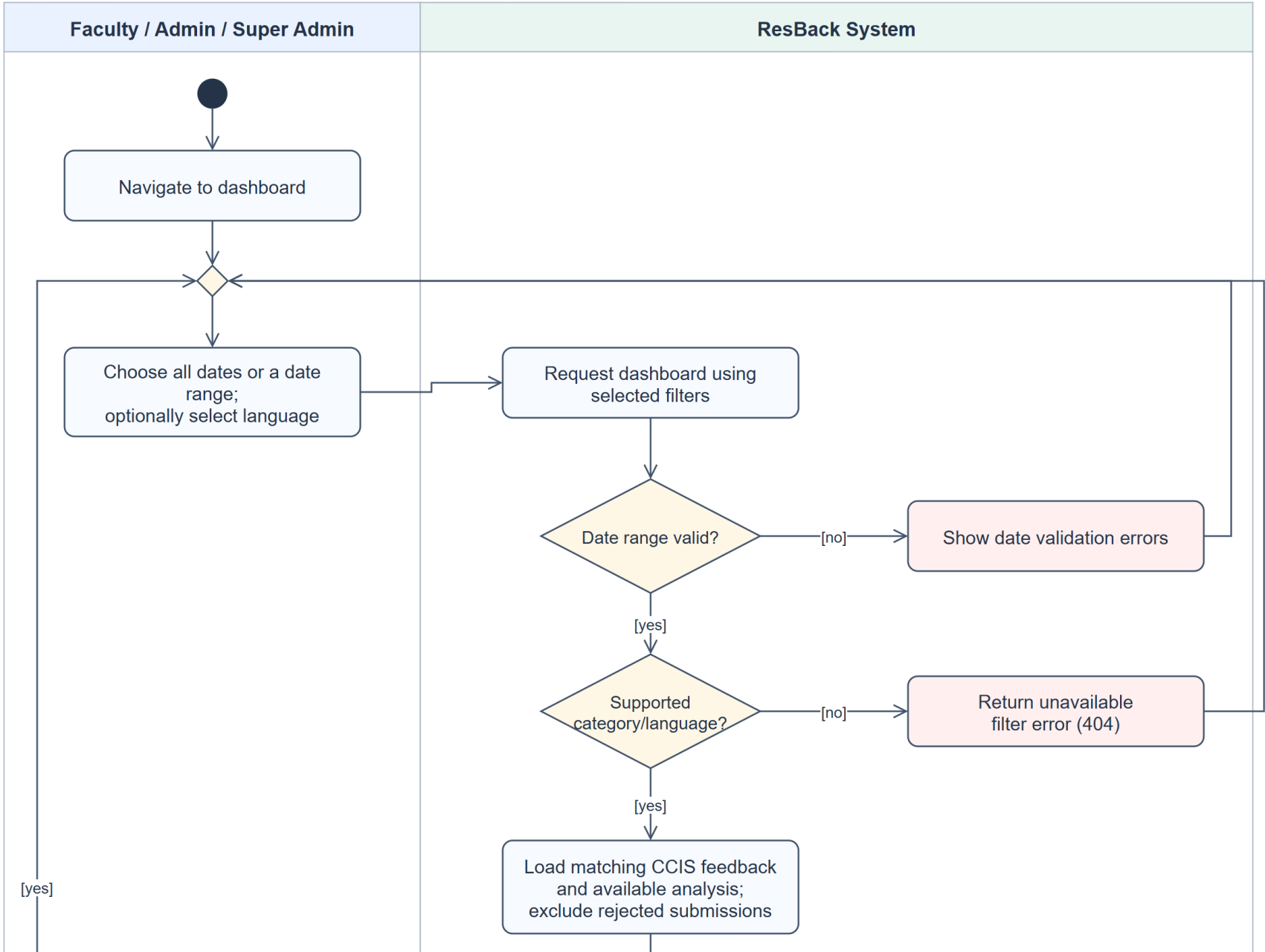


Figure 14 Activity diagram for view dashboard and ranked concerns enlarged continuation panel 1 of 2

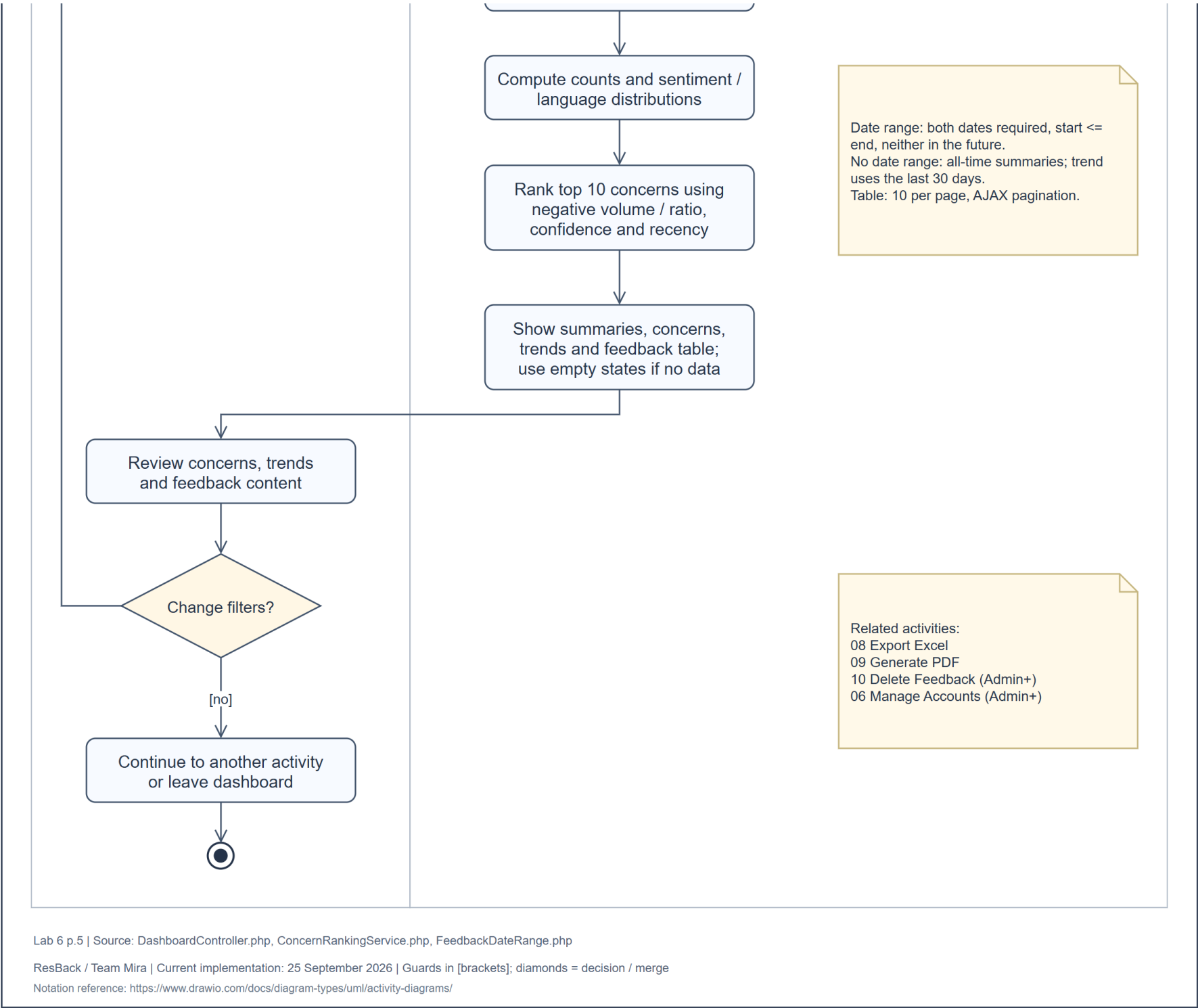


Figure 14 Activity diagram for view dashboard and ranked concerns enlarged continuation panel 2 of 2

Figure 15 Activity diagram for manage accounts

act 06 | Manage Accounts

Precondition: signed in, active, and Admin/Super Admin. Operations are rechecked on the server.

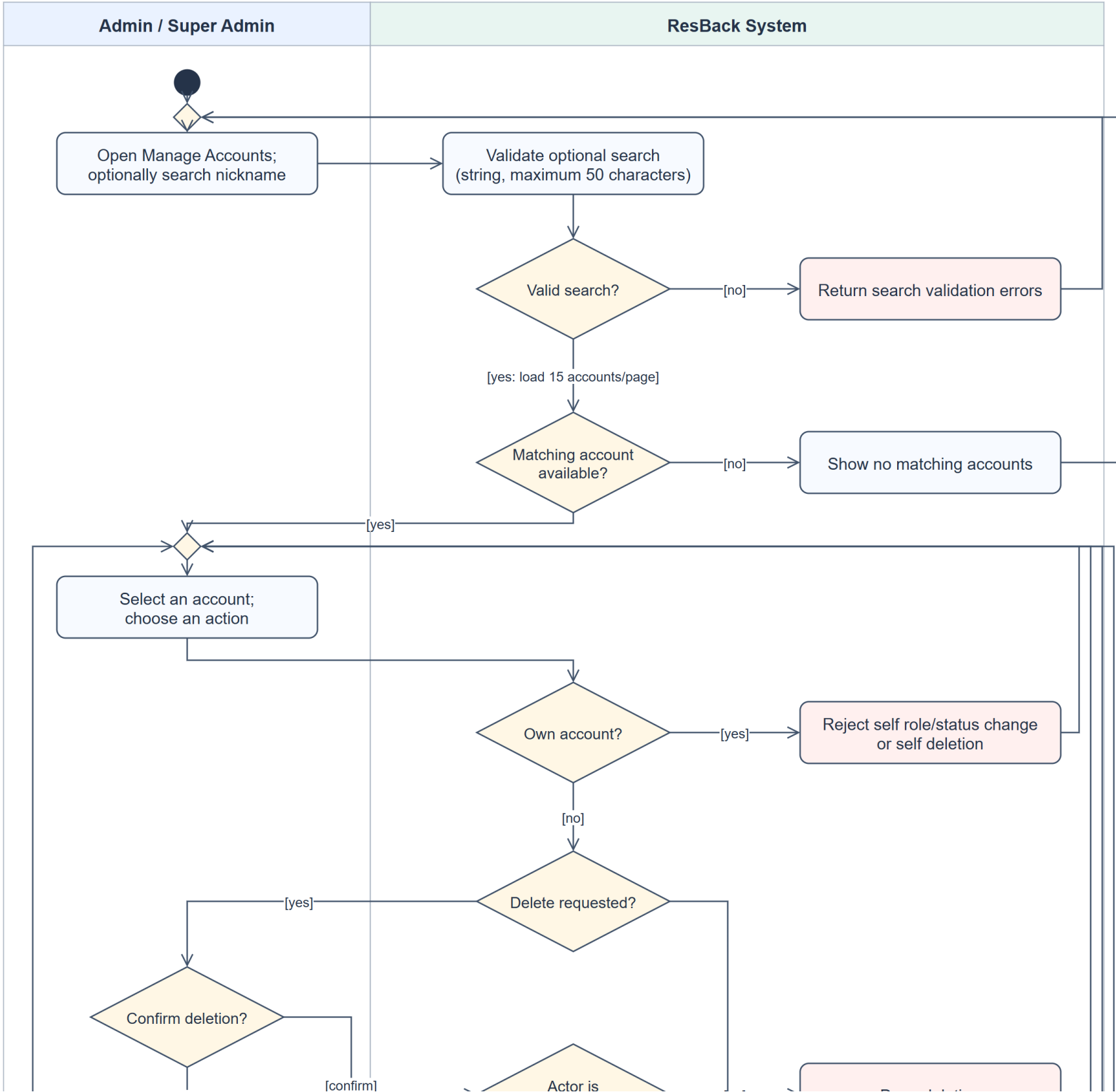


Figure 15 Activity diagram for manage accounts enlarged continuation panel 1 of 2

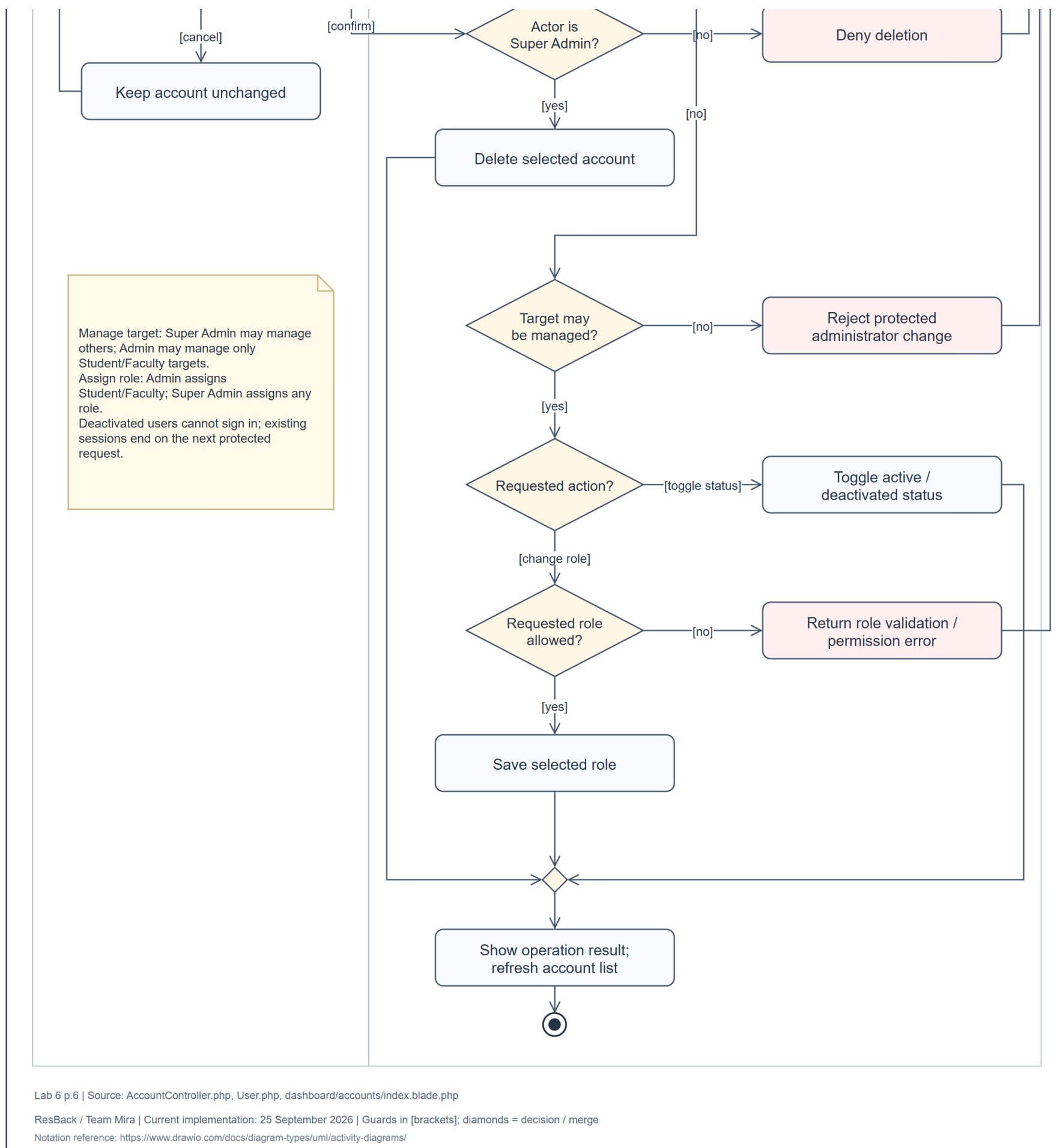


Figure 15 Activity diagram for manage accounts enlarged continuation panel 2 of 2

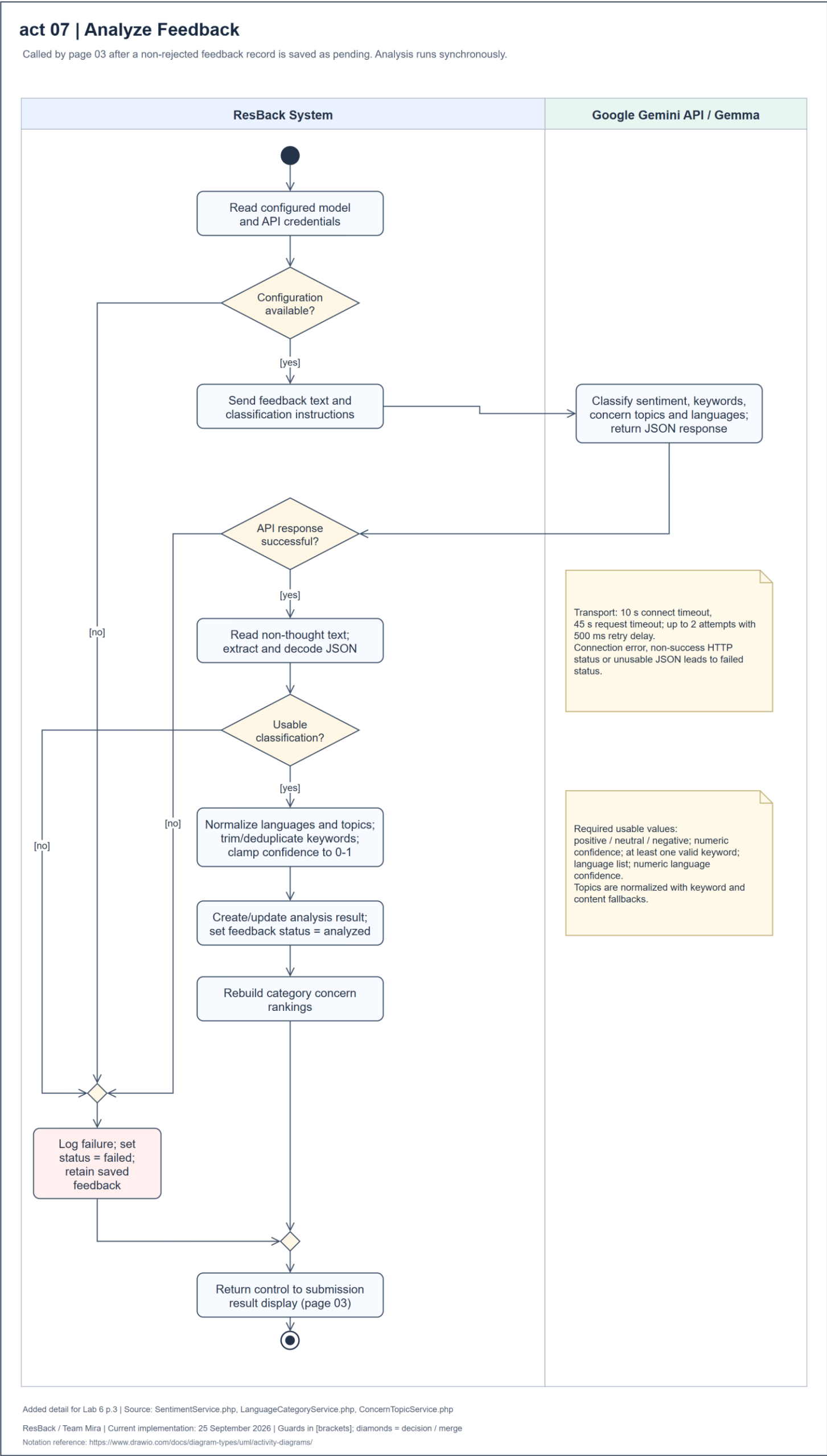


Figure 16 Activity diagram for analyze feedback

act 07 | Analyze Feedback

Called by page 03 after a non-rejected feedback record is saved as pending. Analysis runs synchronously.

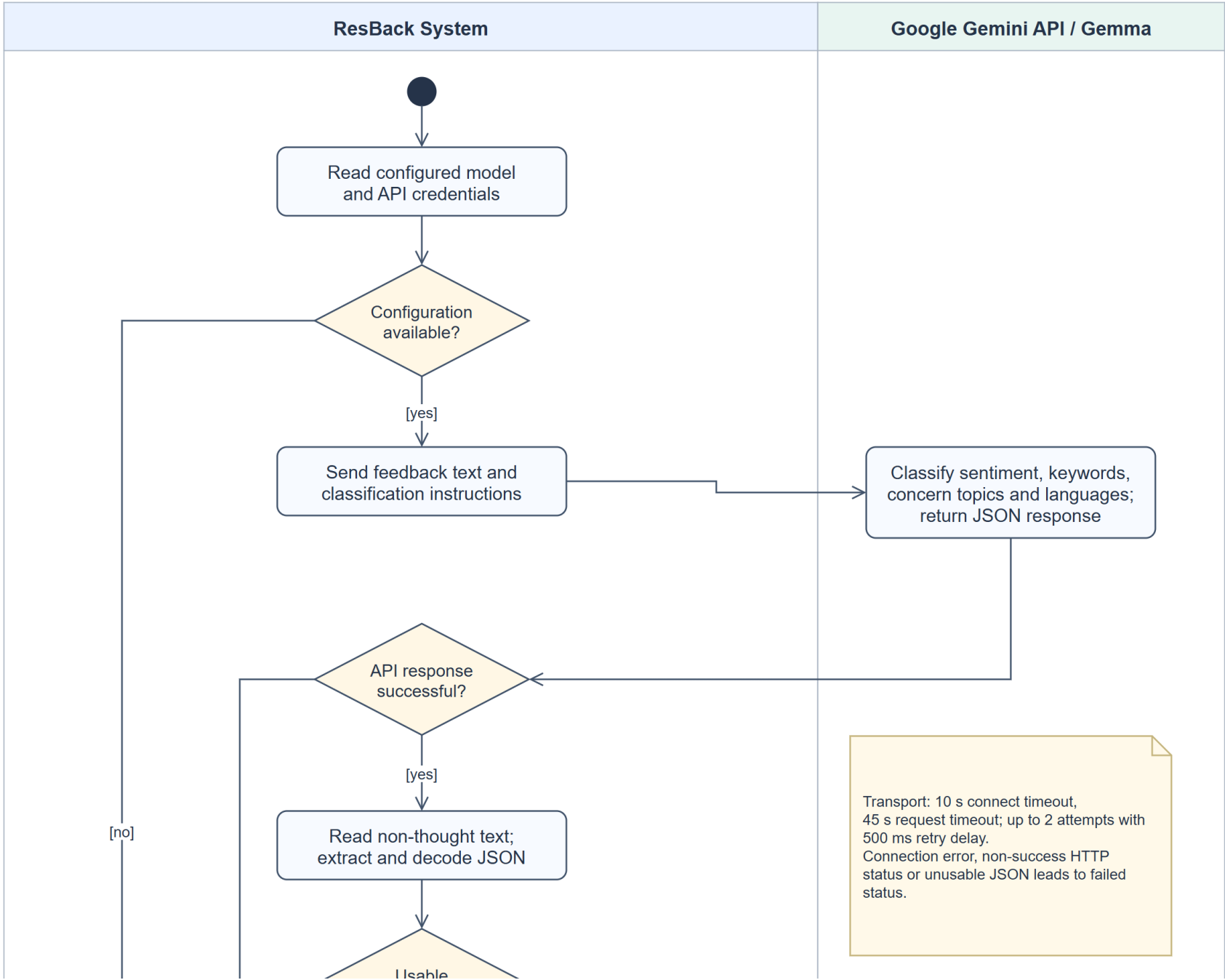


Figure 16 Activity diagram for analyze feedback enlarged continuation panel 1 of 2

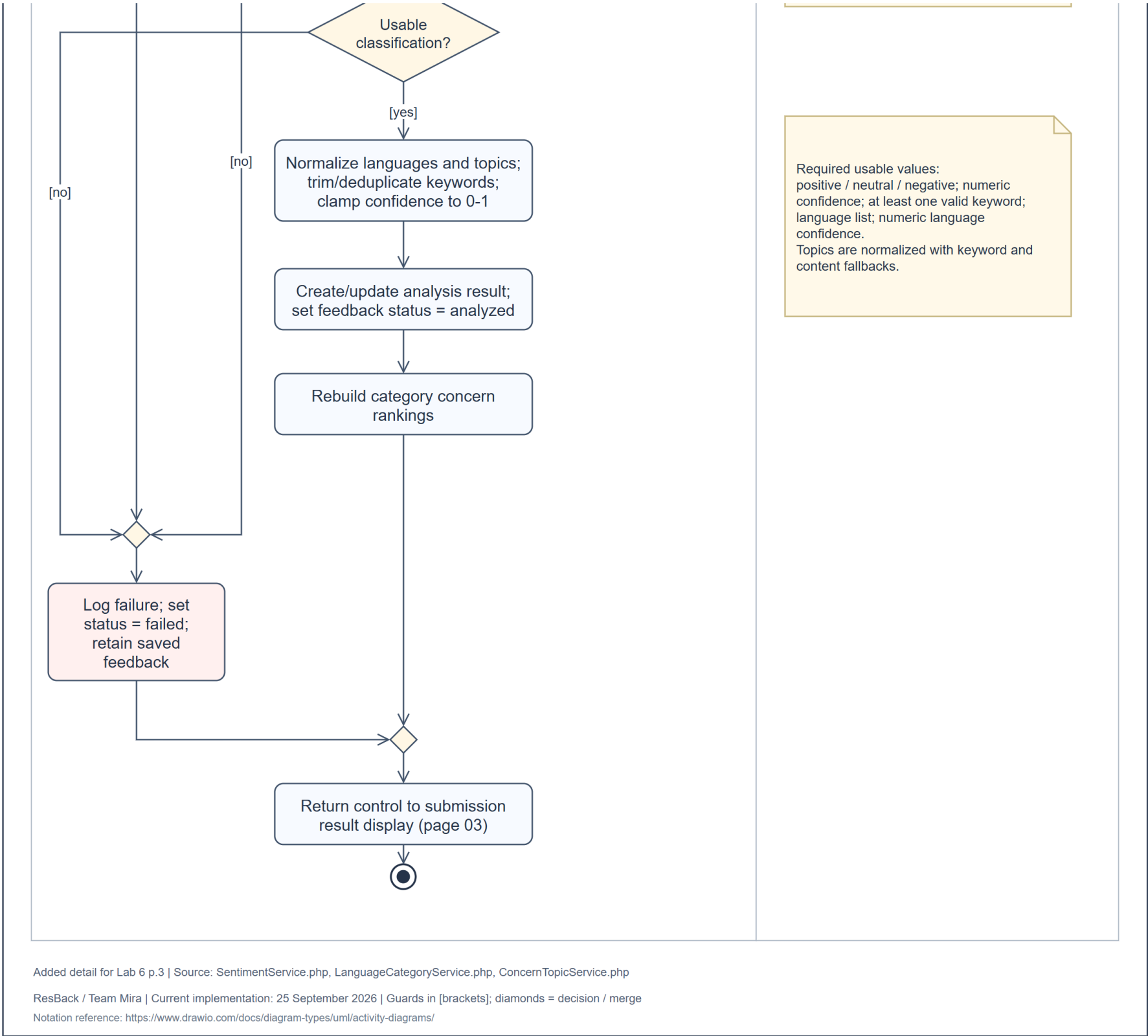
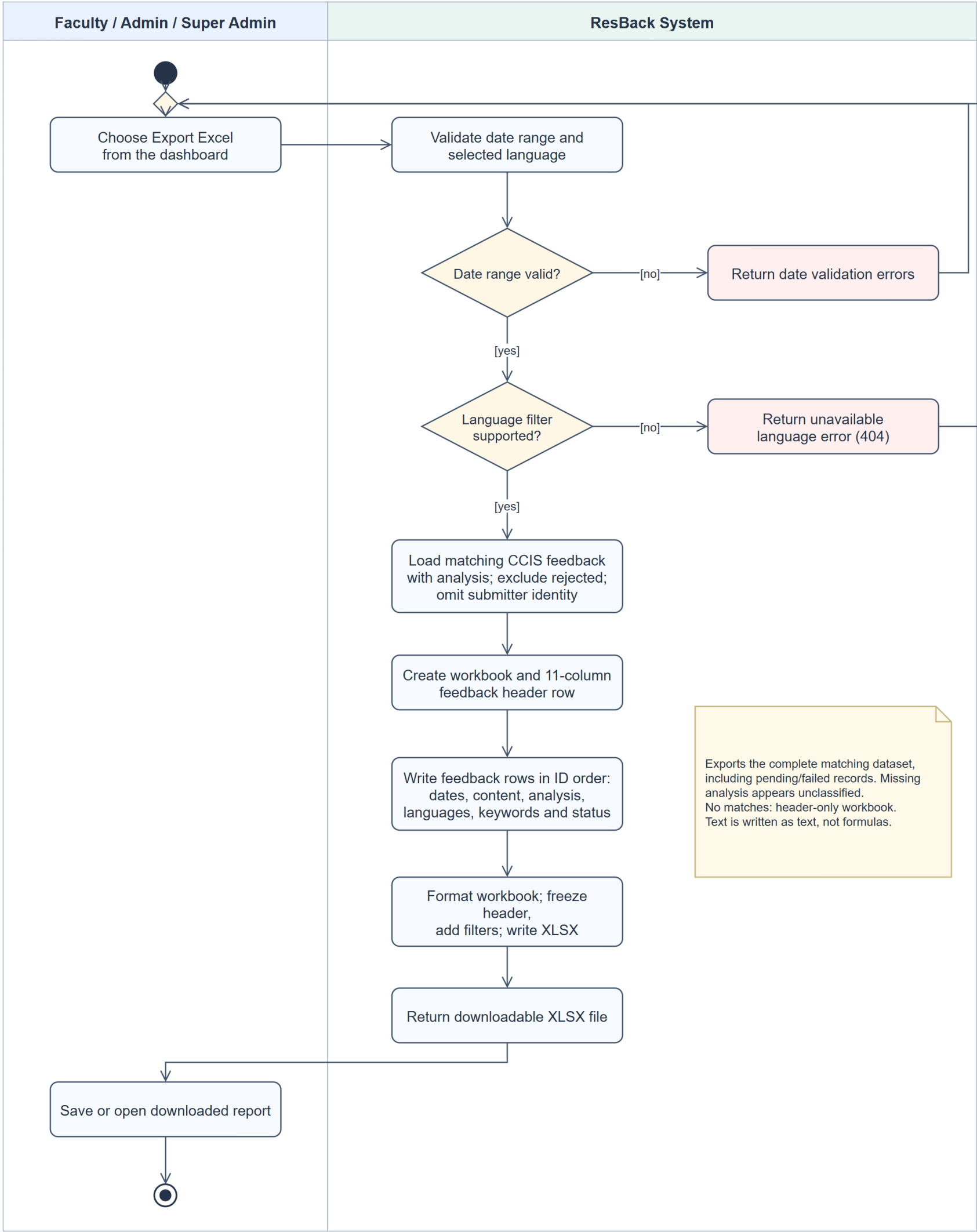


Figure 16 Activity diagram for analyze feedback enlarged continuation panel 2 of 2

act 08 | Export Feedback to Excel

Precondition: signed in, active and authorized for reports. Uses the dashboard date and language filters.

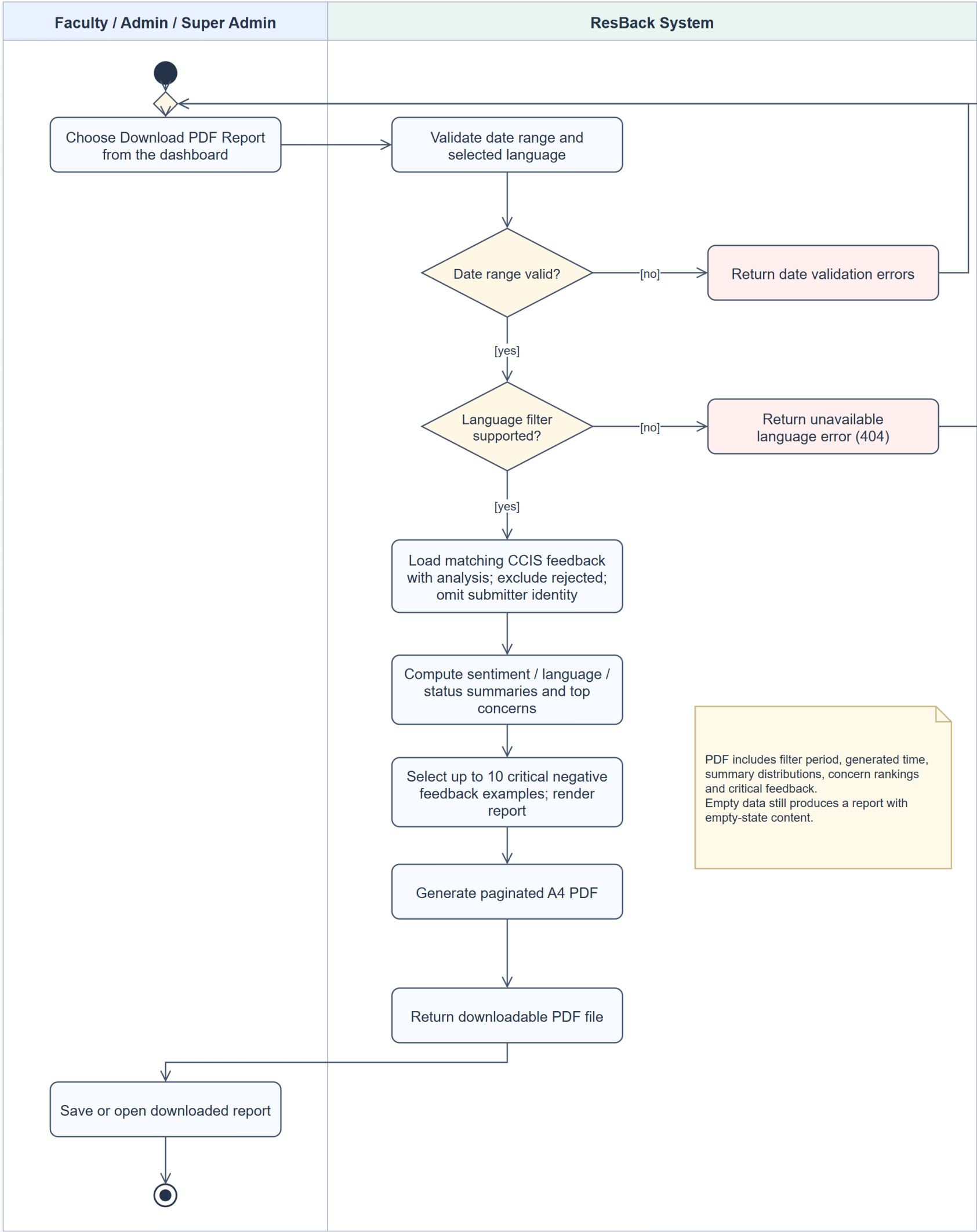


Added detail for Lab 6 p.5 | Source: FeedbackExportController.php, FeedbackDateRange.php
ResBack / Team Mira | Current implementation: 25 September 2026 | Guards in [brackets]; diamonds = decision / merge
Notation reference: <https://www.drawio.com/docs/diagram-types/uml/activity-diagrams/>

Figure 17 Activity diagram for export feedback to Excel

act 09 | Generate a PDF Analytics Report

Precondition: signed in, active and authorized for reports. Uses the dashboard date and language filters.

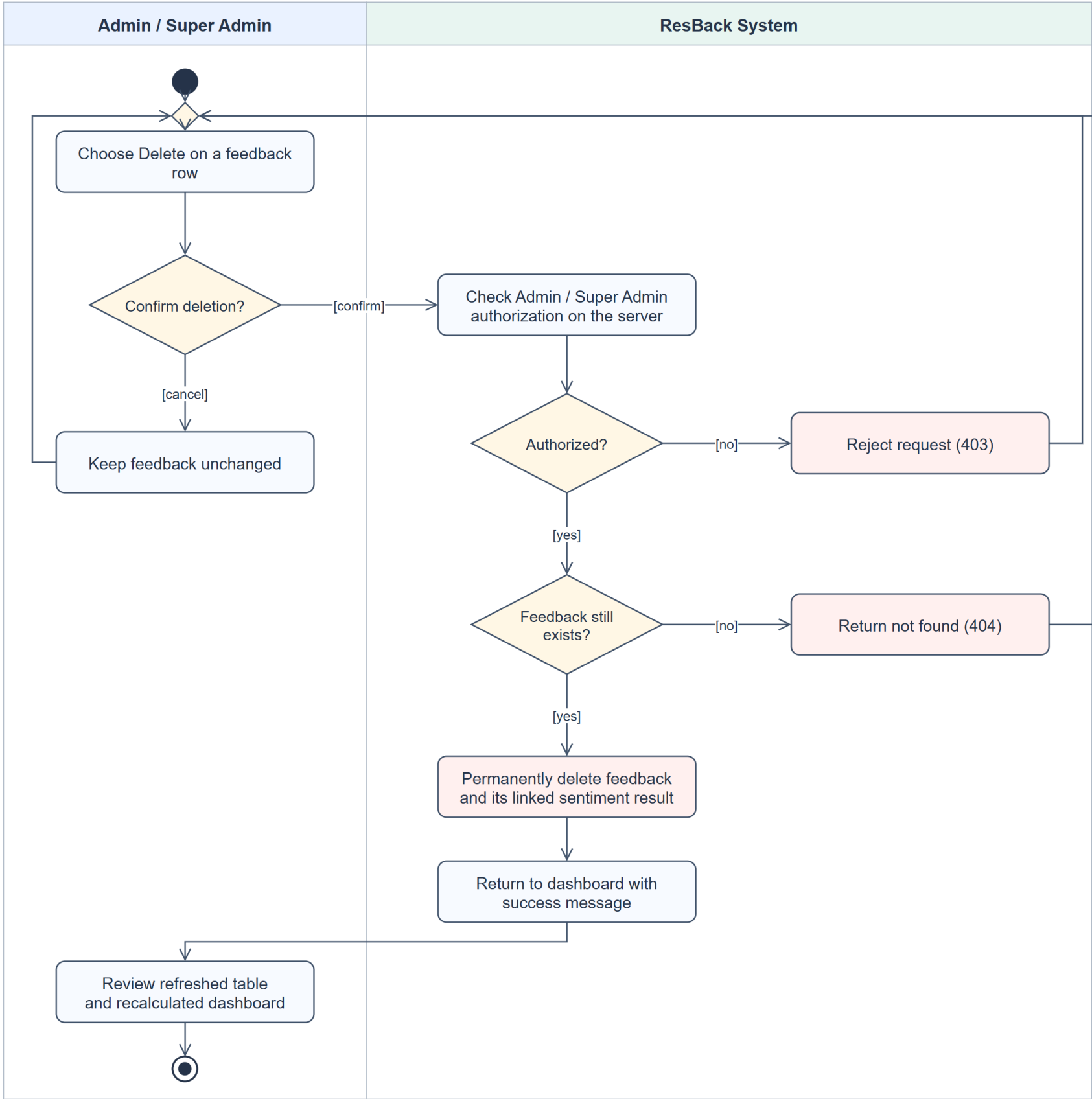


Added detail for Lab 6 p.5 | Source: FeedbackReportController.php, FeedbackDateRange.php
ResBack / Team Mira | Current implementation: 25 September 2026 | Guards in [brackets]; diamonds = decision / merge
Notation reference: <https://www.drawio.com/docs/diagram-types/uml/activity-diagrams/>

Figure 18 Activity diagram for generate a PDF analytics report

act 10 | Delete Feedback

Precondition: signed in and active. Faculty and Student deletion requests are forbidden (403).



Added detail for Lab 6 p.4/p.5 | Source: FeedbackController::destroy, dashboard/partials/feedback-table.blade.php

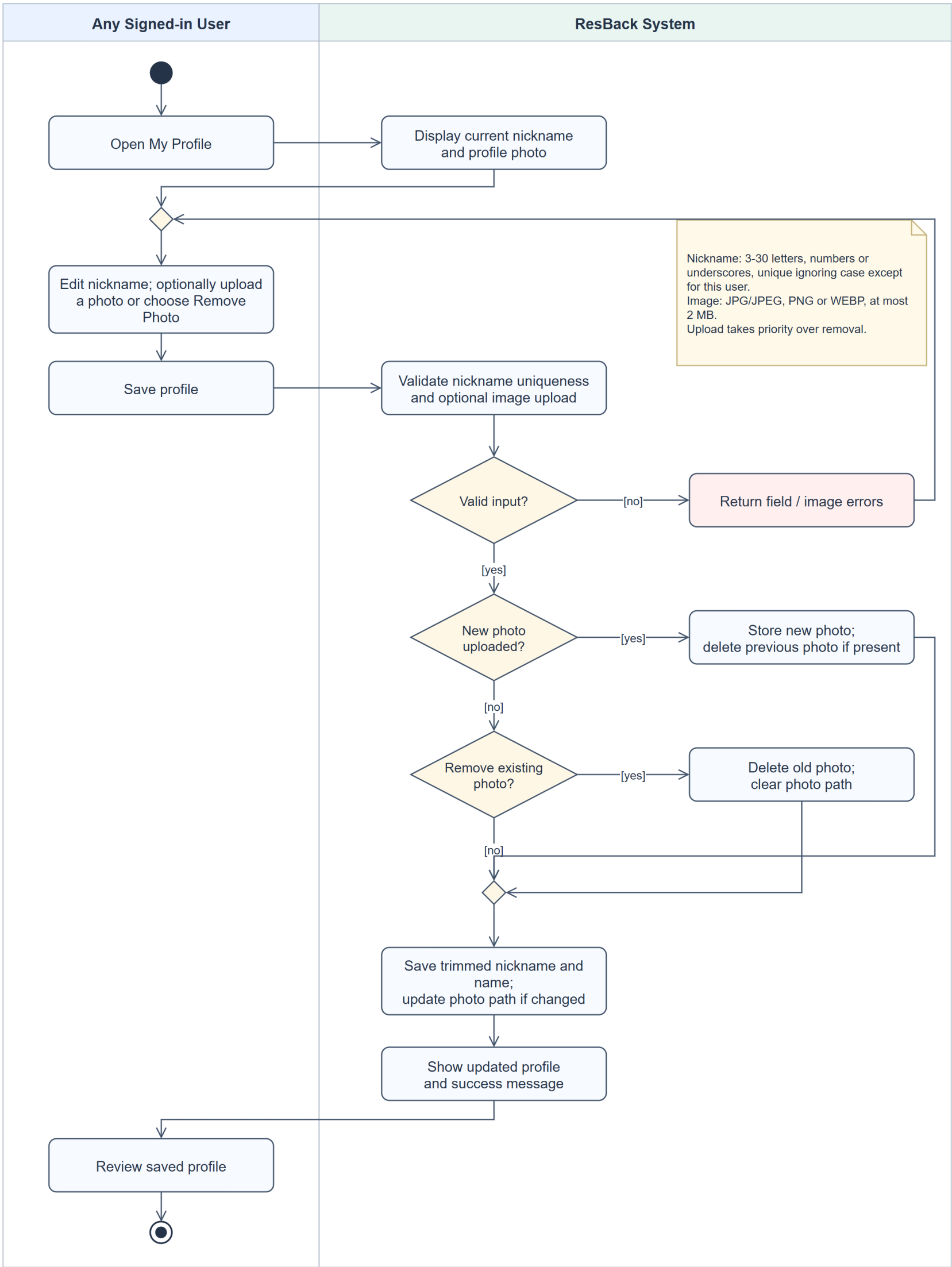
ResBack / Team Mira | Current implementation: 25 September 2026 | Guards in [brackets]; diamonds = decision / merge

Notation reference: <https://www.drawio.com/docs/diagram-types/uml/activity-diagrams/>

Figure 19 Activity diagram for delete feedback

act 11 | Edit My Profile

Precondition: account is active. Users edit their own nickname and photo; email/password editing is not implemented here.

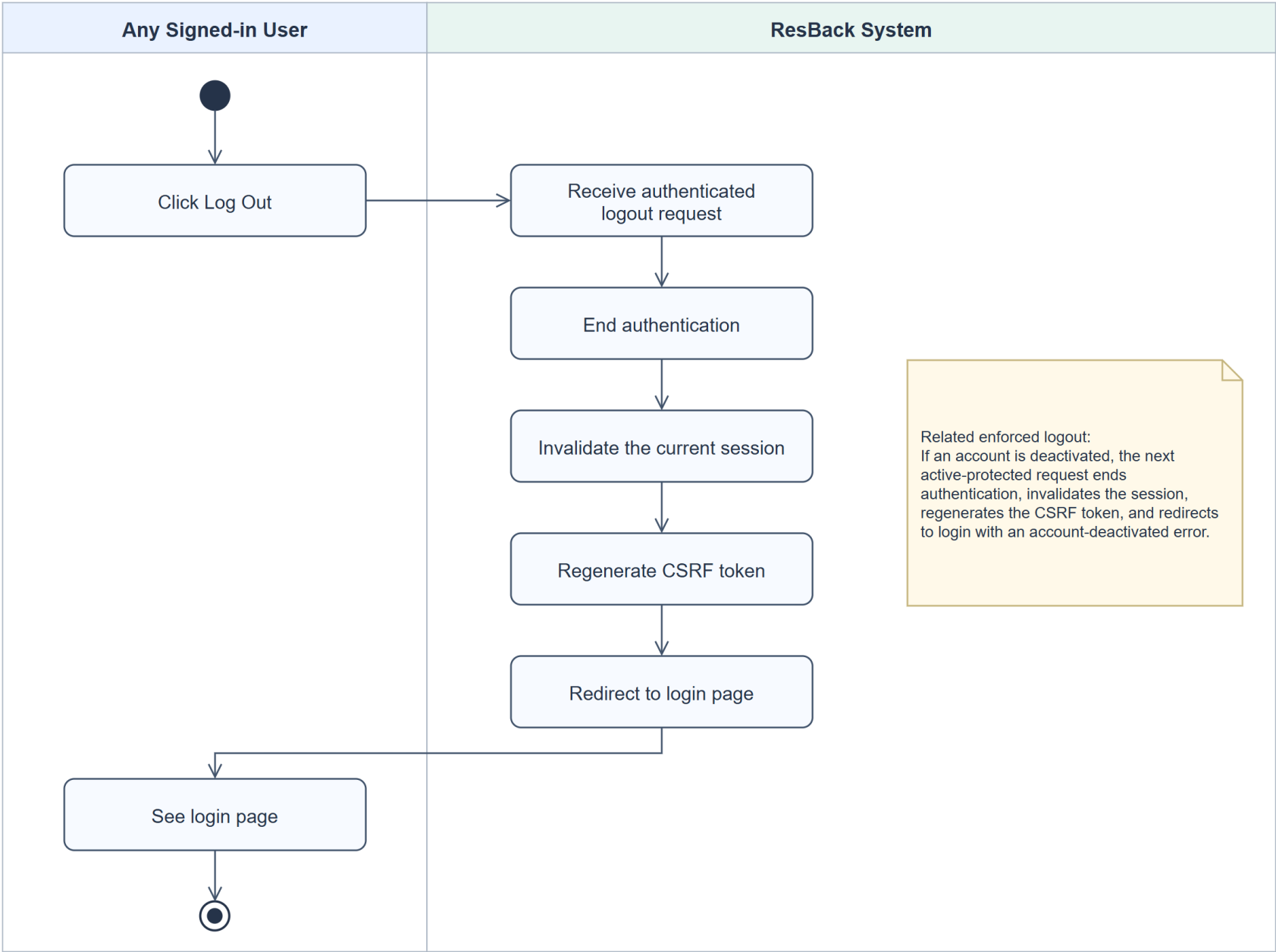


New activity | Source: ProfileController.php, profile/edit.blade.php, User.php
ResBack / Team Mira | Current implementation: 25 September 2026 | Guards in [brackets]; diamonds = decision / merge
Notation reference: <https://www.drawio.com/docs/diagram-types/uml/activity-diagrams/>

Figure 20 Activity diagram for edit my profile

act 12 | Log Out

Precondition: user has an authenticated session. Logout is available to every role.



New activity | Source: LoginController::logout, ActiveAccountMiddleware.php
ResBack / Team Mira | Current implementation: 25 September 2026 | Guards in [brackets]; diamonds = decision / merge
Notation reference: <https://www.drawio.com/docs/diagram-types/uml/activity-diagrams/>

Figure 21 Activity diagram for log out

APPENDIX F CLASS DIAGRAMS

Figures 22–27 present six class views. Plus, hash, and minus symbols identify public, protected, and private members. The source sheets retain their original 01–06 labels.

Figure 22. The core domain model shows record attributes, operations, ownership, analysis relationships, and deletion behavior. User roles are stored values rather than subclasses.

Figure 23. Authentication, accounts, and profiles connect registration, login, routing, account administration, and profile editing to the User model.

Figure 24. Feedback submission and protection connect validation, rate limiting, duplicate detection, screening, and analysis. Screening occurs before external inference.

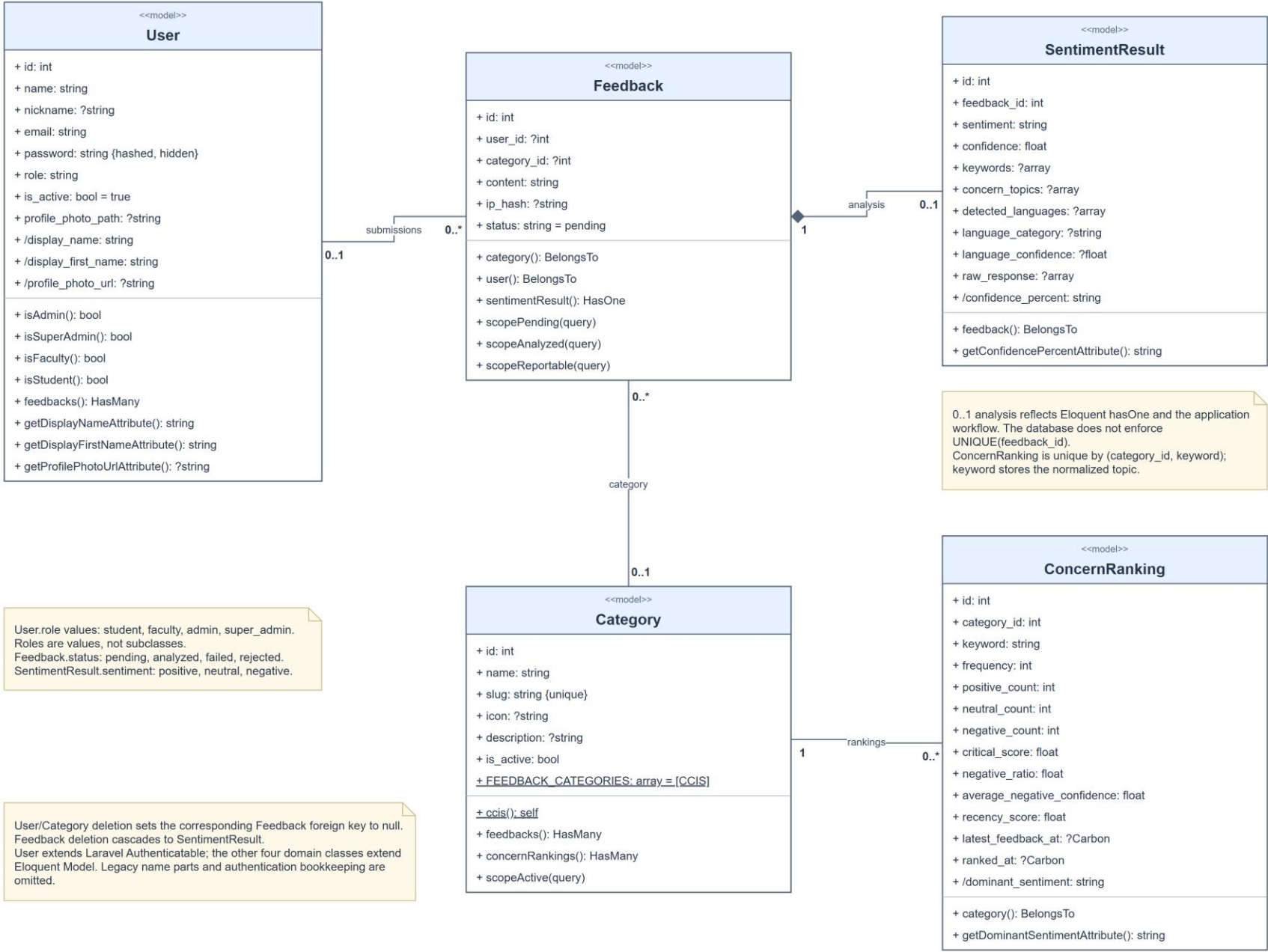
Figure 25. Sentiment analysis and concern ranking connect model-output normalization, language and topic categories, optional date boundaries, and ranking reconstruction.

Figure 26. Dashboard and report generation share filters, ranked concerns, models, and export utilities. Review outputs omit submitter identification and rejected feedback.

Figure 27. Account-access middleware checks authentication, active status, and roles. Controller authorization adds account-management restrictions.

class 01 | Core Domain Model

Application relationships and persisted attributes. Models use Eloquent dynamic properties; inherited ORM members and routine timestamps are omitted.

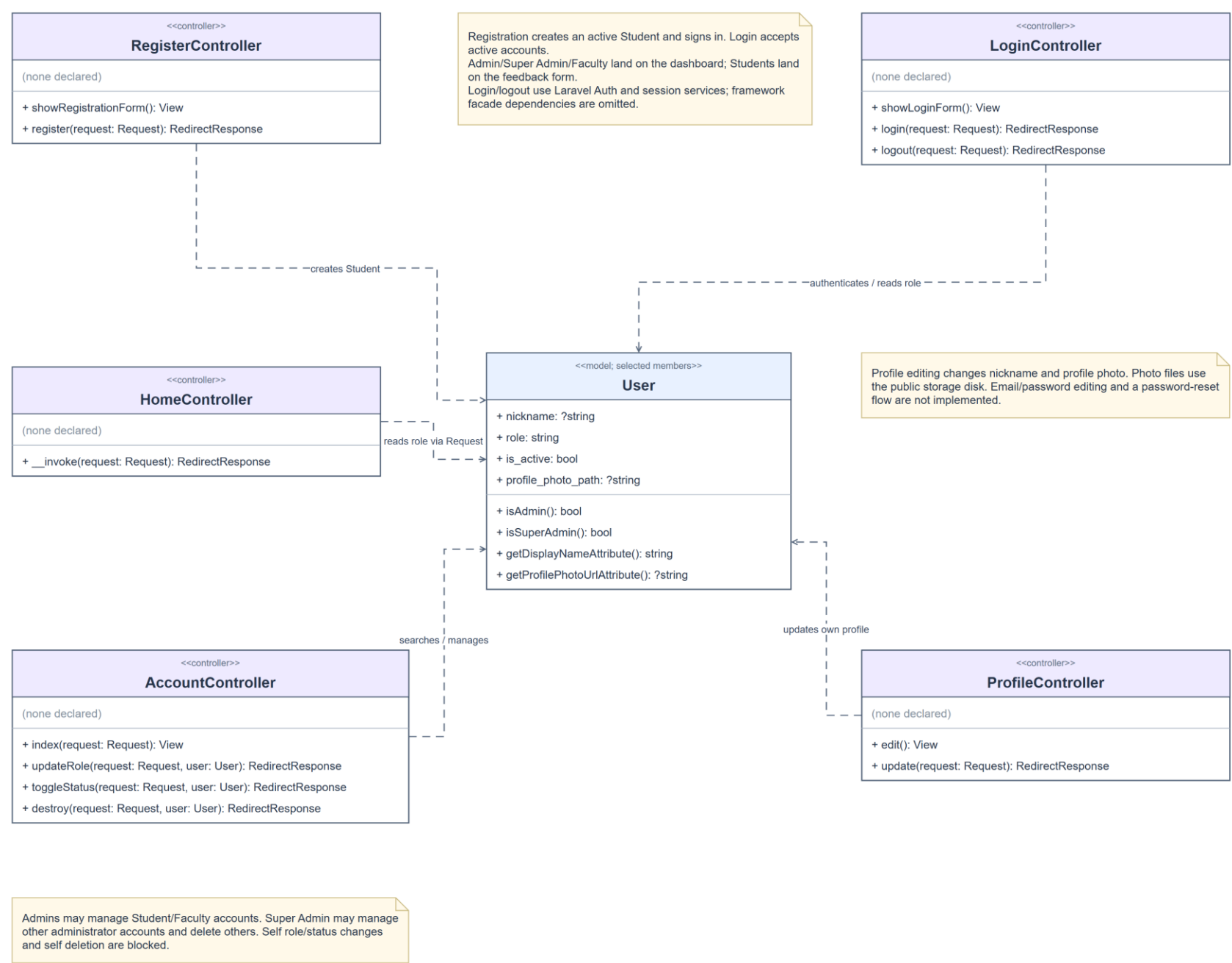


+ public # protected - private / derived ? nullable Underlined = static / constant Hollow triangle = inheritance
Solid line = association Solid diamond = composition Dashed arrow = dependency Selected implementation views; omitted details are noted.
ResBack / Team Mira | Code inspected: 25 September 2026 | UML notation: drawio.com/docs/diagram-types/uml/class-diagrams/

Figure 22 Class diagram for core domain model

class 02 | Authentication, Accounts and Profiles

Selected User members are repeated for context. All five controllers extend App\Http\Controllers\Controller; inherited members are omitted.

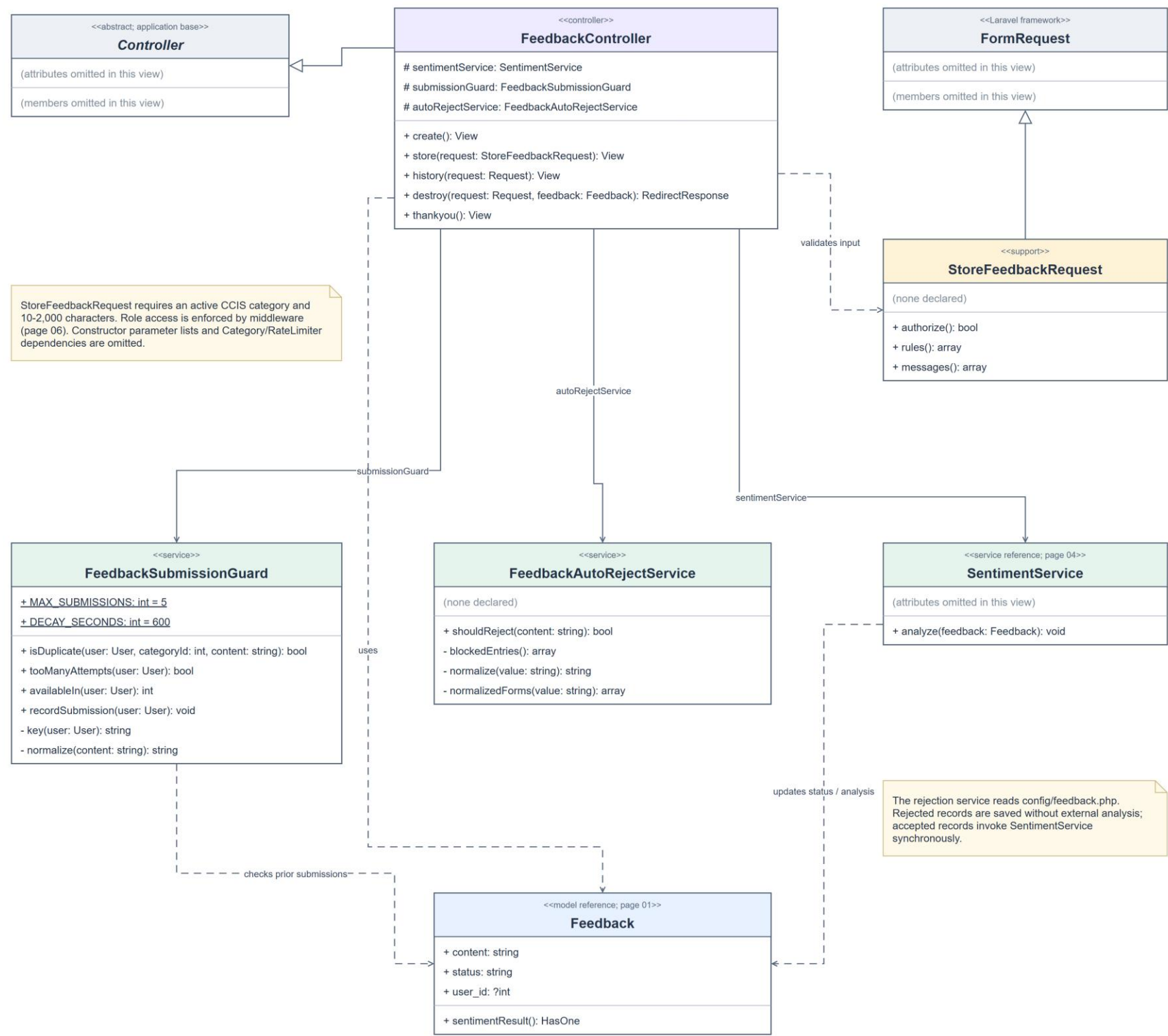


+ public # protected - private / derived ? nullable Underlined = static / constant Hollow triangle = inheritance
Solid line = association Solid diamond = composition Dashed arrow = dependency Selected implementation views; omitted details are noted.
ResBack / Team Mira | Code inspected: 25 September 2026 | UML notation: drawio.com/docs/diagram-types/uml/class-diagrams/

Figure 23 Class diagram for authentication accounts and profiles

class 03 | Feedback Submission and Protection

Constructor-injected services are stored associations. Request/model uses are dependencies. Attribute visibility matches the PHP source.

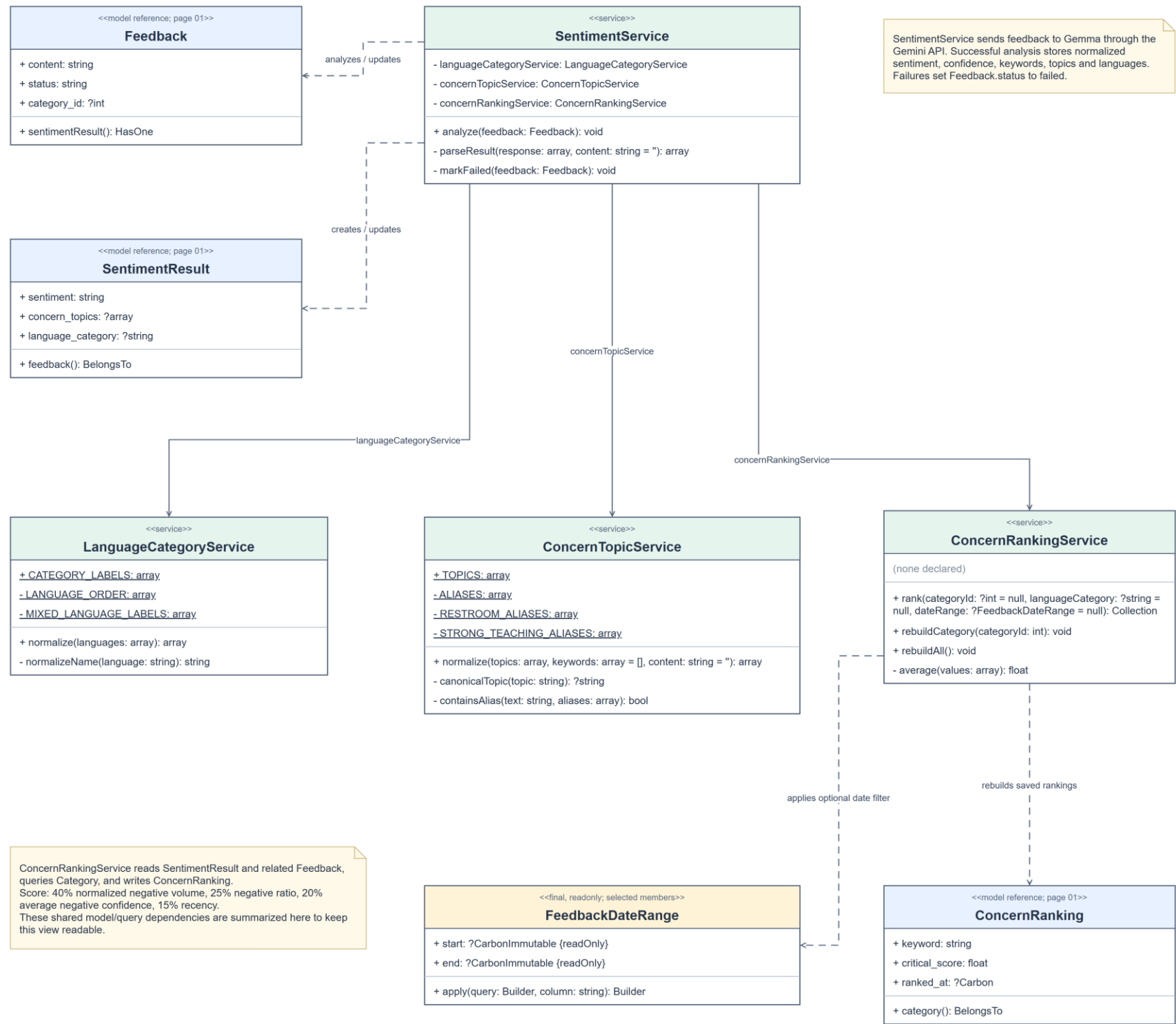


+ public # protected - private / derived ? nullable Underlined = static / constant Hollow triangle = inheritance
Solid line = association Solid diamond = composition Dashed arrow = dependency Selected implementation views; omitted details are noted.
ResBack / Team Mira | Code Inspected: 25 September 2026 | UML notation: drawio.com/docs/diagram-types/uml/class-diagrams/

Figure 24 Class diagram for feedback submission and protection

class 04 | Sentiment Analysis and Concern Ranking

Private service references are constructor-injected. Selected model dependencies are drawn; routine framework HTTP/DB calls are omitted.

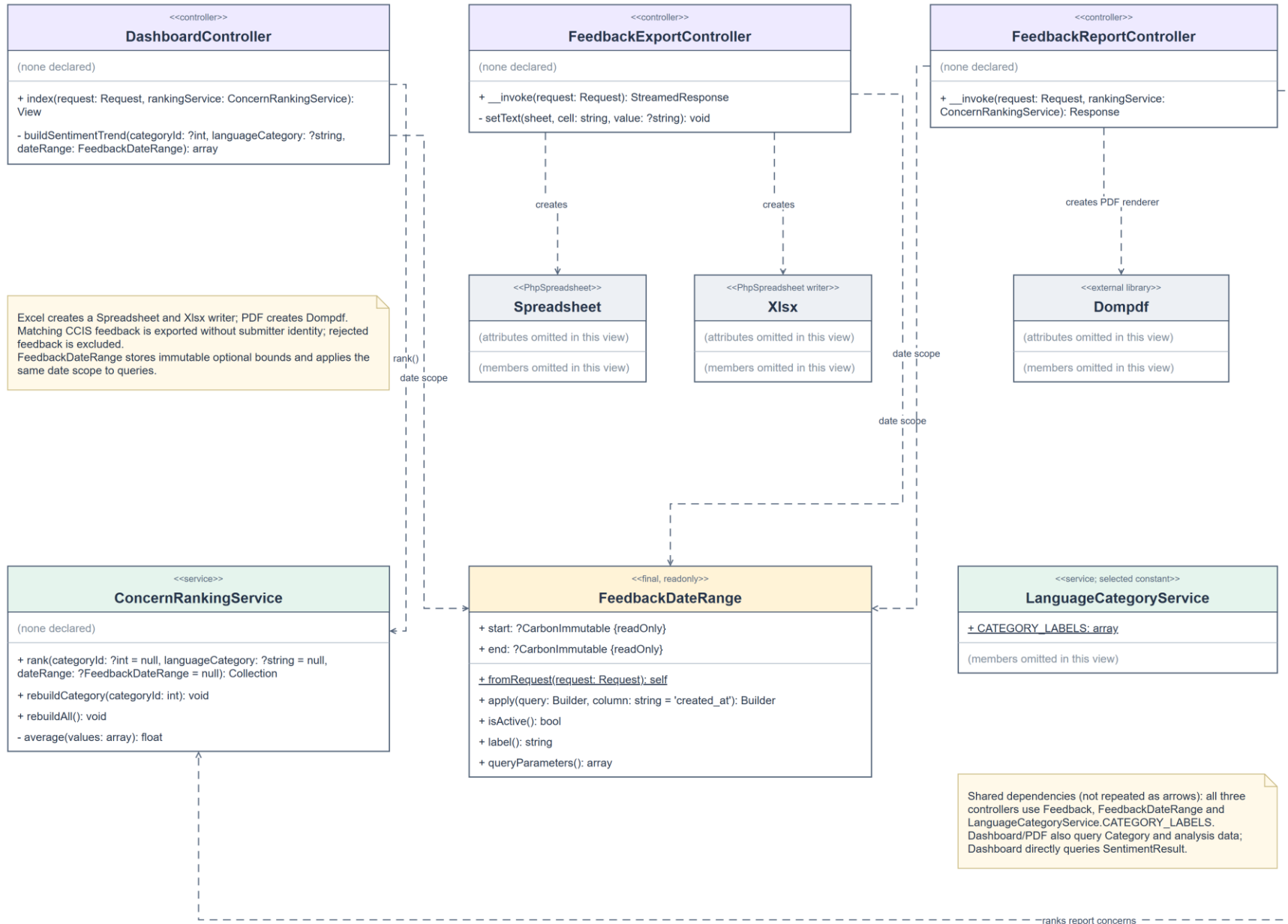


+ public # protected - private / derived ? nullable Underlined = static / constant Hollow triangle = inheritance
Solid line = association Solid diamond = composition Dashed arrow = dependency Selected Implementation views; omitted details are noted.
ResBack / Team Mira | Code inspected: 25 September 2026 | UML notation: drawio.com/docs/diagram-types/uml/class-diagrams/

Figure 25 Class diagram for sentiment analysis and concern ranking

class 05 | Dashboard and Report Generation

All controllers extend Controller. Method parameters create dependencies; these services are not stored controller attributes



* public # protected - private / derived ? nullable Underlined = static / constant Hollow triangle = inheritance
 Solid line = association Solid diamond = composition Dashed arrow = dependency Selected implementation views; omitted details are noted

ResBack / Team Mira | Code inspected: 25 September 2026 | UML notation: drawio.com/docs/diagram-types/uml/class-diagrams/

Figure 26 Class diagram for dashboard and report generation

class 06 | Account Access Middleware

Route aliases are configured in bootstrap/app.php. Dependencies below use the authenticated User obtained from Request.

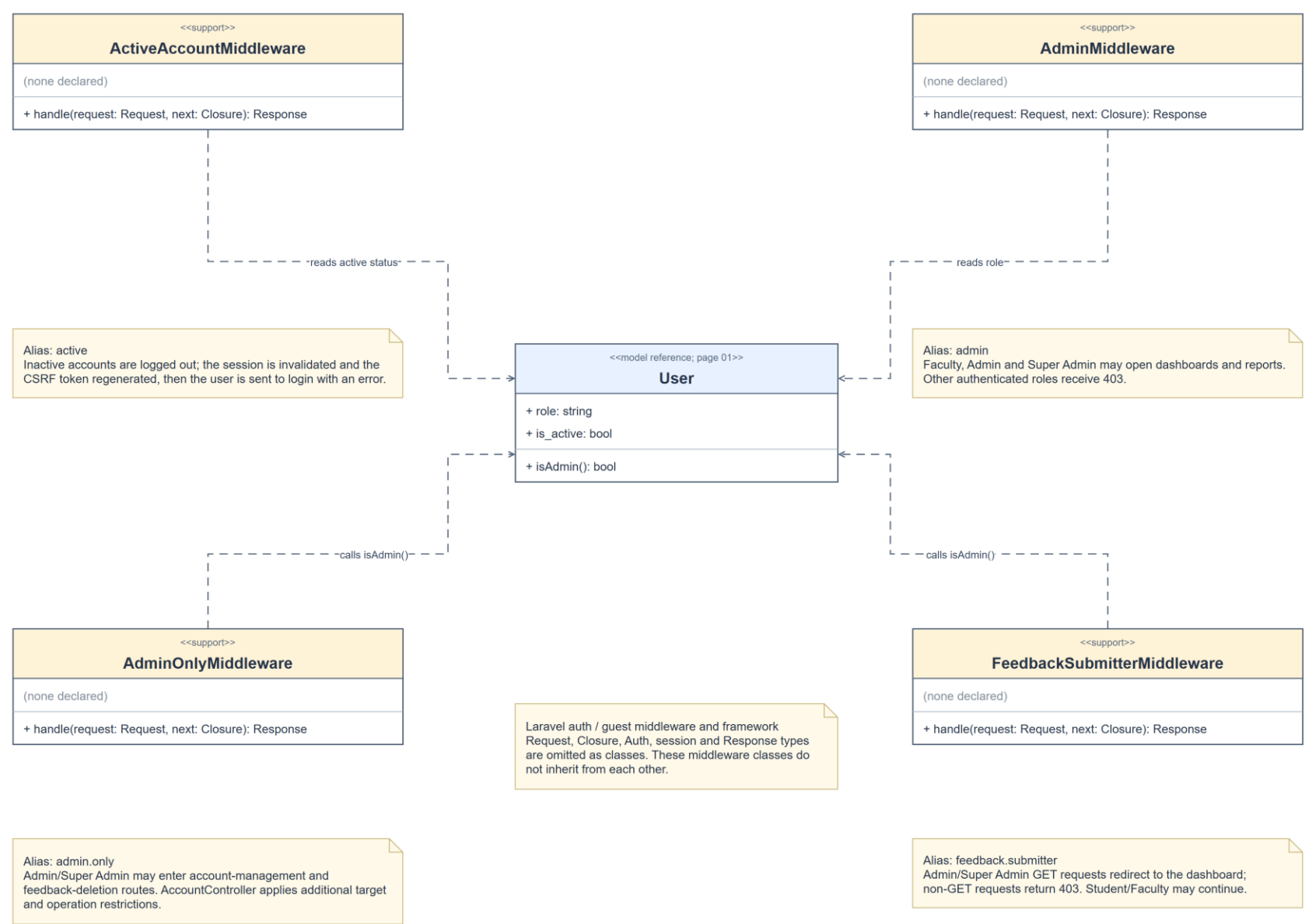


Figure 27 Class diagram for account access middleware

APPENDIX G COMPONENT DIAGRAM

Figure 28 shows the logical modules inside the Laravel application: Web Presentation, Identity and Accounts, Feedback Management, Sentiment Analysis, Concern Ranking, and Dashboard and Reporting. These are logical components of the existing application rather than separately deployed services.

Provided and required interfaces identify service contracts. Delegation connectors link application boundaries to internal components. The diagram also includes external inference, relational storage, profile-photo storage, and report libraries.

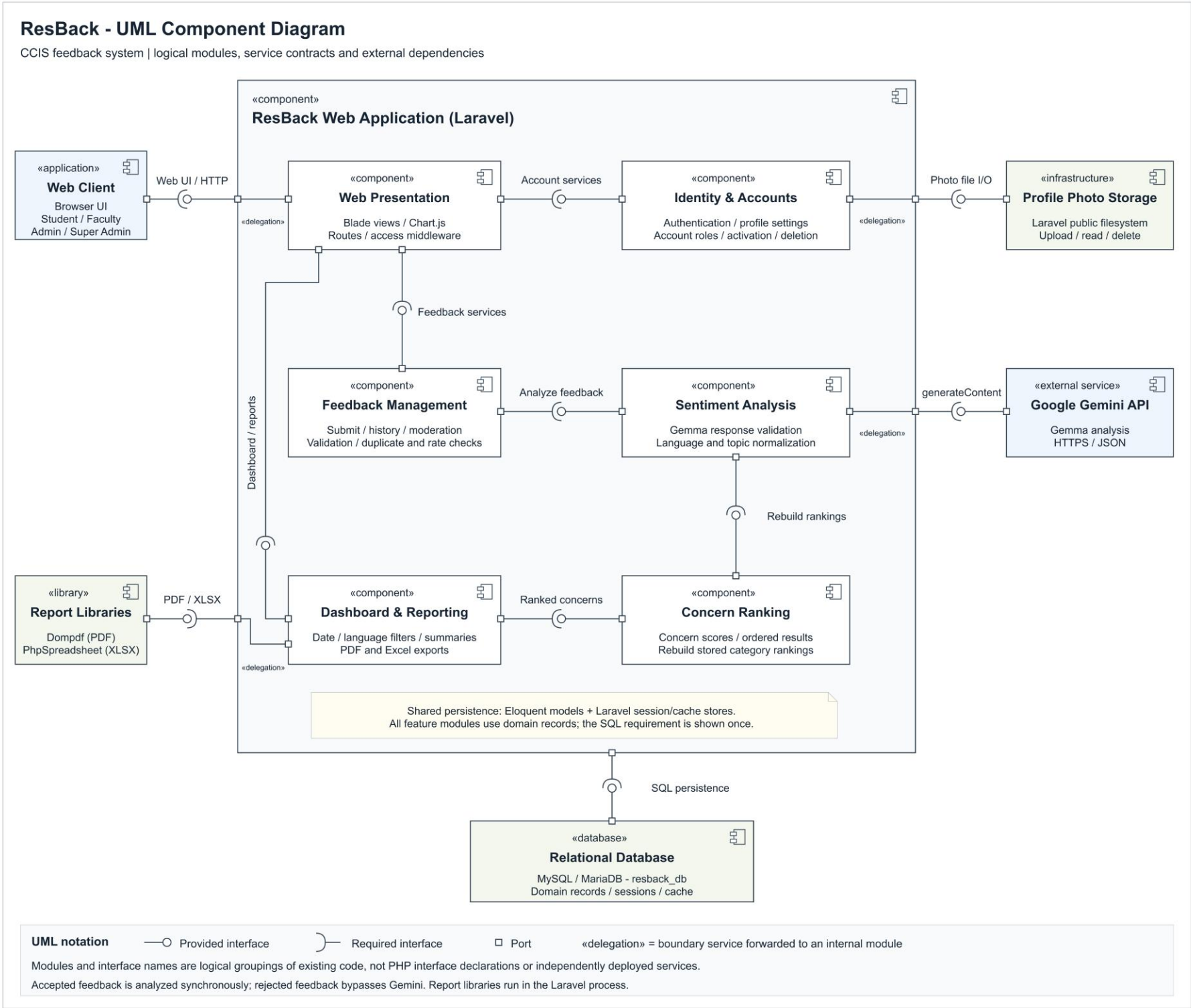


Figure 28 ResBack UML component diagram

APPENDIX H PACKAGE DIAGRAM

Figure 29 groups the application into packages for presentation, account access, feedback handling, analysis, reporting, and persistence. Dashed dependency arrows point toward the package being used. The Google API remains an external dependency.

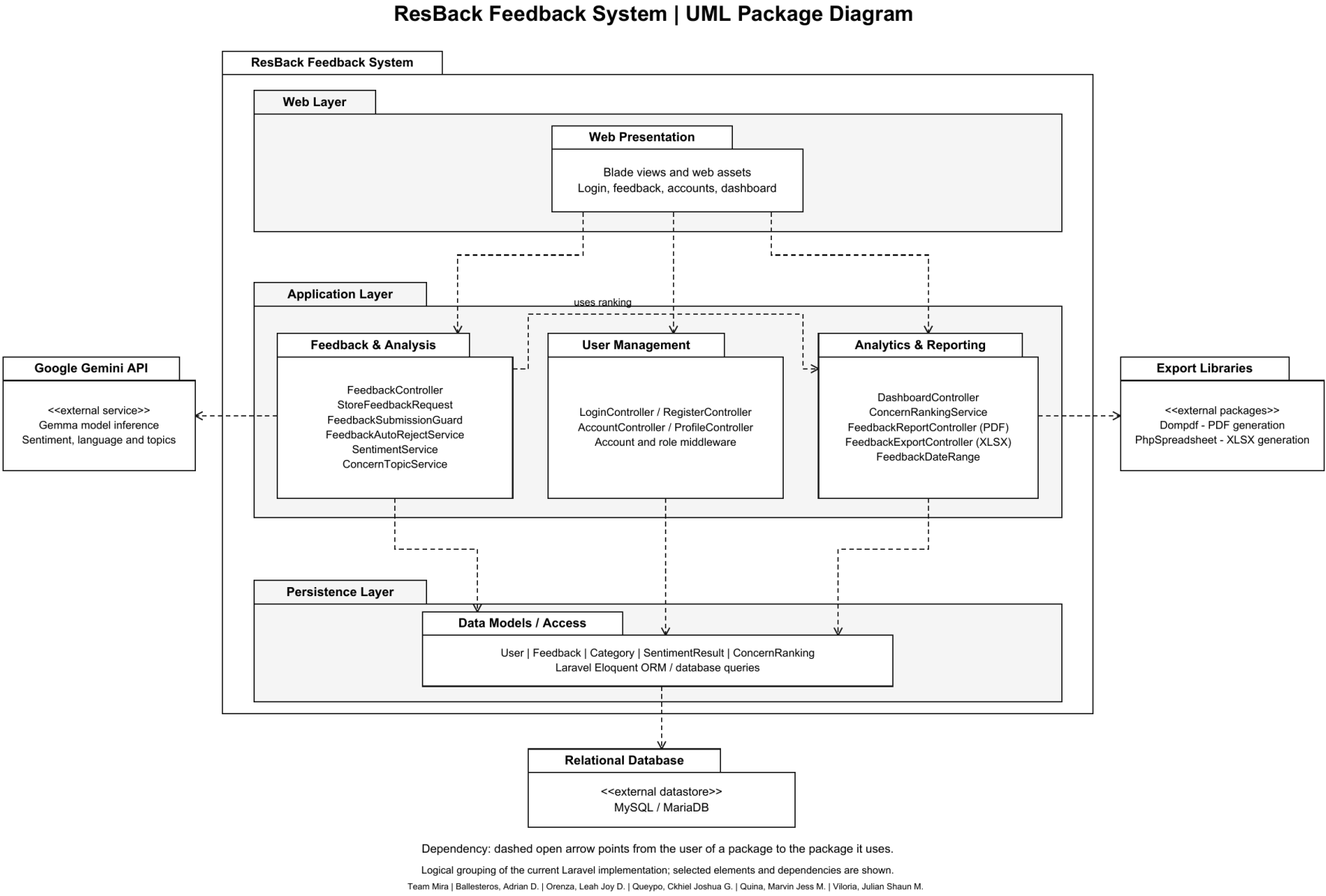


Figure 29 ResBack UML package diagram

APPENDIX I DEPLOYMENT DIAGRAMS LOCAL AND ONLINE

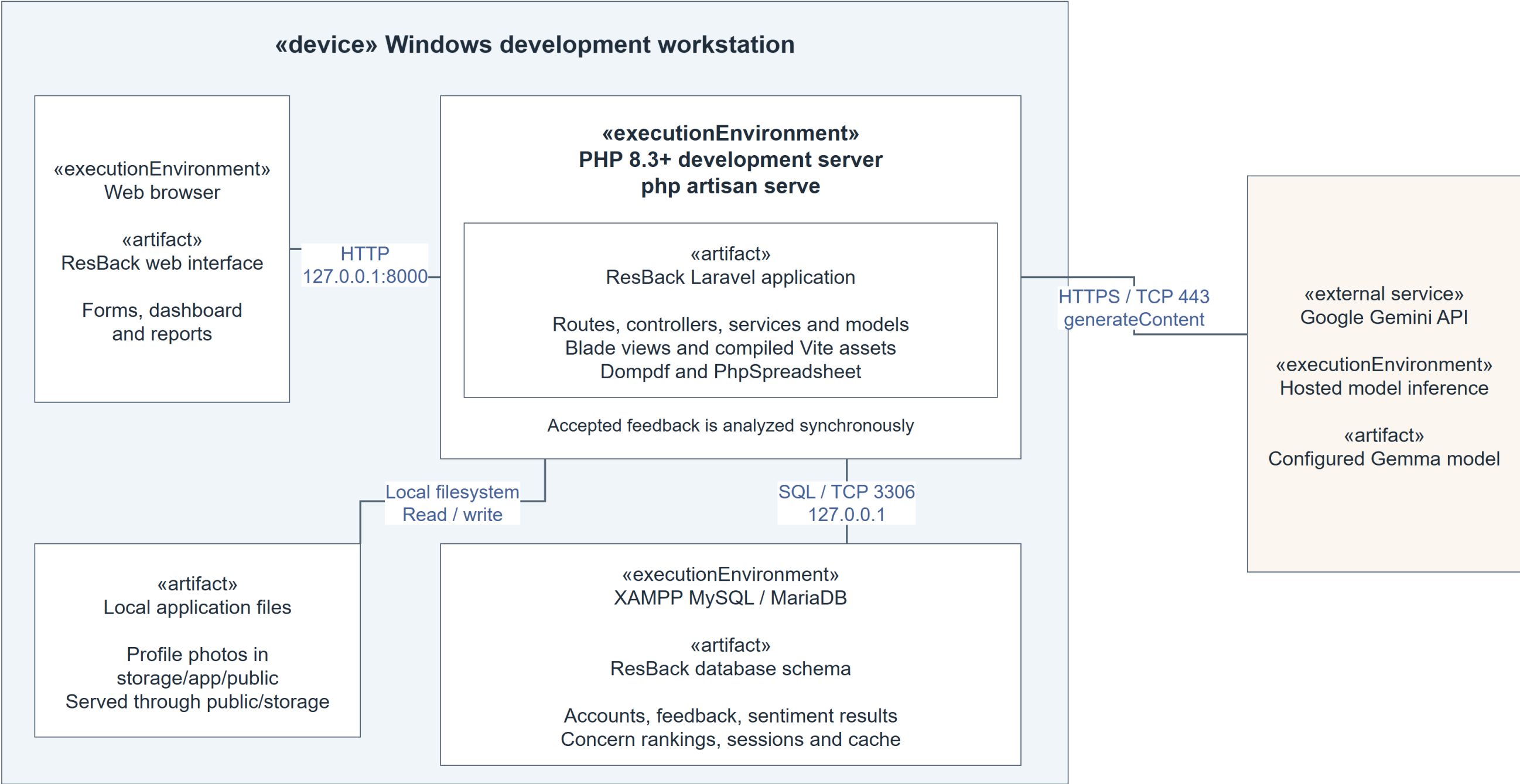
Figure 30 shows the local development deployment: the web browser, PHP development server, XAMPP MySQL or MariaDB, application files, and external Google Gemini API. Web and database services run locally, while Gemma inference requires internet access.

Figure 31 shows the proposed online deployment on Render with Supabase PostgreSQL and profile-photo storage. This remains a target architecture and requires database compatibility, persistent-storage integration, deployment configuration, and operational verification.

The local and online diagrams are separate views. Solid communication paths identify the protocols between execution environments and services.

ResBack Local Deployment Diagram

Windows workstation with PHP development server and XAMPP database



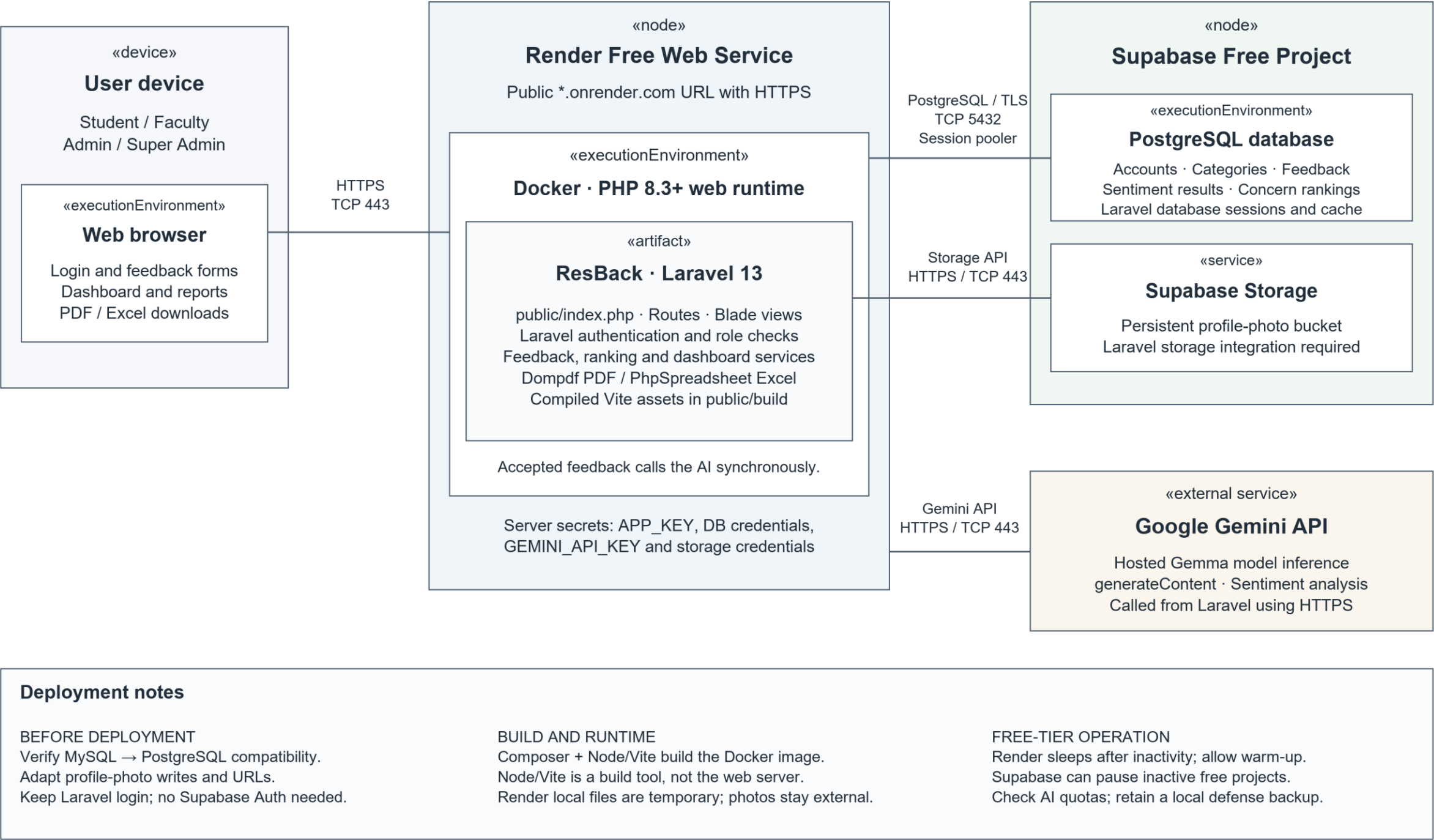
Local web and database services run on the workstation. Model inference requires an internet connection to the external API.

Figure 30 ResBack UML local deployment diagram

ResBack Deployment Diagram

PROPOSED FINAL-DEFENSE DEPLOYMENT | Render Free + Supabase Free

Target architecture only. Deployment and PostgreSQL / storage integration are not yet implemented.



Solid lines = runtime communication paths. Provider internals are abstracted. Proposed architecture · 03 October 2026

Figure 31 Proposed online deployment diagram